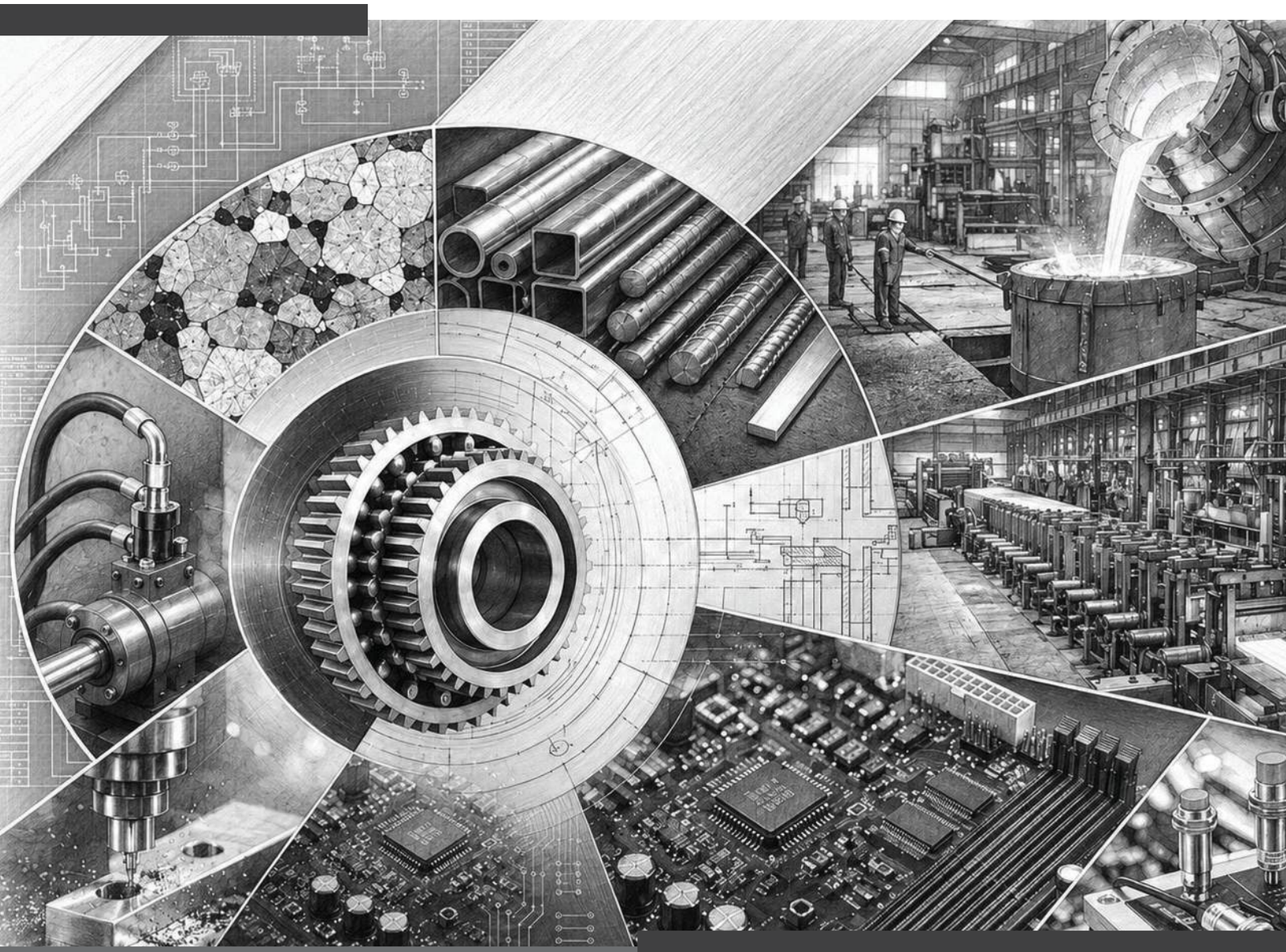
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MECHATRONICS

SEMESTER ONE **2026/27**

STUDENT BOOK

GRADE 1

Mechatronics

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Occupational Health, Safety and Environment

1

EZZSTEEL APPLIED TECHNOLOGY SCHOOLS

Unit



Unit OVERVIEW

In this unit, you will develop the knowledge and practical skills needed to apply occupational health and safety procedures in an industrial workplace. You will begin by learning the fundamental meanings of hazard, risk and prevention and how these concepts influence the decisions you make before and during work. You will then explore national occupational safety requirements, workplace responsibilities, relevant European directives, technical safety standards, product markings and other safety information used in industrial environments. You will also examine occupational health and work-related stress and consider how working conditions, workload, fatigue, noise, heat and work organisation can affect both health and safe performance.

As the unit progresses, you will apply this knowledge to industrial situations. You will identify and record hazards, assess risks using likelihood and severity, select appropriate preventive and protective measures, and examine workplace and work-equipment safety requirements. You will also learn how to select, fit and inspect Personal Protective Equipment (PPE) according to the hazards and work activities involved. Your practical work will include steel-workshop hazard identification, workplace risk assessment, interpretation of safety information, PPE selection and inspection, and workplace and equipment-safety checks. You will be expected to record your findings clearly, justify the controls you select and follow the authorised safety procedures for each task. Electrical hazards may be considered as examples when inspecting workplaces and equipment. Detailed electrical protection and isolation procedures are addressed separately within the relevant electrical-safety competency.



Primary evidence

Steel-workshop hazard records; risk and corrective-action records; PPE selection, fitting and inspection records; workplace and equipment-safety observations; assessor observations and applied workplace tasks

INTENDED LEARNING OUTCOMES (ILOs)

ILO1 Define hazard, risk and prevention and explain the difference between a hazard and the level of risk, associated with it.

ILO2 Interpret relevant national safety laws and European directives when considering occupational safety requirements in an industrial workplace.

ILO3 Analyse work-related stress and occupational-health risks and recognise appropriate preventive, reporting and workplace-control measures.

ILO4 Identify appropriate personal protective equipment (PPE) for a given hazard, work activity or exposed part of the body.

ILO5 Analyse workplace and work-equipment safety requirements and identify conditions that require preventive or corrective action.

ILO6 Apply appropriate preventive safety measures in simulated workplace situations and explain the relationship between an identified risk and the control selected.

UNIT MAP

Sequence	Theory lessons	Practical sessions
1	Foundations of Occupational Health and Safety	Classify hazards and unsafe conditions from photographed or simulated workplace situations and begin completing a hazard record.
2	National Safety Legislation and Workplace Responsibilities	Analyse workplace cases, identify relevant safety responsibilities and record appropriate corrective actions.

Sequence	Theory lessons	Practical sessions
3	European Directives and Technical Safety Standards	Inspect equipment labels, safety markings and manufacturer information and identify the precautions required for safe use.
4	Occupational Health and Work-Related Stress	Analyse workplace health and stress scenarios and recommend suitable preventive measures, reporting routes or controls.
5	Industrial Hazard Identification and Recording	Complete structured hazard identification and recording in a simulated steel-workshop environment.
6	Risk Assessment and Preventive Measures	Complete a risk assessment and corrective-action record using likelihood, severity, risk priority and appropriate controls.
7	Personal Protective Equipment and Equipment-Safety Inspection	Select, fit and inspect PPE and complete workplace and equipment-safety checks.
8	Unit Revision	
9	Formative Assessment	

Before you begin any task in a workshop or production area, you need to understand what could cause harm and what you can do to prevent that harm from occurring. This is one of the foundations of safe industrial work.

A hazard is something that has the potential to cause harm. The risk depends on the conditions in which you encounter that hazard, including how likely it is that harm will occur and how serious the consequences could be. This means that the same hazard can present very different levels of risk depending on the way a task is organised, the equipment being used, the working environment and the controls already in place. For example, a hazardous substance that is correctly stored and controlled may present a relatively low risk. If the same substance is used incorrectly, without suitable controls or protective equipment, the level of risk can become much greater. Safe working therefore requires you to do more than simply notice a hazard. You need to consider the conditions of exposure, assess the risk and decide what action is needed.

Prevention is central to this process. Wherever possible, risks should be eliminated or reduced at their source. If a hazard cannot be completely removed, you should consider safer alternatives,

engineering or collective controls, safe working procedures and, where necessary, appropriate PPE. PPE is an important form of protection, but it should not automatically be your first response when a risk can be controlled more effectively in another way. Safety decisions are also guided by more than personal experience. Industrial workplaces operate within a system of national legislation, European requirements, technical standards, manufacturer instructions and safety information. These requirements help define responsibilities, establish how risks should be managed and provide information about the safe design, use and operation of equipment. Throughout this unit, you will learn how to interpret this information and connect it to the decisions you make in practical workplace situations.

Occupational safety is closely connected to occupational health. Health at work involves your physical and psychological well-being as well as protection from immediate injury. Working conditions such as excessive noise or heat, fatigue, unsuitable workload, unclear responsibilities, poor communication or badly organised work can affect your health and may also reduce your ability to work safely. Recognising these conditions and knowing when and how to report them is therefore an important part of professional safety practice. Much of your work in this unit will be practical. You will inspect simulated industrial and steel-workshop situations, identify and record hazards, consider their possible consequences, assess risk and select suitable preventive measures. You will also examine workplace and work-equipment safety conditions and learn how to choose, fit and inspect PPE according to the work being carried out.

As you work through the unit, try to develop a simple professional habit as indicated in the figure below:



LESSON 1

Foundations of Occupational Health and Safety

WHY SAFETY STARTS BEFORE THE TASK



When you enter a workshop or production area, you may be working near sharp materials, machinery, chemicals, work at height, moving objects and other sources of harm.

Working safely does not begin with putting on PPE. It begins by understanding:

- what could cause harm
- how you or another person could be exposed
- what injury or ill health could result
- how likely it is that harm could occur
- how serious the consequences could be
- what can be done to prevent the event from occurring.

This way of thinking is important because a hazard and a risk are not the same thing. A hazard may remain present, while the level of risk changes according to the conditions in which you work.

WARM-UP

SAME HAZARD, DIFFERENT RISK

Study these two situations.

Situation A

A sharp edge on a metal sheet is positioned behind a suitable barrier. The area is controlled, and workers cannot normally reach the edge.

Situation B

The same sharp metal edge is exposed beside an area where people regularly walk and handle materials.

Discuss the following questions with a partner:

1. Has the hazard itself changed?
2. What is different between the two situations?
3. In which situation is contact with the sharp edge more likely?
4. In which situation could the risk be greater?
5. What action could prevent someone from being exposed?

Record your starting idea below:

.....

.....

By the end of this lesson, I will be able to…

1. Define hazard, risk and prevention
2. Distinguish a hazard/unsafe condition from the possible harm and from a preventive measure.
3. Distinguish a hazard from a risk
4. Explain why prevention is required before work begins

KEY TERMS

Occupational health and safety • Hazard • Exposure • Harm • Risk • Likelihood • Severity • Safety • Occupational health • Prevention

SUBTOPIC 1.1 What Is Occupational Health and Safety?

Safety is an essential part of professional industrial work. Protecting people during work is both a formal workplace responsibility and an important professional duty. An accident at work can result in cuts, burns, fractures or other injuries. More serious incidents may cause permanent disability or death. The effects are not limited to the injured person.

Accidents can also affect families, other workers and the organization through lost working time, reduced productivity and additional costs. Occupational health and safety therefore aim to create working conditions in which hazards are recognized, and risks are controlled before they result in harm.



Definition - SAFETY

Safety is a condition in which people are not exposed to unacceptable levels of danger.

This does not mean that every hazard has disappeared. In many workplaces, hazards are part of the activity or equipment. Safe work depends on recognizing those hazards, understanding the risks they create and reducing the risks to an acceptable level.

OCCUPATIONAL HEALTH

Your health at work involves more than simply avoiding an accident.

Occupational health includes your:

- **physical well-being**
- **mental well-being**
- **social well-being**

A healthy workplace should also consider whether the work is suitable for the physical and psychological capabilities of the people carrying it out.

Good occupational-health practice therefore aims to:

1. adapt work, where appropriate, to the capabilities of the individual
2. protect people from health risks created by their working conditions
3. support physical, mental and social well-being at work
4. prevent or reduce occupational risks before they cause harm

Definition - OCCUPATIONAL HEALTH

Occupational health is concerned with protecting and supporting your physical, mental and social well-being in relation to the work you perform and the conditions in which you perform it.

THINK ABOUT

A worker finishes a shift without suffering an accident. Does this automatically mean that the workplace was healthy and safe?

No conclusion should be made from the absence of an accident alone. A person could still have been exposed to excessive noise, harmful substances or another unsafe condition without an immediate visible injury. Safety therefore requires you to recognize potential harm before an accident occurs.



Workplace Well-being



1) Physical well-being
Safe posture, healthy body, and personal protection.



2) Mental well-being
Concentration, calmness, focus, and mental well-being.



3) Social well-being
Teamwork, communication, and supportive interaction with colleagues.



Figure 1.1. Occupational health includes physical, mental and social well-being at work.

SUBTOPIC 1.2 FROM HAZARD TO RISK

Two of the most important terms you will use throughout this unit are hazard and risk. They are closely connected, but they do not mean the same thing.

HAZARD

A hazard exists because something has the potential to cause harm. The hazard may be associated with an object, substance, activity or situation.

Definition - HAZARD - INCIDENT - ACCIDENT

HAZARD

A hazard is an inherent property or characteristic of an object, substance, activity or situation that has the potential to cause harm to people, property or the environment. A hazard can exist even when nobody is currently being exposed to it.

INCIDENT

An unplanned event that could result in, or does result in, injury, ill health, damage or other loss.

ACCIDENT

An unplanned event that results in injury or ill health.

REMEMBER

Not every incident causes harm. An incident may occur without injury or damage, while an accident results in harm such as injury or ill health.

The difference is mainly whether harm occurred:

- Incident: An unplanned event that could cause harm or loss but may not actually cause harm.
- Accident: An unplanned event that results in injury or ill health.

Example:

- A worker slips on a wet floor but stays on their feet → Incident
- A worker slips, falls, and breaks their arm → Accident



So, in simple terms:

Incident = an unplanned event; harm may or may not occur.

Accident = an unplanned event that results in harm.

Hazard or hazardous factor	How harm could occur	Possible consequence
Sharp metal edge	A hand contacts the edge during handling	Cut, laceration or more serious injury
Ladder or work at height	A person loses balance or falls	Fracture or other fall injury
Sulphuric acid	Contact with the substance or inhalation of vapor	Chemical burns or respiratory harm
Loose material in a walking route	A person steps on or strikes the material	Trip, impact or crushing injury

Notice that the **injury is not a hazard.**

For example:

Hazard: sharp metal edge

Possible harm: cut or laceration

Writing "cut" as the hazard would therefore be incorrect. The cut is what might happen because someone becomes exposed to the hazard.

EXPOSURE:

A hazard does not automatically cause an injury simply because it exists. There must normally be a way in which a person can come into contact with it. This is known as exposure.



Definition - EXPOSURE

Exposure describes the condition or event through which a person comes into contact with, or is affected by, a hazard.

For example: Hazard → Exposure → Harm

Sharp metal edge → Hand contacts the edge → Cut

or

Chemical substance → Splash reaches the skin → Chemical burn



When you identify a hazard, ask yourself:

Who could be exposed, and how could the exposure happen?

HARM:

Definition - HARM

Harm is the injury, ill health, damage or other negative consequences that could result from exposure to a hazard.

This gives you three different ideas:

Question	What you are identifying
What could cause harm?	Hazard
How could someone come into contact with it?	Exposure
What could happen to the person?	Harm

Being able to separate these three ideas will help you describe workplace conditions clearly.

WHAT IS RISK?

A hazard tells you what has the potential to cause harm. Risk asks a different question:

How likely is that harm to occur under these particular conditions, and how serious could the result be?

Definition - RISK

Risk is the likelihood that a hazard will actually result in harm under particular conditions of use or exposure, taking account of the seriousness of the possible consequence.

Two ideas are therefore important when thinking about risk:

1. LIKELIHOOD

How likely is it that exposure or a harmful event will happen?

2. SEVERITY

How serious could the resulting injury or ill health be?

You will learn formal risk-assessment methods later in this unit. For now, remember the basic relationship: Risk depends on how likely exposure is and how serious the resulting harm could be. If either the likelihood or the possible severity increases, the level of concern may also increase.
THE HAZARD CAN STAY THE SAME WHILE THE RISK CHANGES.
Consider an industrial solvent.

CONDITION A:

The solvent is:

- inside a closed container
- kept in a ventilated storage area
- stored correctly
- protected from unnecessary access

The chemical still has hazardous properties, but the opportunity for exposure is limited.

CONDITION B:

The same solvent is:

- in an open container
- being handled in a poorly ventilated area
- used without suitable controls

The chemical hazard itself has not changed. What has changed is the condition of exposure. Because exposure is more likely in Condition B, the risk is greater.

REMEMBER

Hazard = the potential to cause harm.
Risk = the possibility of that harm occurring under the conditions present.

1.1 CHECK YOUR UNDERSTANDING

For each statement, decide whether it describes a hazard, exposure, or harm.

Statement	Hazard / Exposure / Harm
A sharp edge on a metal sheet	
A worker's hand contacts the sharp edge	
A deep cut to the hand	
A corrosive chemical	
Vapor from the chemical is inhaled	
Respiratory injury	

Hazard source	Person exposed during a task
(a) Cutting tasks using a butcher's knife.	(b) Cut-resistant gloves protect the hands.

Figure 1.2. Harm occurs when a person is exposed to a hazard. (Di Gennaro, Chiappetta and Chillemi (2023), New Course in Mechanical Technology, Volume 1, Hoepli, Fig. A1.2).

SUBTOPIC 1.3 PREVENTION — ACT BEFORE HARM OCCURS

Recognising a hazard is only the beginning. Once you know that something could cause harm, you need to think about how the unwanted event can be prevented.

Definition - PREVENTION

Prevention is the set of measures, procedures and organisational arrangements used to anticipate a harmful event and reduce the likelihood that it will occur.

Prevention therefore focuses on acting **before harm happens**.

Preventive action may involve:

- removing a source of danger where this is possible.
- replacing something hazardous with a safer alternative.
- reducing exposure to the source
- using suitable technical controls.
- organizing the work so that people are not unnecessarily exposed.

You will study different types of preventive and protective measures in greater detail later in this unit.

PREVENTION BEFORE STARTING WORK

Before you begin a task, use the following questions

1. WHAT COULD CAUSE HARM?

Identify the hazard.

2. WHO COULD BE EXPOSED?

Think about yourself and anyone else who may enter or work in the area.

3. HOW COULD EXPOSURE OCCUR?

Consider the task, the location and the condition of the equipment or material.

4. WHAT COULD THE CONSEQUENCE BE?

Identify the possible injury or ill health.

5. WHAT SHOULD BE DONE BEFORE WORK CONTINUES?

Consider what preventive action is required. This is the beginning of professional safety thinking.

PUTTING THE THREE CORE TERMS TOGETHER

Term	What it tells you	Example
Hazard	What has the potential to cause harm?	Sharp metal edge.
Risk	How likely harm is under the conditions present and how serious it could be.	High possibility of hand contact while manually handling an exposed sheet edge.
Prevention	What can be done to stop or reduce the chance of the event?	Prevent access to the sharp edge or remove/control the hazardous condition before handling.

DID YOU KNOW?

A workplace does not become safe simply because an accident has not happened yet. A hazard may remain unnoticed for a long period before the conditions arise that allow it to cause harm. This is why hazard recognition and prevention must take place before an incident.

INDUSTRY EXAMPLE :

Situation:

A worker needs to move materials close to a sheet-metal storage area.

Observation:

One sheet has a sharp exposed edge projecting into the working area.

Step 1: Identify the hazard

The sharp metal edge is the hazard.

Step 2: Identify the exposure

A worker could contact the edge while walking past or handling nearby material.

Step 3: Identify the harm

Possible harm includes a cut or laceration.

Step 4: Think about the risk

Contact is more likely because the edge projects into an area where people are working.

Step 5: Think preventively

The unsafe condition should be controlled before normal work continues. The important point is that you have moved from simply saying: "That is dangerous." to explaining: what the hazard is, how exposure could occur, and what harm might result. That is much more useful in a professional workplace record.



Activity 1.1 CHECK AND APPLY THE CORE SAFETY CONCEPTS

Purpose: Check your understanding of occupational health and hazard terminology, then apply the ideas of hazard, exposure, risk and prevention to simple industrial situations.

TIME	GROUPING	RESOURCES	EVIDENCE
40 mins	pairs	Activity sheet, lesson notes, scenario cards or images	Completed responses and paired explanation

TASK

PART A: COMPLETE THE IDEA

Complete the following statement using suitable words:

Occupational health is concerned with complete physical, mental and ..... well-being and is more than simply the absence of ..... or infirmity.

PART B: TRUE OR FALSE?

A hazard is an inherent property or characteristic of an object, activity, substance, or situation that has the potential to cause harm.

True ☐ **False** ☐

Correct the statement if it is false:

.....

PART C: HAZARD, EXPOSURE AND HARM

Complete the table:

Situation	Hazard	How could exposure occur?	Possible harm
A sharp metal edge is exposed beside a handling area.			
A ladder is being used for work at height.			
A corrosive substance is being handled.			

PART D: SAME HAZARD, DIFFERENT RISK

A container of industrial solvent is considered under two conditions.

A: The container is closed and stored in a controlled, ventilated location.

B: The same product is open and being used in an area with poor ventilation.

Answer the following questions.

1. Is the chemical hazard different between A and B?

.....

2. What condition has changed?

.....

3. In which situation is exposure more likely?

.....

4. In which situation is the risk greater?

.....

5. Suggest one preventive action.

.....

PART E: SAY IT PROFESSIONALLY

Rewrite the following statement so that it clearly identifies the hazard, exposure, and possible harm.

Weak statement:

"The sharp metal is dangerous."

Your improved statement:

.....

✓

CHECKPOINT Before you submit

- I can identify the hazard, not just the injury.
- I can explain how exposure could occur.
- I can identify the possible harm.
- I understand that the same hazard can present different levels of risk.
- I can suggest a reasonable preventive action.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN

Explain why occupational health and safety matters in industrial work.	WITH HELP ☐	INDEPENDENT ☐
Define a hazard.	WITH HELP ☐	INDEPENDENT ☐
Distinguish an incident from an accident.	WITH HELP ☐	INDEPENDENT ☐
Distinguish a hazard from a risk.	WITH HELP ☐	INDEPENDENT ☐
Link a hazard to exposure and possible harm.	WITH HELP ☐	INDEPENDENT ☐
Distinguish a hazard/unsafe condition from the possible harm and from a preventive measure.	WITH HELP ☐	INDEPENDENT ☐
Explain how likelihood and severity influence risk.	WITH HELP ☐	INDEPENDENT ☐
Define prevention and explain why it should begin before work starts.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION

- What changed in my understanding of the words hazard and risk?

.....

- Which part of the lesson do I need to practise again?

.....

- Where could I use this knowledge in a workshop?

.....

EXIT TICKET

- One hazard I can now identify correctly:

.....

- One example of exposure to that hazard:

.....

- One possible consequence:

.....

- One preventive action that could be considered before work begins:

.....

TRANSFER From knowledge to performance

In Practical Session 1, you will apply these ideas to photographs or a controlled simulated steel-workplace area.

You will:

observe → identify → classify → record → explain

- Your aim is to recognize and record hazards correctly.

1 SESSION — Classify hazards and unsafe conditions from photographed or simulated workplace situations and begin completing a hazard record

Skill code: PC1.1 Duration: 60 minutes Location: Training workshop, laboratory, photographed workplace exercise or authorised simulated steel-workshop area

SAFETY GATE: OBSERVE BEFORE YOU ACT

Before you begin, make sure you understand:

- the area you are allowed to inspect.
- any area that is restricted.
- the PPE required for the activity.
- whether equipment is isolated or for observation only.
- which objects or equipment you must not touch.
- how you should report an unsafe condition.

Performance task	• Inspect the workplace images or designated simulated steel-workshop area. Identify the hazards and unsafe conditions that you can observe and record them clearly using the hazard-record form. • For every observation, think about: • What is the hazard? • Who could be exposed? • How could exposure occur? • What harm could result? • What preventive action may be needed before work continues?
Performance standard	• Identify at least six credible hazards or unsafe conditions. • Distinguish the hazard from the injury or consequence. • Identify who may be exposed. • Explain how exposure could occur. • Identify a reasonable possible consequence. • Record your findings using clear occupational-safety terminology. • Explain at least one of your findings verbally.
Required evidence	• A completed workplace hazard record. • Annotations on the permitted photograph, image or area plan where provided. • A short verbal explanation of one identified hazard.

PREPARATION

PPE



Wear only the PPE specified for your training area.

- Safety footwear where required
- Safety glasses where required
- High-visibility clothing where required
- Other PPE specified by your instructor or site procedure

.....

.....

TOOLS & EQUIPMENT



- Clipboard or writing surface
- Pen or pencil
- Workplace photographs or controlled inspection area
- Hazard-identification record
- Area plan, where provided

PRE-START CHECKS



Before beginning, confirm:

- I know where I am permitted to inspect.
- I know which areas are restricted.
- I know which equipment or objects I must not touch.
- I understand that this is an observation activity.
- I know how to report a serious unsafe condition.
- I have the correct recording sheet.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Listen to the safety and inspection instructions and confirm the boundaries of the activity.	You can explain what you may and may not do.
2	Observe the workplace systematically before beginning to write. Look at the work area, materials, access routes, equipment, and activities shown.	Initial observation notes.
3	Find conditions or factors that have the potential to cause harm.	At least six potential hazards or unsafe conditions.
4	For each hazard, identify who could be exposed, how exposure could occur and what harm could result.	Complete hazard descriptions.

Step	Action	Evidence/checkpoint
5	Enter each finding clearly in your hazard record.	Completed hazard-record fields.
6	Check that you have not written the injury itself as the hazard.	Corrected terminology where required.
7	Select one observation and explain it using the sequence hazard → exposure → harm.	Short verbal explanation.
8	Complete your self-check and submit your hazard record.	Completed evidence.

SESSION 01 • HAZARD IDENTIFICATION RECORD

Use one row for each workplace condition you identify.

No	Observed condition	Hazard	Who could be exposed?	How could exposure occur?	Possible harm	Preventive action/ reporting required
Example	Sharp metal edge projecting into the handling area	Sharp edge	Worker handling material or walking nearby	Hand or body contacts exposed edge	Cut or laceration	Report and prevent contact until the condition is made safe
1						
2						
3						
4						
5						
6						

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO



The trainer demonstrates one staged example:

A sharp metal edge projects from a sheet stored beside a pedestrian or material-handling route.

1. I first stay within the authorised inspection area and observe without touching the material.
2. "The sharp metal edge is the hazard because it has the potential to cause harm."
3. "A worker handling nearby material or walking close to the sheet could be exposed."
4. "Exposure could occur if a hand, arm or other part of the body contacts the exposed edge."
5. "Possible harm includes a cut or laceration."
6. "I record the observed condition separately from the possible injury."
7. "I check that I have identified the hazard, not simply written the consequence."
8. "I consider whether the location or working conditions make exposure more likely."
9. "I record that preventive action is required to prevent contact with the edge."
10. "I report the condition through the authorised route. I do not move or modify the material unless I have been authorised to do so."

WE DO



The whole group reviews two or three safely staged examples of unsafe workplace conditions, such as:

- A disconnected trailing cable across a walking route
- A sharp metal offcut in a handling area
- Materials obstructing a marked walkway
- An unsecured empty training cylinder
- An incorrectly positioned ladder
- A clean simulated liquid spill

YOU DO



You will inspect assigned photographs, simulated areas or controlled workshop zones.

You must contribute at least:

- two hazard observations
- two explanations of how exposure could occur
- two possible consequences
- one suggested preventive or reporting action
- one verbal hazard → exposure → harm explanation

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/ comment
Remained within the authorised inspection area and followed all safety instructions.	☐	☐	
Observed workplace conditions without touching or operating equipment.	☐	☐	
Identified credible hazards or unsafe conditions.	☐	☐	
Distinguished the hazard from the possible harm or consequence.	☐	☐	
Identified who could be exposed to the hazard.	☐	☐	
Explained clearly how exposure could occur.	☐	☐	
Identified a reasonable possible injury or ill-health consequence.	☐	☐	
Recognized that the same hazard may present a different level of risk when exposure conditions change.	☐	☐	
Identified an appropriate preventive or reporting action without exceeding learner authority.	☐	☐	
Recorded findings clearly using appropriate occupational-safety terminology.	☐	☐	
Explained one finding verbally using the sequence hazard → exposure → harm.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Safety and procedure	Consistently anticipates inspection boundaries, maintains a safe position and appropriately alerts the trainer to potentially serious conditions.	Follows the authorised inspection route and all instructions without unsafe action.	Requires occasional reminders about boundaries, observation method or reporting.	Enters a restricted area, touches or operates equipment, or undertakes an unauthorised action.
Hazard identification	Identifies obvious and less-obvious credible hazards and clearly distinguishes the source of harm from the consequence.	Identifies the required number of credible hazards and describes them correctly.	Identifies mainly obvious conditions or occasionally confuses the hazard with the injury.	Misses significant hazards or repeatedly records injuries rather than hazards.
Exposure and harm	Clearly explains who could be exposed, how exposure could occur and how different conditions could change the possibility of harm.	Correctly links each hazard to an exposure route and reasonable possible consequence.	Links some hazards to exposure and harm, but uses vague or incomplete descriptions.	Cannot explain how exposure could occur or gives unrelated consequences.
Prevention and reporting	Suggests clear, practical preventive or reporting actions while recognising the limits of learner authority.	Identifies an appropriate preventive or reporting action for the observed condition.	Suggests broadly relevant action, but requires support to connect it to the observed hazard.	Recommends unsafe action or attempts to correct a condition outside authorised learner activity.
Communication and records	Produces concise, technically precise records and gives a clear hazard → exposure → harm explanation.	Completes required fields legibly and explains one finding using appropriate terminology.	Records contain omissions, vague terminology or partially developed explanations.	Records are incomplete, unclear or do not demonstrate understanding of the task.

2 LESSON

NATIONAL SAFETY LEGISLATION AND WORKPLACE RESPONSIBILITIES

WHY SAFETY RESPONSIBILITIES MUST BE CLEAR

Working safely is not based only on personal judgement. Industrial workplaces operate within rules that define how safety and health must be organized, who has responsibility for different decisions, what information workers must receive and what must happen when an unsafe condition, accident or occupational-health problem is identified. As a technician, you are unlikely to be responsible for interpreting complex legislation yourself. However, you do need to understand how legal requirements affect the work you carry out.



You need to know:

- where workplace safety requirements come from.
- who has responsibility for organizing and supervising safe work.
- what safety information and training you should receive.
- when a condition must be reported.
- why you must work within your level of competence and authority.
- what can happen when safety requirements are ignored.

WARM-UP

WHO SHOULD ACT?

You are preparing to assist with a workshop task. A new worker has been assigned to use a machine. The worker has received a general site induction, but has not received training for the machine or the specific hazards associated with the task.

The supervisor knows that the worker has not completed the required training.

Discuss with a partner:

1. Should the worker begin the task simply because production needs to continue?
2. Who has responsibility for organizing and supervising the work?
3. What information should the worker receive before carrying out the task?
4. What should the worker do if they are unsure how to perform the task safely?
5. What could happen if the required safety arrangements are ignored?

Record your starting decision:

.....

.....

BY THE END OF THIS LESSON, I WILL BE ABLE TO:

1. interpret national safety laws/requirements relevant to a workplace situation.

2. differentiate employer and employee responsibilities and identify the required safe action.

KEY TERMS:

Legislation • Primary law • Regulation • Decree • Secondary legislation • Employer • Manager • Supervisor • Workers' Safety Representative • Prevention and Protection Service • Occupational Health Doctor • Information • Training • Specific training • Health surveillance • Reporting • Corrective action • Non-compliance

SUBTOPIC 2.1 FROM NATIONAL LAW TO WORKPLACE REQUIREMENTS

Occupational health and safety are both professional responsibilities and legal requirements. National laws establish the main duties that organizations and people must follow. These broad duties are then supported by more detailed rules that explain how requirements should be applied in practice. You do not need to memorize every legal document to work safely. You do need to understand that workplace rules and procedures are connected to a wider system of legal responsibility.



LEVELS OF NATIONAL REQUIREMENTS:

A national legislative system can contain several levels.

PRIMARY LEGISLATION

The main sources of law are normally acts passed by the national parliament. These establish the broad legal requirements that must be followed.

IMPLEMENTING REGULATIONS AND DECREES

Governments may issue regulations and decrees that turn broad legislative principles into more specific technical and procedural requirements.

For example, a law may establish that workers must be protected from occupational risks, while implementing requirements explain how particular safety responsibilities are organized.

SECONDARY LEGISLATION

More detailed requirements may also be established through:

- ministerial decrees
- administrative orders

These do not have the same position as primary legislation, but they can be important in determining what an organization must do in day-to-day practice.

FROM LAW TO ACTION

You can think of the process as a sequence:

National law



Implementing regulations and requirements



Workplace safety arrangements



Safe procedures and instructions



Safe work

A technician normally interacts with the final parts of this chain — workplace procedures, training, instructions and controls — but these arrangements exist within a wider legal system.



Definition - LEGISLATION - PRIMARY LAW - IMPLEMENTING REQUIREMENT

- **LEGISLATION**

Legislation is the body of laws and legally established requirements that govern activities within a jurisdiction.

- **PRIMARY LAW**

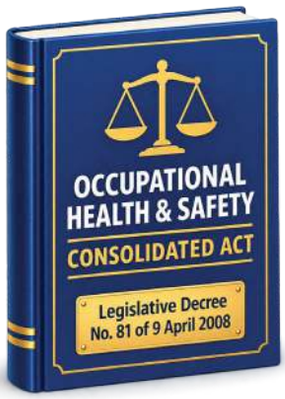
A primary law establishes the main legal requirements and is normally enacted through the national legislative process.

- **IMPLEMENTING REQUIREMENT**

An implementing regulation or decree provides more detailed requirements for putting the principles of a law into practice.

EXAMPLE OF A CONSOLIDATED OCCUPATIONAL-SAFETY LAW:

One national system described in this unit uses a Consolidated Act on Occupational Health and Safety. In that system, Legislative Decree No. 81 of 9 April 2008 brings occupational health and safety requirements together within a single main instrument. Its provisions apply across different sectors and employment arrangements and also extend safety protection to students using laboratories in vocational training.



An important principle is therefore:

Being a learner does not remove the need to follow workplace safety requirements.

When you enter an authorized workshop or industrial training area, you must treat the safety arrangements as part of the work, not as optional classroom guidance.

Safety Critical- USE THE REQUIREMENT THAT APPLIES TO YOUR WORKPLACE

- The detailed legal example above shows how a national occupational-safety system can be organized.
- When you work in an actual industrial organization, follow the legislation, authorized procedures and reporting arrangements that apply to that workplace.
- Do not independently replace an authorized local requirement with a rule taken from another workplace or jurisdiction.

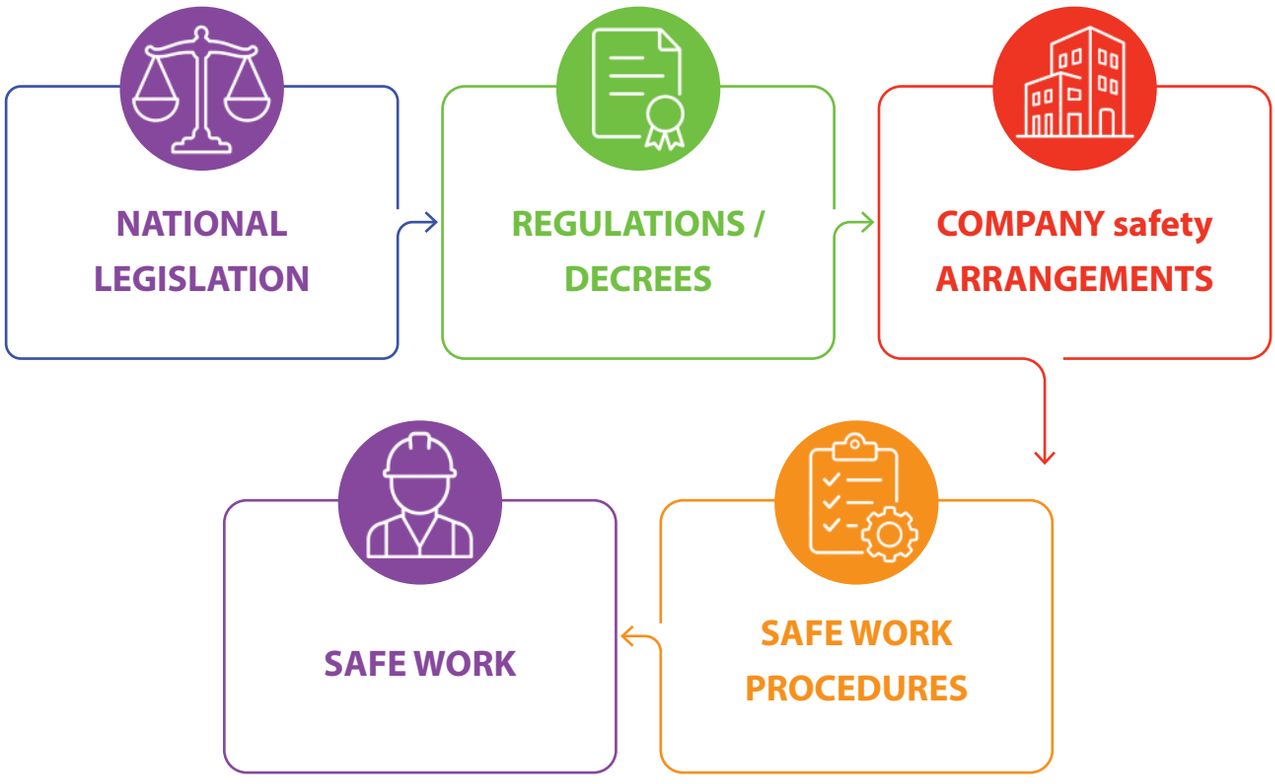


Figure 2.1 — National safety requirements are translated into workplace arrangements and instructions that guide safe work.

SUBTOPIC 2.2 WHO IS RESPONSIBLE FOR WORKPLACE SAFETY?

Safe work depends on people having clearly defined responsibilities. A safety system does not place every responsibility on one person. Different roles organize work, supervise activities, provide specialist advice, represent workers and monitor occupational health.

Understanding these roles helps you know:

- who makes particular safety decisions
- who supervises your work
- who provides specialist support
- who should receive particular safety information

THE EMPLOYER

The employer has overall legal responsibility for the workplace safety system.

In the national system, two important responsibilities cannot be delegated:

1. conducting and documenting the workplace risk assessment.

2. appointing the Head of the Prevention and Protection Service.

The employer therefore has a central role in ensuring that an organized system for managing occupational risk exists.

THE MANAGER

The manager puts the employer's requirements into practice by organizing and supervising work activities.

A manager uses the authority and professional competence associated with the role to make sure work is properly organized.

Think of the manager's role as helping convert the organization's safety requirements into workable arrangements.

THE SUPERVISOR

The supervisor oversees the way work is actually carried out.

The supervisor:

- oversees correct execution of the work;
- checks that instructions are followed;
- acts within the powers assigned to the role.

This makes the supervisor particularly important to you as a learner or technician.

If you do not understand an instruction or notice that the actual work differs from the authorised arrangement, the supervisor is one of the people you may need to inform.



Definition - SUPERVISOR

A supervisor oversees the correct execution of work within the authority assigned to the role and helps ensure that workplace instructions are followed.

WORKERS' SAFETY REPRESENTATIVE

The Workers' Safety Representative represents workers on occupational health and safety matters.

This role can include:

- consultation about risk assessment
- access to relevant safety information
- representing worker concerns relating to safety and health

The representative is therefore different from the supervisor. A supervisor oversees work. A workers' representative represents workers' safety interests.

PREVENTION AND PROTECTION SERVICE

The Prevention and Protection Service supports the organization in managing occupational risks. It is led by a qualified Head of Prevention and Protection Service.

The service coordinates prevention and protection activities and requires appropriate professional competence and continuing training.

OCCUPATIONAL HEALTH DOCTOR

The Occupational Health Doctor supports the occupational-health aspects of the safety system.

The role includes:

- assisting with risk assessment.
- carrying out health surveillance.
- conducting appropriate medical examinations and tests.
- monitoring whether workers remain fit in relation to occupational risk factors.

Definition - HEALTH SURVEILLANCE

Health surveillance uses appropriate medical examinations and tests to monitor workers' health and fitness in relation to occupational exposures.

PUTTING THE ROLES TOGETHER

Role	Main function
Employer	Holds overall responsibility and ensures the risk-assessment and prevention system is established
Manager	Organises and supervises work according to the employer's requirements
Supervisor	Oversees the correct execution of day-to-day work and ensures instructions are followed
Workers' Safety Representative	Represents workers on health and safety matters
Head of Prevention and Protection Service	Coordinates prevention and protection activities
Occupational Health Doctor	Supports risk assessment and carries out occupational-health surveillance

THINK ABOUT

A technician notices an unsafe condition. Should the technician personally take over the responsibilities of the employer, manager, or safety specialist? The answer is **No.** Your responsibility is to recognise that a problem exists, follow the authorised safety process and provide accurate information to the person responsible for deciding what action is required.

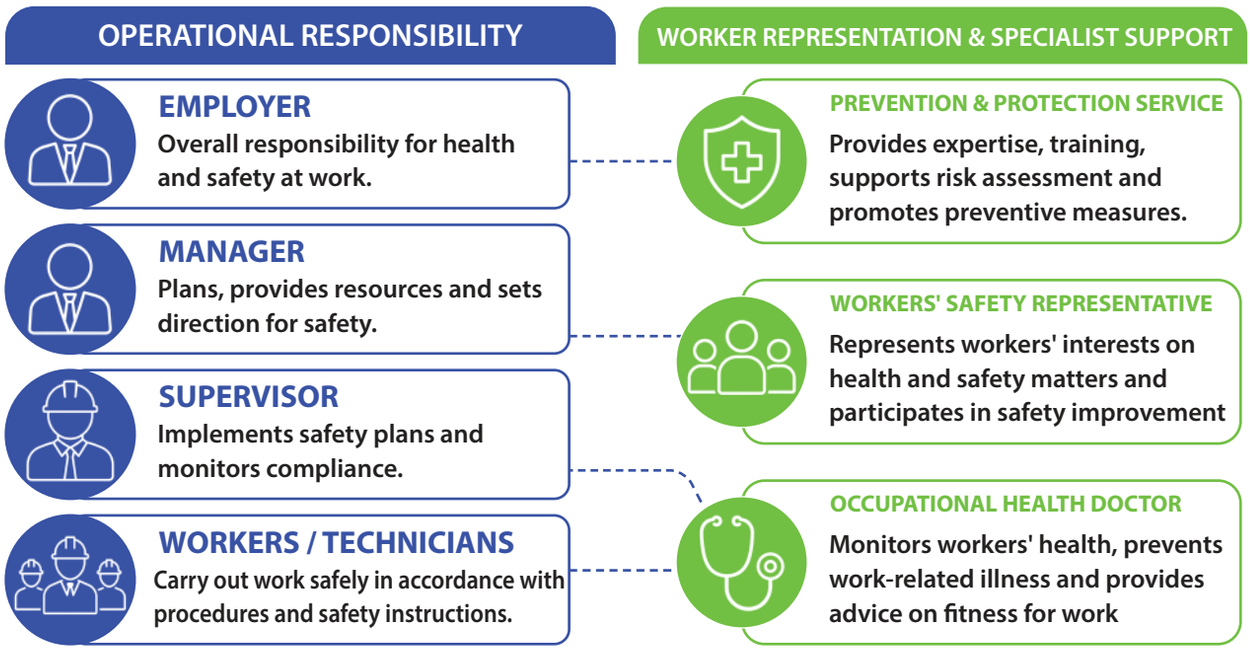


Figure 2.2 — Workplace safety combines operational responsibility, worker representation and specialist prevention and occupational-health support.

SUBTOPIC 2.3 INFORMATION, TRAINING AND REPORTING

Having safety rules is not enough. Workers must also receive the information and training needed to understand workplace hazards and carry out their duties safely.

INFORMATION YOU NEED

Safety information can include:

- general occupational health and safety risks within the organization
- emergency arrangements
- the people responsible for prevention and protection
- occupational-health contacts
- the particular risks associated with your own work
- relevant safety requirements
- hazards associated with chemical substances
- preventive and protective measures used by the organization
- the names of designated emergency personnel

Receiving this information helps you understand **what hazards exist and what arrangements are already in place**.

TRAINING

Training develops your understanding of how occupational safety works.

General safety training can cover:

- risk and harm
- prevention and protection
- the structure of the workplace safety system
- the rights and duties of people involved in workplace safety
- risks associated with particular jobs
- the measures used to prevent or control those risks

SPECIFIC TRAINING

Some activities require more focused training because they involve particular hazards, equipment or professional competence.

Examples include training for:

- safe manual handling
- operating work equipment and machinery
- correct selection and use of particular PPE
- rope-access and work-positioning systems
- safe handling and storage of hazardous substances

The important principle is:

Receiving general safety information does not automatically prepare you to carry out every specialist task.

WORKER / EMPLOYEE RESPONSIBILITIES

As a technician, you have responsibilities that support the workplace safety system.

Your responsibility	What this means in practice
Follow authorized instructions	Follow approved safety procedures and instructions for the task.
Use PPE correctly	Use the required PPE correctly and according to the supplied instructions.
Report hazards and defects	Report unsafe conditions, hazards, damaged equipment or missing protective measures through the authorized process.
Cooperate with safety arrangements	Follow workplace safety procedures and cooperate with the people responsible for health and safety.
Work within your competence and authority	Do not carry out tasks or make safety decisions beyond your training, competence or assigned authority.
Attend required training	Complete the information, instruction and task-specific training required for your work.

REMEMBER

Your responsibility is not to replace the employer, manager or safety specialist. Your role is to follow authorized arrangements, work within your competence, use the required controls, and report problems through the correct process.

INFORMATION, TRAINING OR SPECIFIC TRAINING?

Example	Category
You are told who the occupational-health contact is.	Information
You learn how risk, prevention and protection are connected.	Training
You are trained to operate a particular machine safely.	Specific training
You are told about the hazards associated with your work area.	Information
You receive instruction and practice for handling a particular hazardous substance.	Specific training

REPORTING ACCIDENTS AND OCCUPATIONAL DISEASES

Occupational safety systems also include formal reporting arrangements. Compulsory occupational insurance systems may protect workers affected by:

- workplace accidents
- occupational diseases

This is important because occupational harm is not limited to sudden accidents. A worker can also develop an illness because of harmful occupational exposure.

Definition - OCCUPATIONAL DISEASE

An occupational disease is an illness associated with harmful conditions or exposures arising from work.

EXAMPLE OF FORMAL NOTIFICATION REQUIREMENTS

The national system provides an example of how formal reporting duties may be defined. It requires:

- an injury causing more than three days of incapacity to be reported to the national insurance authority within 48 hours of the employer becoming aware of it.
- a fatal workplace accident to be reported within 24 hours.
- a diagnosed occupational disease to be reported within five working days of receipt of the medical certificate.
- serious workplace accidents to be notified to the relevant public-safety authority.

These requirements demonstrate that **reporting is a formal responsibility**, not an informal choice. For your practical work, always use the authorized reporting process provided for the training or workplace environment.

2.1 CHECK YOUR UNDERSTANDING

A worker develops an occupational disease after repeated harmful exposure.

Statement:

"Occupational insurance covers only sudden workplace accidents."

True ☐ **False** ☐



Figure 2.3 — Workers need appropriate information, training and task-specific preparation to work safely and respond correctly when problems occur.

SUBTOPIC 2.4 WHEN A SAFETY REQUIREMENT IS NOT MET

A safety requirement has little value if it is ignored during the actual work. **Non-compliance** occurs when an applicable requirement, instruction or authorised safety arrangement is not being followed.

Examples might include:

1. required information has not been provided.
2. required training has not been completed.
3. a known unsafe condition has not been corrected.
4. an authorized safe-working instruction is not being followed.
5. a required safety responsibility has not been carried out.

WHY NON-COMPLIANCE MATTERS

Failure to manage workplace safety can have serious consequences. Workplace accidents can lead to:

- temporary injury
- permanent disability
- death
- pain and loss for workers and families
- lost productivity
- healthcare costs
- regulatory penalties

The purpose of safety legislation is therefore not simply to create paperwork. It helps ensure that responsibilities are defined and action is taken before uncontrolled conditions result in harm.

CORRECTIVE AND IMPROVEMENT ACTIONS

Two types of action can be distinguished:

1. CORRECTIVE ACTION

Corrective action is taken to eliminate or control an unacceptable condition.
Example:
A required safety arrangement has not been provided before work begins. The issue must be corrected before the work is allowed to continue under the approved process.

2. IMPROVEMENT ACTION

An improvement action reduces risk further when an existing arrangement is already considered acceptable.
The difference is important:
Corrective action deals with something that is not acceptable. Improvement action makes an already controlled situation better.

TYPES OF SAFETY ACTION

The source groups prevention and protection measures into three broad forms:

1. ORGANISATIONAL

These concern how work is organized.

Examples include:

- information
- training
- health surveillance

2. PROCEDURAL

These concern how a task should be performed. An example is an authorized safe-work instruction.

3. TECHNICAL

These involve engineering measures used to control a hazard. You will study control selection in greater detail later in the unit. At this stage, focus on identifying what requirement has not been met and who is responsible for ensuring that appropriate action is taken.

INDUSTRIAL EXAMPLE :

Situation

A new technician has been asked to begin using a machine. The technician has received general workplace information, but has not yet completed the specific training needed for safe operation of that equipment.

- **What is the problem?**
  The worker has not received the task-specific preparation required for the activity.
- **Which part of the safety system is relevant?**
  Information, instruction, and training.
- **Who has responsibility?**
  The organization must provide appropriate arrangements. Managers organize the work, and supervisors oversee its correct execution.
- **What should the technician do?**
  Do not pretend to have competence that has not been developed. Inform the authorized supervisor and wait for the required instruction or training before proceeding.
- **What is the corrective action?**
  The required training and safe-work arrangements must be completed before the technician carries out the activity independently.

Safety Critical - RESPONSIBILITY DOES NOT MEAN ACTING OUTSIDE YOUR AUTHORITY

Finding a non-compliance does not automatically authorize you to correct equipment, change a workplace system or make a legal decision.

Activity 2.1 WHO IS RESPONSIBLE AND WHAT SHOULD HAPPEN?

Purpose: Apply your understanding of national safety requirements, workplace roles, worker information and reporting to simple industrial situations.

TIME	GROUPING	RESOURCES	EVIDENCE
30 mins	Individual → pairs	Lesson notes, role cards and workplace scenarios	Completed classification and case-analysis sheet

TASK

PART A: BUILD THE LEGAL FRAMEWORK

Match each requirement to the correct level. Choose from:

Primary law • Implementing regulation/decree • Secondary legislation

Requirement	Level
An act passed through the national legislative process	
A government requirement that provides detailed implementation of a law	
A ministerial or administrative requirement used in day-to-day compliance	

PART B: WHO IS THIS?

This person oversees the correct execution of work, checks that instructions are followed, and acts within the powers assigned to the role.



Which role is being described?

- ☐ Employer
- ☐ Manager
- ☐ Supervisor
- ☐ Workers' Safety Representative

Explain one reason for your answer:

PART C: TRUE OR FALSE?

Compulsory occupational insurance protects workers only against accidents and does not cover occupational disease.

True ☐ **False** ☐

PART D: WORKPLACE CASE

A trainee has been assigned to assist with a workshop machine.

You find that:

- the trainee has received general workplace information.
- the trainee has not received specific training for the machine.
- the task is due to start immediately.
- the supervisor has been informed that the training is incomplete.

Complete the analysis:

1. What safety requirement has not yet been met?

.....

2. Which workplace role should oversee the correct execution of the work?

.....

3. What should the trainee do before beginning the task?

.....

4. What corrective action is required?

.....

5. Why would simply beginning the task be an example of unsafe non-compliance?

.....

PART E: REPORT THE ISSUE CLEARLY

Write a short professional statement that could be given to the supervisor.

Use the following structure:

Observed condition → Requirement not met → Safe action needed

.....

.....

CHECKPOINT Before you submit

- I can identify the level of a national safety requirement.
- I can distinguish between employer, manager and supervisor responsibilities.
- I understand that worker protection can cover occupational diseases as well as accidents.
- I can identify a missing safety requirement in a workplace case.
- I can state the responsible corrective action.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN

Explain how national legislation leads to workplace safety requirements.	WITH HELP ☐	INDEPENDENT ☐
Distinguish primary law from implementing and secondary requirements.	WITH HELP ☐	INDEPENDENT ☐
Describe the roles of the employer, manager and supervisor.	WITH HELP ☐	INDEPENDENT ☐
Explain the functions of the Workers' Safety Representative, Prevention and Protection Service and Occupational Health Doctor.	WITH HELP ☐	INDEPENDENT ☐
Distinguish information, training and specific training.	WITH HELP ☐	INDEPENDENT ☐
Explain why accidents and occupational diseases must be reported.	WITH HELP ☐	INDEPENDENT ☐
Identify a non-compliance and state appropriate responsible action.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION

1. Which workplace safety role was most difficult for me to distinguish from another role?
2. Why is safety training an organizational responsibility rather than simply a personal choice?
3. What should I do if I am asked to carry out work for which I have not received the required preparation?

EXIT TICKET

Complete the sequence:

Safety requirement:

.....

↓

Responsible workplace role:

.....

↓

Required workplace action:

.....

↓

My responsibility as a learner/technician:

.....

➔

TRANSFER From knowledge to performance

- In Practical Session 2, you will analyze workplace cases rather than simply recalling definitions.
- For each case, you will:
- identify the requirement → identify the responsible role → record the unsafe condition → determine the required corrective action → communicate your finding
- You are not expected to interpret complex legislation or make decisions outside your authority.

2 SESSION Analyze workplace cases, identify relevant safety responsibilities and record appropriate corrective actions

Skill code: PC1.1 Duration: 60 minutes Location: Classroom, training workshop or controlled simulated workplace

SAFETY GATE: ANALYSE, DO NOT INTERVENE

This practical session tests your ability to recognize when a safety requirement has not been met and identify who should take action.

You are **not authorized** by this activity to:

- operate equipment
- repair a defect
- change a workplace procedure
- approve a risk assessment
- provide medical advice
- alter an official safety record

Your task is to:

observe or read → identify → record →report



Performanctask

Your task is to analyze a workplace case. For each safety issue you identify:

1. record the unsafe condition or hazard.

2. identify the safety requirement involved.

3. identify the workplace role responsible for the relevant action.

4. state what corrective action is required.

5. state what you should do as a learner or technician.

WORKPLACE CASE.

THE NEW MAINTENANCE TECHNICIAN.

A new technician joins a mechanical maintenance area. During the first shift, you observe the following: .

- the technician has received general information about the company.
- the technician has not yet received the specific training required for one of the machines.
- the technician is unsure who the Prevention and Protection Service contact is.
- the supervisor is aware that the task-specific training has not been completed.
- the task has nevertheless been placed on the day's work schedule.
- an unsafe condition close to the work area has been mentioned verbally but has not yet been formally recorded.

The technician says, "I will just start and ask someone if I have a problem.".

Performance standard	• identify at least four relevant safety issues within the case. • distinguish an unsafe condition from the responsibility associated with it. • correctly identify the relevant workplace role. • recognize missing information, training or supervision. • state a logical corrective action. • remain within the limits of learner authority. • communicate one finding clearly using professional terminology .
Required evidence	• completed workplace-case analysis. • hazard/unsafe-condition record. • responsibility and corrective-action entries. • assessor observation. • one short verbal explanation.

PREPARATION

RESOURCES



- Workplace-case sheet
- Safety-role reference card
- Hazard/unsafe-condition record
- Pen or pencil
- Industrial image or simulated scenario where used

PRE-START CHECKS



- I understand the limits of the activity.
- I know that identifying a problem does not authorize me to repair it.
- I can distinguish an employer, manager and supervisor.
- I know how to record an unsafe condition clearly.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Read the complete workplace case before deciding what is wrong.	Key facts identified.
2	Mark each condition that may represent a missing or unmet safety requirement.	Minimum four relevant issues.
3	Describe the unsafe condition or hazard clearly.	Clear factual statement.
4	Identify the safety requirement connected to the issue.	Appropriate requirement recorded.
5	Identify the role responsible for organizing, supervising or supporting the required action.	Role correctly identified.
6	State what corrective action is required.	Action addresses the unmet requirement.
7	State what you should and should not do personally.	Learner authority respected.
8	Present one finding to the trainer.	Concise verbal explanation.

CASE-ANALYSIS RECORD

No	Observed condition	Hazard	Who could be exposed?	How could exposure occur?	Possible harm
Example	Technician does not know the prevention-and-protection contact	Required safety information has not been fully communicated	The technician and nearby workers	An emergency or accident occurs, and the technician cannot contact help in time	Delayed medical response, worsening of injuries, or uncontrolled hazards
1					
2					
3					
4					

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO



The trainer demonstrates one case:

A worker is assigned to operate equipment, but has not completed the required specific training.

The trainer verbalises:

1. "I first identify the fact, not an assumption: the required training has not been completed."
2. "The safety issue concerns information, instruction and role-appropriate training."
3. "The worker should not pretend that general induction is the same as task-specific preparation."
4. "The manager has an organisational role, while the supervisor oversees the correct execution of the work."
5. "The condition should be reported through the authorised workplace route."
6. "The corrective action is to provide the required preparation before the task is performed independently."
7. "As a learner, I record and report the issue. I do not authorise myself to change the training requirement."

WE DO



The group analyses two short cases.

- **CASE A**

A supervisor notices that a worker does not understand an instruction, but allows the task to continue without clarification.

- **CASE B**

A worker has received general safety information, but does not know the particular hazards associated with the task they are about to perform.

YOU DO



Each pair analyses the main Session 2 workplace case independently.

You must contribute:

- at least two identified safety issues
- one correct workplace role explanation
- one corrective-action statement
- one verbal report

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/ comment
Identified factual safety issues from the case without adding unsupported assumptions.	☐	☐	
Connected the issue to an appropriate workplace safety requirement.	☐	☐	
Distinguished employer, manager, and supervisor responsibilities.	☐	☐	
Recognised the purpose of specialist safety and occupational-health roles.	☐	☐	
Identified missing information or training correctly.	☐	☐	
Recorded an unsafe condition or hazard clearly.	☐	☐	
Proposed a corrective action that addressed the identified issue.	☐	☐	
Remained within the limits of learner authority.	☐	☐	
Identified the appropriate need to report or escalate the issue.	☐	☐	
Communicated one finding using appropriate safety terminology.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Interpretation of requirements	Correctly identifies both obvious and less-obvious unmet requirements and clearly explains why they matter.	Correctly identifies the main applicable safety requirements.	Identifies some requirements, but confuses duties or uses vague reasoning.	Cannot identify the relevant safety requirement.
Roles and responsibilities	Distinguishes operational and specialist roles accurately and explains how they interact.	Correctly identifies the responsible workplace role for each main issue.	Correct role is identified inconsistently or requires prompting.	Assigns responsibilities incorrectly or assumes personal authority.
Hazard and condition recording	Produces concise, factual and technically clear records.	Records relevant conditions and hazards clearly.	Records contain vague wording or minor omissions.	Records are unclear, unrelated or incomplete.
Corrective action	States a precise corrective action and explains why it addresses the unmet requirement.	States appropriate corrective action.	Suggests a partly appropriate action, but needs guidance.	Recommends ineffective, irrelevant or unauthorized action.
Communication and authority	Communicates clearly, prioritizes the key issue and explicitly recognizes limits of authority.	Gives a clear verbal explanation and follows the reporting principle.	Explanation is understandable, but incomplete.	Fails to report appropriately or claims authority not held.

3 LESSON

EUROPEAN DIRECTIVES AND TECHNICAL SAFETY STANDARDS

WHY PRODUCT MARKINGS AND TECHNICAL INFORMATION MATTER

When you work with industrial machinery or equipment, you may see markings, symbols, rating information, warning labels and references to technical standards. These are not simply decorations on the equipment. They can tell you:



- what type of equipment you are dealing with.
- what safety requirements apply to its design.
- whether a manufacturer has declared conformity with relevant requirements.
- whether special conditions apply to its use.
- where to find additional safety information.
- what precautions you need to follow before using or working near it.

As a technician, you are not expected to certify a machine or carry out a legal conformity assessment yourself. You are expected to recognize important safety information, use the correct documentation and report anything that is missing, unclear or inconsistent.

WARM-UP



CAN YOU USE THE MACHINE?

You are preparing to work near an industrial machine.

On the machine you can see:

- the manufacturer's name
- a model number
- a serial number
- a CE marking
- electrical rating information
- a warning symbol
- A set of operating instructions is also available, but the model number printed on the document is slightly different from the model number on the machine.

Discuss with a partner:

1. What information would you check first?
2. What does the CE marking tell you?
3. Does the CE marking alone tell you that the machine is safe for every task?

4. Why is the model-number difference important?
5. Should you simply assume that the instructions are suitable?
6. What should you do if the information cannot be confirmed?

Record your initial idea:



BY THE END OF THIS LESSON, I WILL BE ABLE TO

1. Interpret relevant EU directives and technical/safety standards.
2. Use safety markings, labels and equipment documentation to identify required precautions and user responsibilities.

KEY TERMS

Product directive • EHSR • Conformity assessment • Declaration of Conformity • Declaration of Incorporation • CE marking • Harmonised standard • Technical standard • Technical regulation • CEN • CENELEC • ETSI • ISO • IEC • ATEX • Ex marking • Manufacturer instructions • Safety marking

SUBTOPIC 3.1 THE EU WORKPLACE SAFETY FRAMEWORK

European safety legislation helps protect both **workers** and **users of products and equipment**. It includes requirements for safe working conditions as well as requirements that apply to products placed on the market. These two areas have different purposes and responsibilities.
Council Directive 89/391/EEC, known as the **Framework Directive,** provides a general framework for protecting workers' health and safety at work. It places key responsibilities on employers to identify and assess workplace risks, take preventive and protective measures, and provide workers with appropriate information, instruction and training.
The Framework Directive is supported by more specific directives addressing particular workplace risks and situations, such as manual handling, chemical agents, noise and vibration.

Workplace-safety directives	Product-safety directives
Focus on workers and working conditions	• Focus on **products placed on the market**
Establish duties mainly for employers and workplaces	• Establish duties **mainly for manufacturers and other economic operators**
Require workplace risks to be identified, assessed and controlled	• Require applicable product requirements to be addressed and conformity demonstrated
Cover prevention, protection, information, instruction and training	• Cover product design, conformity assessment, technical documentation and, where required, CE marking

For example, under the Framework Directive, an employer must take measures necessary to protect workers and provide appropriate information and training.

For a product covered by EU harmonization legislation, the manufacturer is responsible for meeting the applicable product requirements and carrying out the required conformity procedures. This can include preparing technical documentation, issuing an EU Declaration of Conformity and affixing the CE marking where required.

IMPORTANT DISTINCTION

A product's conformity marking does **not** replace workplace safety responsibilities. A machine may meet applicable product requirements and still require appropriate workplace controls, instructions, training and safe working procedures.

SUBTOPIC 3.2 FROM EU PRODUCT DIRECTIVES TO SAFE PRODUCT DESIGN

Industrial equipment placed on the European market may be subject to requirements that control its design, manufacture and conformity. An important approach to European product regulation is based on two connected parts:

1. PRODUCT DIRECTIVES

These establish the essential safety outcomes that a product must achieve.

Example: The Machinery Directive **2006/42/EC** requires machinery to meet essential health and safety requirements before it is placed on the EU market.

For example, the machine must be designed so that users are protected from hazards such as moving machine parts.

2. HARMONISED TECHNICAL STANDARDS

These provide detailed technical methods and specifications that can be used to achieve those requirements. This arrangement allows the legal requirement to state what level of safety must be achieved, while technical standards provide recognized engineering methods that can help achieve it.

EXAMPLE: EN ISO 12100 provides principles and a methodology for machinery risk assessment and risk reduction. It helps manufacturers determine hazards, assess risks and select appropriate protective measures.

ESSENTIAL HEALTH AND SAFETY REQUIREMENTS



Definition - EHSRs

Essential Health and Safety Requirements (EHSRs) are the minimum health and safety outcomes that a product covered by the relevant directive must satisfy.

The manufacturer must determine which requirements apply to the product. The general conformity process can then involve

1. Applicable product directive
2. Essential Health and Safety Requirements
3. Relevant technical standards
4. Product design and manufacture
5. Conformity assessment
6. Declaration of Conformity
7. CE marking

CONFORMITY ASSESSMENT

Before an applicable product is placed on the market, the manufacturer carries out the required conformity assessment. Depending on the product and applicable requirements, this may also involve an independent notified body. The purpose is to demonstrate that the applicable safety requirements have been addressed. When the required process has been completed, conformity is documented, and the appropriate product marking is applied.

IMPORTANT RELATIONSHIP

A directive and a technical standard do not perform the same function:

Product directive	Harmonised technical standard
Establishes required safety outcomes	Provides detailed technical specifications or methods
Forms part of the regulatory framework	Supports engineering implementation
Defines what must be achieved	Helps demonstrate how requirements can be met
May lead to conformity-assessment requirements	Can provide a presumption of conformity with relevant requirements

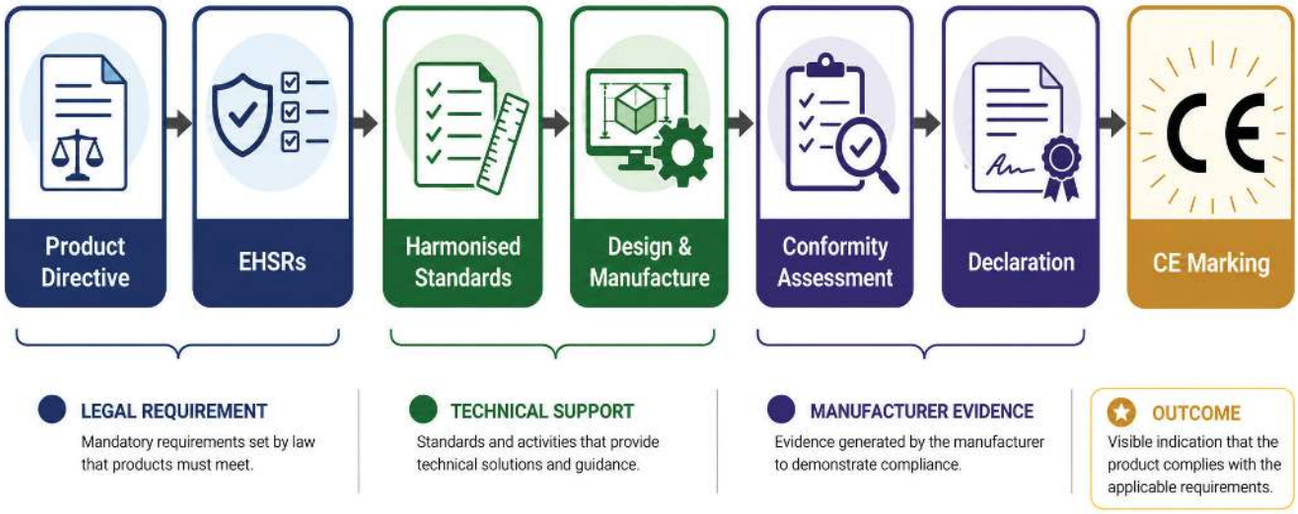
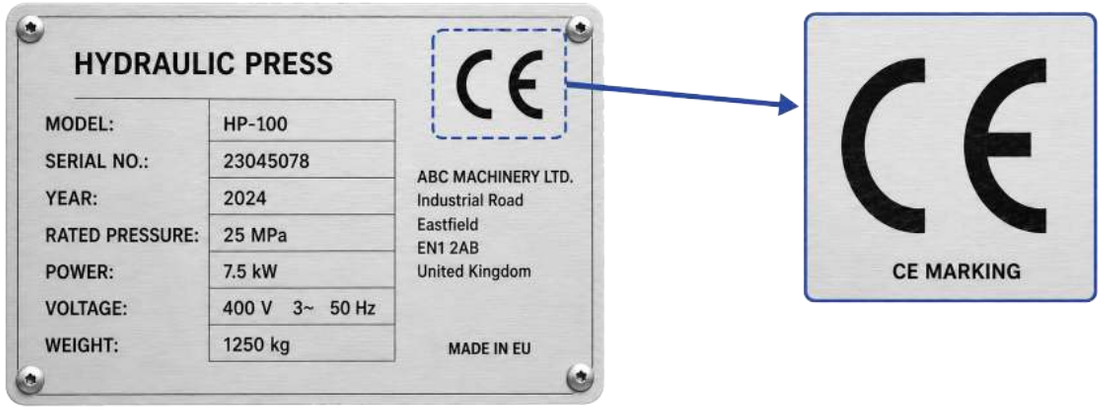


Figure 3.1 — Simplified conformity pathway showing how applicable EU product requirements may be translated into technical design, relevant standards, conformity assessment and manufacturer evidence before the appropriate conformity marking is applied, where required

SUBTOPIC 3.3 CE MARKING AND KEY EU PRODUCT DIRECTIVES

WHAT DOES CE MARKING MEAN?

You will see the CE marking on many types of industrial equipment.



Definition - CE MARKING

The CE marking is placed on a product by the manufacturer to declare that the applicable EU requirements have been addressed and that the required conformity-assessment procedures have been completed.

A very important distinction is:

CE is a conformity marking. It is not a general quality mark. The marking follows a specified geometric form. It has a minimum height of 5 mm and is associated with the identification of the manufacturer or authorised representative.

WHAT CE MARKING DOES NOT TELL YOU

Seeing a CE marking does not remove your responsibility to work safely.
Before using industrial equipment, you may still need to consider:

- whether you are looking at the correct equipment.
- whether the equipment is in suitable condition.
- whether guards or protective devices are present.
- whether you have the correct instructions.
- whether you have received the required training.
- whether the equipment is suitable for the intended task.
- whether additional workplace controls apply.



A marking is therefore one part of the safety information available to you.

KEY PRODUCT-DIRECTIVE EXAMPLE

Directive	What it concerns	Example of relevance
Machinery Directive 2006/42/EC	Machinery containing connected components, with at least one moving part and a drive system other than direct human or animal effort	Production machinery, machine assemblies

You do not need to memorize every directive number immediately. Your first task is to understand what type of equipment each directive addresses and why the information matters.

PARTLY COMPLETED MACHINERY

Not every machine-related assembly is a complete machine ready for independent operation. The Machinery Directive recognizes partly completed machinery. This means a subassembly may be intended only to be incorporated into other machinery.

Partly completed machinery is accompanied by:

- assembly Instructions
- a Declaration of Incorporation rather than the full Declaration of Conformity used for completed machinery.
- This distinction matters because a component intended for incorporation into another machine must not automatically be treated as a complete machine ready for independent use.

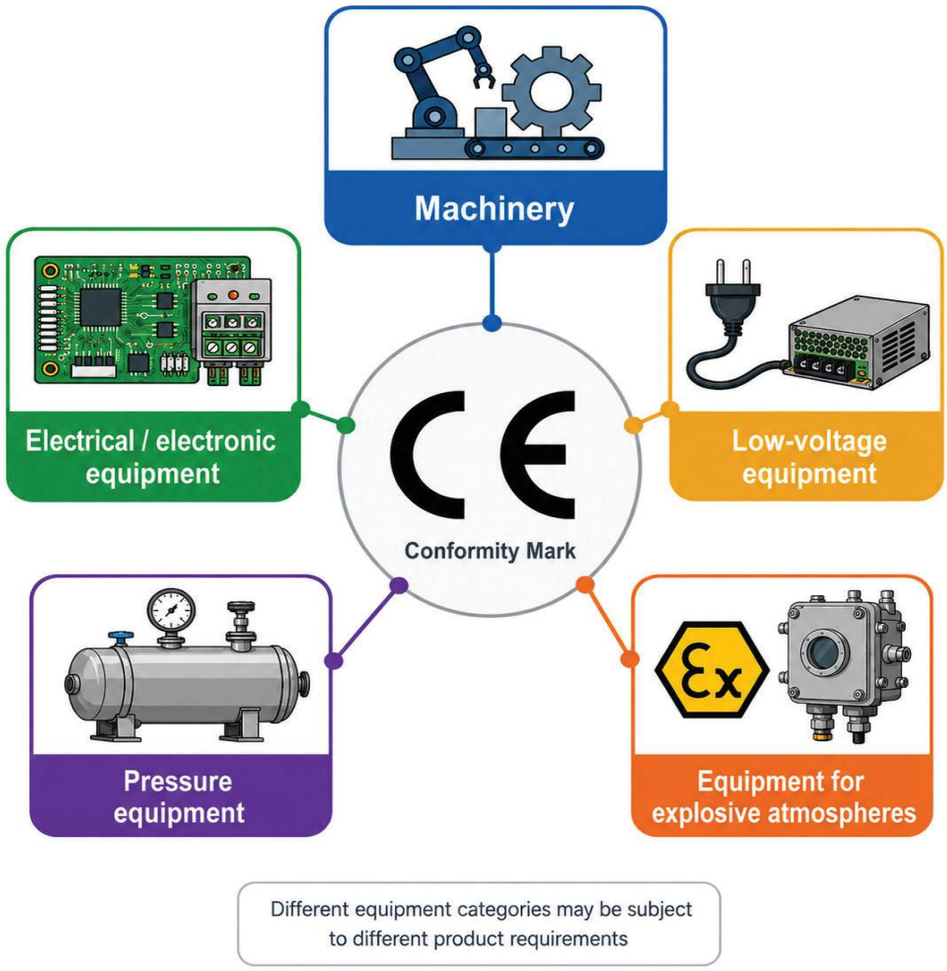


Figure 3.2 — Different types of industrial equipment may be subject to different product requirements before conformity is declared.



(a) (b) (c)

Figure 3.3 — Examples of partly completed machinery: a) servo valves; b) rotary tables (fitted without safety devices); c) hydraulic power units with solenoid valves.

THE Ex MARKING

Some equipment is designed for use in areas where explosive atmospheres may be present. The ATEX product requirements use a specific Ex marking in addition to the applicable CE marking. If you see an Ex marking, do not simply interpret it as:
"This equipment is explosion-proof in every situation."
Instead, recognise that the marking tells you that special equipment requirements apply.

You must still follow:

- the equipment information
- the authorized workplace classification
- manufacturer instructions
- site procedures
- task-specific controls

SAFETY CRITICAL: DO NOT MAKE YOUR OWN CONFORMITY DECISION

As a technician, do not:

- certify equipment yourself
- add or reproduce a CE marking
- alter an Ex marking
- declare an unmarked product compliant
- assume that a similar-looking product has the same approval
- release equipment for use because a marking appears to be present

Your role is to inspect the information provided, record what you observe, and report discrepancies through the authorized route.

SUBTOPIC 3.4 HOW TECHNICAL SAFETY STANDARDS SUPPORT INDUSTRIAL WORK

Legislation establishes requirements, but engineers and manufacturers also need detailed technical methods. This is one of the main purposes of technical standards.

Definition - TECHNICAL STANDARD

A technical standard is a documented technical specification, guideline or assessment method developed through a recognized standardization process.

Standards can support consistency in:

- design
- manufacture
- testing
- terminology
- measurement
- inspection
- safety
- quality
- compatibility



THREE LEVELS OF STANDARDISATION

Technical standards can be developed or adopted at different levels.

1. INTERNATIONAL

Important organizations include:

ISO

International Organization for Standardization

IEC

International Electrotechnical Commission

IEC has a particular focus on electrical, electronic and related technologies.

2. EUROPEAN

Important European standardization bodies include:

CEN

European Committee for Standardization

CENELEC

European Committee for Electrotechnical Standardization

ETSI

European Telecommunications Standards Institute

These bodies develop European technical specifications within their respective fields.

3. NATIONAL

National standards organizations develop or adopt standards for use within their national systems. The same standard may therefore carry several identifiers showing that it has been adopted at international, European and national levels.

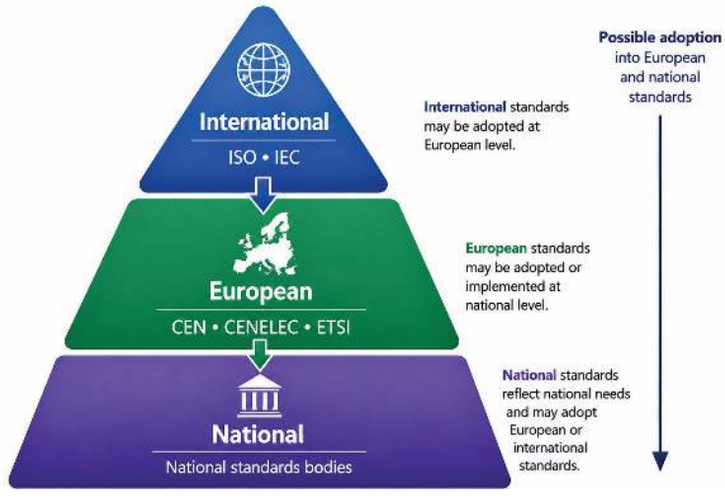


Figure 3.4 — Technical standards may be developed at international, European or national level, with international and European standards often adopted or implemented through national standardization systems.

IS A TECHNICAL STANDARD A LAW?

Not automatically. A technical standard and a legal requirement are different. Compliance with a standard may normally be voluntary unless that standard becomes part of a binding requirement.

For example, it may be:

- referenced by legislation
- incorporated into a technical regulation
- required by an applicable specification or authorized organizational requirement

Definition - TECHNICAL REGULATION

A technical regulation is a legally binding technical requirement.

The key distinction is:

A standard provides a recognized technical approach. Technical regulation imposes a binding technical requirement.

HARMONISED MACHINERY-SAFETY STANDARDS

For machinery, harmonized standards are grouped according to their scope.

Symbol	Meaning	Example
Type A	General machinery-safety design principles	EN ISO 12100— general principles for design, risk assessment and risk reduction
Type B1	Particular safety aspects relevant across different machine types	EN ISO 13849-1 — safety-related parts of control systems
Type B2	Particular safety devices or protective functions	EN ISO 13850 — emergency stop; EN ISO 13857 — safety distances
Type C	Detailed safety requirements for a specific category of machine	EN ISO 16092-1 — machine-tool presses

Type C requirements address a particular category of machine and take precedence over relevant Type A and B requirements where they differ.

REMEMBER

You are not expected to memorize every standard number.

When a standard is identified in equipment information, your task is to recognize:

1. what the standard relates to

2. why it is relevant

3. whether its requirements affect the task you are performing

3.1 CHECK YOUR UNDERSTANDING:

Which type of machinery standard deals specifically with safety devices?

☐ Type A

☐ Type B2

☐ Type B1

☐ Type C

SUBTOPIC 3.5 READING MARKINGS, LABELS AND EQUIPMENT INFORMATION

Knowing that directives and standards exist is useful. The practical skill is being able to find the information that affects your work. Before using equipment, you may need to inspect:

- identification information
- conformity markings
- safety markings
- warnings
- operating ratings
- manufacturer instructions
- conformity documentation
- precautions for safe use

START WITH PRODUCT IDENTIFICATION

Ask: Am I looking at the correct product and the correct document?

Check information such as:

- manufacturer
- product or equipment description
- model or type
- serial or identification number
- applicable rating information
- markings and warnings

Then compare the information with the documentation provided. A document for a similar item is not automatically suitable for the item in front of you.

CHECK THE MARKINGS

Look for markings that are relevant to the product.

Examples may include:

- CE marking
- Ex marking where applicable
- warning symbols
- manufacturer identification
- operating or rating information

A marking should be sufficiently clear to be interpreted. If an important marking is missing, unreadable or inconsistent with the documentation, do not invent the missing information. Record what you observe and report it.

USE THE MANUFACTURER INFORMATION

Manufacturer information may identify:

- required operating conditions
- protective devices
- warnings relating to use
- limitations
- training requirements
- safety precautions
- assembly requirements

For work equipment, conformity assessment is connected not only to product design but also to guards, protective devices, training information and precautions that must be followed during use.

DIFFERENT LABELS COMMUNICATE DIFFERENT INFORMATION

You may also encounter safety labels on substances used around industrial equipment.

Chemical hazard communication uses:

- **pictograms** to show the type of hazard
- **H-statements** to describe the nature and severity of the hazard
- **P-statements** to communicate precautionary measures

For example:

H-statement = What is the hazard?

P-statement = What precaution should I follow?

The **flame over a circle** pictogram indicates **oxidizing properties**: a substance or mixture that can promote combustion. Do not confuse it with the normal flame pictogram used for flammable substances.

INDUSTRIAL EXAMPLE

Situation:

You are preparing to assist with an industrial machine.

The equipment label shows:

Model: MX-200

Serial: 14582

CE marking: present

The operating instructions provided are labelled:

Model: MX-200A



1. Step 1 — Identify what matches

The equipment carries identification and conformity information.

2. Step 2 — Identify the discrepancy

The model number on the equipment is **MX-200**.

The document refers to **MX-200A**.

They are not identical.

3. Step 3 — Do not make an unsupported assumption



Do not say:

"They look almost the same, so the manual must be correct."

Do not say:

"The machine is illegal."

Neither statement has been demonstrated by your inspection.

4. Step 4 — Record the fact

A suitable factual statement is:

"The equipment is identified as Model MX-200. The supplied operating instructions are identified as Model MX-200A. Applicability of the instructions requires authorized confirmation before they are relied upon for the task."

5. Step 5 — Report

Refer the discrepancy to the responsible supervisor or authorized technical person. The key professional principle is:

Identify **compare** **record** **report**



Activity 3.1 CONNECT THE DIRECTIVE, STANDARD, MARKING AND SAFE ACTION

TIME	GROUPING	RESOURCES	EVIDENCE
20 min	Individual → pairs	Lesson notes, equipment-marking cards and reference sheet	Completed classification and equipment-information analysis

TASK:

PART A — PUT THE PROCESS IN ORDER

Number these stages from 1 to 5.

Stage	Order
CE marking applied where required	
Applicable EHSRs identified	
Conformity assessment completed	
Product directive identified	
Declaration of Conformity prepared	

PART B — CHECK YOUR KNOWLEDGE

1. Essential Health and Safety Requirements are:

a. optional quality improvements

b. essential health and safety requirements

c. maintenance schedules

d. worker training records



Answer: .....

2. Standards relating specifically to machinery safety devices are generally:

a. Type A

b. Type B1

c. Type B2

d. Type C

Answer: .....

3. The Electromagnetic Compatibility Directive 2014/30/EU is an EU product directive.

True ☐ **False** ☐

4. Complete the statement.

European technical specifications may be developed through standardisation bodies including:

CEN, ... **and** .....

5. CE marking is a manufacturer's general quality award showing that the product is better than competing products.

True ☐ **False** ☐

PART C — MATCH THE PRODUCT TO THE DIRECTIVE

Choose from:

Machinery • EMC • Low Voltage • Pressure Equipment • ATEX

Industrial item/situation	Directive
Pressure vessel operating above the specified pressure threshold	
Machine with powered moving parts	
Equipment intended for a potentially explosive atmosphere	
Electrical/electronic apparatus that must avoid unacceptable electromagnetic disturbance	
Electrical equipment within the specified voltage range	

PART D — INTERPRET THE MARKING

1. What does CE marking communicate?

.....

.....

2. Does CE marking mean you can automatically use the equipment for any task?

☐ **Yes** ☐ **No**

Explain:

.....

.....

3. What does an Ex marking alert you to?

.....

.....

4. A flame-over-circle pictogram appears on a chemical safety label.

What does it indicate?

☐ **Flammable**

☐ **Oxidizing**

☐ **Pressurized gas**

☐ **Acute toxicity**



PART E — DOCUMENT CHECK

You inspect the following:

Machine label:

Model: ST-450

Serial: 74218

Manufacturer instructions:

Model: ST-450

Serial reference: 74281

Answer:

1. What discrepancy do you identify?

.....

2. Should you simply assume the document belongs to the machine?

☐ **Yes**

☐ **No**

3. Write a professional observation.

.....

.....

4. What should happen next?

.....

✓

CHECKPOINT Before you submit

☐ I can explain the purpose of an EHSR.

☐ I can recognize CE as a conformity marking rather than a quality award.

☐ I can match the main product-directive examples to equipment types.

☐ I can identify Type B2 as relating to safety devices.

☐ I can distinguish a standard from a technical regulation.

☐ I can recognize the purpose of Ex marking.

☐ I can identify a mismatch between equipment and documentation.

☐ I can record a discrepancy without making an unsupported legal judgement.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN…

Explain how product directives and EHSRs relate to equipment safety.	WITH HELP ☐	INDEPENDENT ☐
Explain what CE marking means.	WITH HELP ☐	INDEPENDENT ☐
Recognize important EU product-directive examples.	WITH HELP ☐	INDEPENDENT ☐
Explain the purpose of harmonized standards.	WITH HELP ☐	INDEPENDENT ☐
Distinguish Type A, B1, B2 and C machinery standards.	WITH HELP ☐	INDEPENDENT ☐
Recognize important equipment and safety markings.	WITH HELP ☐	INDEPENDENT ☐
Compare equipment identification with manufacturer documentation.	WITH HELP ☐	INDEPENDENT ☐
Identify safe-use information and report a discrepancy.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION

- What is one thing that CE marking tells me?
- What is one thing that CE marking does not tell me?
- Why must the product identity match the documentation I use?

EXIT TICKET

Complete the relationships:

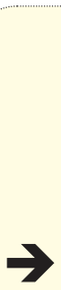
Directive → .....

Harmonized standard → .....

CE marking → .....

Manufacturer instructions → .....

My responsibility → .....



TRANSFER FROM KNOWLEDGE TO PERFORMANCE

In Practical Session 3, you will inspect controlled examples of:

- equipment identification labels
- CE and other relevant safety markings
- manufacturer documentation
- technical-standard references
- safe-use information

You will:

inspect → identify → compare → interpret → record → report

3 SESSION Inspect equipment labels, safety markings and manufacturer information and identify the precautions required for safe use

Skill code: PC1.1 Duration: 120 minutes/ 2 h
Location: Training workshop, laboratory or authorised simulated industrial area

SAFETY GATE: INSPECT INFORMATION, NOT LIVE EQUIPMENT

You may inspect only equipment, training items or documentation authorized by your trainer. Unless specifically authorized as part of a separate safe activity, do not:

- energize equipment
- operate machinery
- remove a guard
- open an electrical enclosure
- disconnect components
- alter a label
- remove a nameplate
- change the setting
- reproduce or add a conformity marking



If a marking or document cannot be read safely from your authorized position, report this rather than moving into an unsafe position.

Performance task	Inspect the equipment labels, safety markings and manufacturer information provided at each station. For each item: 1. identify the equipment or product. 2. identify the relevant marking or safety information. 3. identify the directive or standard information provided. 4. compare the equipment with its documentation. 5. identify the safe-use precautions stated or indicated. 6. record any missing, unreadable or inconsistent information. 7. report any discrepancy requiring authorized review.
Performance standard	• inspect at least four controlled equipment/document sets. • correctly identify relevant CE or safety markings where provided. • identify the meaning of an Ex marking where present. • identify the applicable product-directive information where it is provided. • identify relevant technical-standard references where they are provided. • distinguish equipment identification from conformity information. • compare product and document information accurately. • extract at least one relevant safe-use precaution from each assigned item. • record discrepancies factually. • work within the limits of your authority.

Required evidence	• completed Equipment Marking and Documentation Inspection Record. • inspection-station responses. • direct observation. • one verbal explanation. • one correctly recorded product/document discrepancy.

PREPARATION

PPE

Wear only the PPE required for the authorized training area.

- ☐ Safety footwear where required
- ☐ Safety glasses where required
- ☐ High-visibility clothing where required
- ☐ Other site-specific PPE specified by the trainer

TOOLS & RESOURCES

- ☐ Inspection worksheet
- ☐ Pen or pencil
- ☐ Product-directive reference card
- ☐ Technical-standard reference card
- ☐ Manufacturer documents
- ☐ Training equipment or simulated labels
- ☐ Magnifier where authorized and required for small markings
- ☐ Camera/tablet only where photography is permitted

PRE-START CHECKS

- ☐ I know which equipment I may inspect.
- ☐ I know which items are observation-only.
- ☐ I understand that I must not operate or modify equipment.
- ☐ I know the difference between CE marking and a quality mark.
- ☐ I know how to recognize an Ex marking.
- ☐ I understand that I should use the directive/standard information provided, not invent an applicable requirement.
- ☐ I know how to report a discrepancy.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Attend the safety briefing and confirm the inspection boundaries.	You can repeat the prohibited actions.
2	Record the manufacturer, equipment description and identification information provided.	Product can be traced to the inspection record.
3	Identify CE, Ex, warning or other authorized safety markings.	Markings recorded accurately.
4	Identify directive or technical-standard information provided with the item.	Correct reference recorded without guessing.
5	Compare equipment identification with manufacturer documentation.	Match or discrepancy recorded.
6	Identify safe-use precautions, limitations or warnings contained in the information provided.	At least one relevant precaution recorded.
7	Describe missing, unclear or inconsistent information factually.	Clear inspection statement.
8	Present one significant finding to the trainer.	Concise verbal report.

SESSION 03 . INSPECTION STATIONS

STATION A — EQUIPMENT IDENTIFICATION AND CE MARKING

-

Inspect the authorized equipment or simulated nameplate.

Record:

- manufacturer
- product/equipment description
- model or type
- serial or identification number
- relevant rating information
- CE marking
- warning information

Questions:

1. Can you identify the exact item?
2. Is the marking readable?
3. What does the CE marking communicate?
4. What additional information would you still need before using the equipment?
5. Is anything missing or unclear?

STATION B — PRODUCT DIRECTIVE AND SAFETY MARKING

Inspect the provided equipment-marking examples.

These may include:

- CE
- Ex
- electrical ratings
- pressure ratings
- machinery information

Use the reference card provided.

Record:

Equipment type:

.....

Marking observed:

.....

Relevant directive information provided:

.....

What does the marking tell you?

.....

What does it not tell you?

.....

STATION C — TECHNICAL STANDARD REFERENCE

Inspect the supplied equipment documentation.

You may see examples of standards such as:

- EN ISO 12100
- EN ISO 13849-1
- EN ISO 13850
- EN ISO 13857
- IEC 60204-1
- EN ISO 16092-1

Using the reference card, identify:

Standard reference

.....

Standard type where identified

☐ Type A

☐ Type B1

☐ Type B2

☐ Type C

What aspect of safety does it address?

.....

What safe-use information does this lead you to check?

.....

.....

.....

STATION D — MANUFACTURER DOCUMENTATION AND PRECAUTIONS

Compare the training equipment with the supplied documentation.

Check:

- manufacturer
- model/type
- serial or identification reference
- document title
- document revision where supplied
- markings
- instructions
- warnings
- stated limitations
- safe-use precautions

Important principle:

Similar does not necessarily mean identical.

If the document refers to a different item, do not silently use it.

SESSION 03 . EQUIPMENT MARKING AND DOCUMENTATION INSPECTION RECORD

No.	Equipment /item	Marking/ information observed	Directive / standard information provided	Safe-use precaution identified	Discrepancy/ required action
Example	Training machine MX-200	CE; Model MX-200 Serial 14582	Machinery information provided	Follow manufacturer operating precautions and authorized training	Manual supplied refers to MX-200A — report for document verification
1					
2					
3					
4					

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO	WE DO	YOU DO
The trainer demonstrates one equipment/document inspection. The equipment is labelled: Manufacturer: Industrial Systems Ltd **Model:** MX-200 **Serial:** 14582 **CE marking:** present The supplied manufacturer instructions state: **Model:** MX-200A **Serial reference:** 14528 The trainer verbalises: 1. "I inspect the equipment from the authorised position without operating it." 2. "I first identify the exact product." 3. "I can see a CE marking, which is a manufacturer's conformity declaration rather than a general quality award." 4. "I record the model and serial number exactly as shown." 5. "I now compare those details with the supplied documentation." 6. "The model and serial information do not match." 7. I do not assume that MX-200A is the same as MX-200. 8. "I do not say that the machine is illegal because my inspection has not established that." 9. "I record the discrepancy factually." 10. "I report the mismatch and request authorised confirmation of the correct documentation before relying on it for safe-use instructions."	The whole group inspects a second staged item. For example: An authorized training label contains: • CE marking • Ex marking • product identification • manufacturer information	You will rotate through the inspection stations. You must contribute at least: • one equipment-identification check • one marking interpretation • one directive or standard reference identified from the information provided • one safe-use precaution • one product/document comparison • one factual discrepancy statement • one verbal explanation

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/comment
Remained within the authorized inspection boundary and did not operate or alter equipment.	☐	☐	
Identified the equipment or product accurately.	☐	☐	
Recognized CE marking and explained its purpose appropriately.	☐	☐	
Recognized Ex marking where provided.	☐	☐	
Identified product-directive information from the reference material provided.	☐	☐	
Identified technical-standard information from the documentation provided.	☐	☐	
Distinguished Type A, B1, B2 or C standards where appropriate.	☐	☐	
Compared equipment identification with manufacturer documentation accurately.	☐	☐	
Identified relevant safe-use precautions or limitations.	☐	☐	
Recorded missing, unreadable or inconsistent information factually.	☐	☐	
Reported a discrepancy without making an unauthorized conformity judgement.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Safety and inspection procedure	Consistently maintains inspection boundaries, anticipates unsafe interaction and uses documentation without disturbing equipment.	Follows the authorized inspection procedure without operating or altering equipment.	Requires occasional reminders about sequence or boundaries.	Operates, alters, opens or approaches equipment in an unauthorized way.
Marking interpretation	Correctly interprets all relevant markings provided and clearly distinguishes what they do and do not demonstrate.	Correctly identifies the main CE, Ex and safety information provided.	Identifies some markings, but confuses their meaning.	Cannot identify the relevant marking or gives an unsafe interpretation.
Directive and standard interpretation	Correctly connects product information to the provided directive/ standard references and explains their different functions.	Correctly identifies relevant directive and technical-standard information from the supplied references.	Requires prompting or occasionally confuses directives and standards.	Cannot distinguish a directive from a technical standard.
Documentation and precautions	Detects subtle product/ document discrepancies and extracts precise safe-use limitations or precautions.	Correctly compares the product with its documentation and identifies relevant precautions.	Notices obvious information, but misses important mismatches or precautions.	Uses mismatched documentation without question or cannot identify relevant precautions.
Communication and records	Produces concise, traceable and technically precise inspection records and reports discrepancies objectively.	Completes required records clearly and gives an appropriate verbal report.	Records contain omissions or vague statements.	Records are inaccurate, incomplete or include unsupported conformity claims.

LESSON 4

OCCUPATIONAL HEALTH AND WORK-RELATED STRESS

OUR HEALTH AT WORK IS MORE THAN AVOIDING ACCIDENTS

A workplace hazard does not always cause immediate injury. Some working conditions affect your health gradually. Noise, vibration, unsuitable thermal conditions, hazardous substances or poorly organized work may affect you over time rather than causing one sudden event. Your psychological and social working conditions also matter.



Heavy or poorly balanced workloads, excessive time pressure, unclear responsibilities, conflict, isolation or unpredictable working hours can place demands on you that become difficult to manage. Recognizing these conditions is part of working safely. You are not expected to diagnose yourself or another worker. Your role is to:

recognize the workplace condition →

describe the concern factually →

identify an appropriate preventive measure →

report or refer through the authorized route:

WARM-UP

ACCIDENT OR HEALTH CONCERN?

You are preparing to work in a maintenance area. Before you begin, observe the following:

Consider these two situations.

Situation A

A worker slips on material left in a walkway and injures an ankle.

Situation B

A worker spends repeated shifts in a noisy area while also working under heavy time pressure and unpredictable hours.

Discuss with a partner:

1. Which situation could lead to a sudden injury?
2. Which situation may affect health over a longer period?
3. Can one workplace contain both safety risks and health risks?
4. Could the organization of the work itself create a health concern?
5. What should you do if you recognize a condition that may be affecting health?

Record your initial idea:

.....

.....

.....

.....

.....

.....

BY THE END OF THIS LESSON, I WILL BE ABLE TO:

1. Analyze work-related stress and occupational-health risks.
2. Recommend proportionate prevention, reporting and workplace-control measures.

KEY TERMS

• Occupational health • Health risk • Exposure • Psychosocial risk • Work-related stress • Work context • Job content • Workload • Work pace • Working hours • Thermal microclimate • Noise • Isolation • Role clarity • Autonomy • Health surveillance • Prevention • Referral

SUBTOPIC 4.1 RECOGNISING OCCUPATIONAL-HEALTH RISKS

In Lesson 1, you learned that occupational health includes your physical, mental and social well-being. Now you will look more closely at the workplace conditions that can affect that well-being. Not every occupational risk develops in the same way.

"A safe technician does not begin by asking only:"

SAFETY RISKS

Safety risks are generally associated with accidents that can cause sudden physical injury.



Examples include:

- falls from height
- slips, trips and falls
- collisions or crushing
- cuts and abrasions
- falling objects
- vehicles
- fire
- explosive atmospheres
- heat and flames

HEALTH RISKS

Health risks may result from prolonged or repeated exposure to physical, chemical or biological agents. The effect may not always appear immediately. Examples identified in industrial workplaces include:



- unsuitable thermal microclimate
- poor lighting conditions
- noise, ultrasound and infrasound
- mechanical vibration
- electromagnetic fields
- optical radiation
- ionizing radiation
- hazardous chemical agents
- carcinogens and mutagens
- asbestos
- biological agents

For this lesson, we will concentrate on recognizing common workplace health conditions such as noise and thermal conditions and understanding when they require action.

PSYCHOSOCIAL AND ORGANISATIONAL RISKS

The way work is organised can also influence health.

Examples include:

- work-related stress
- night work
- working in isolation
- employment-related factors
- age-related factors
- social and cultural factors

This means you should not limit occupational-health observations to machines, substances and physical conditions. The organization of the work itself can also create a workplace-health concern.

THREE QUESTIONS TO ASK

When you observe a workplace health concern, ask:

1. WHAT CONDITION IS PRESENT?

For example:

- high noise
- uncomfortable thermal conditions
- excessive workload
- poor communication
- unclear responsibilities



2. HOW COULD THE CONDITION AFFECT THE WORKER?

Think about the potential for:

- physical health effects
- psychological strain
- reduced well-being
- difficulty performing work safely

3. WHAT SHOULD HAPPEN NEXT?

Consider:

- prevention
- workplace control
- reporting
- occupational-health referral where appropriate



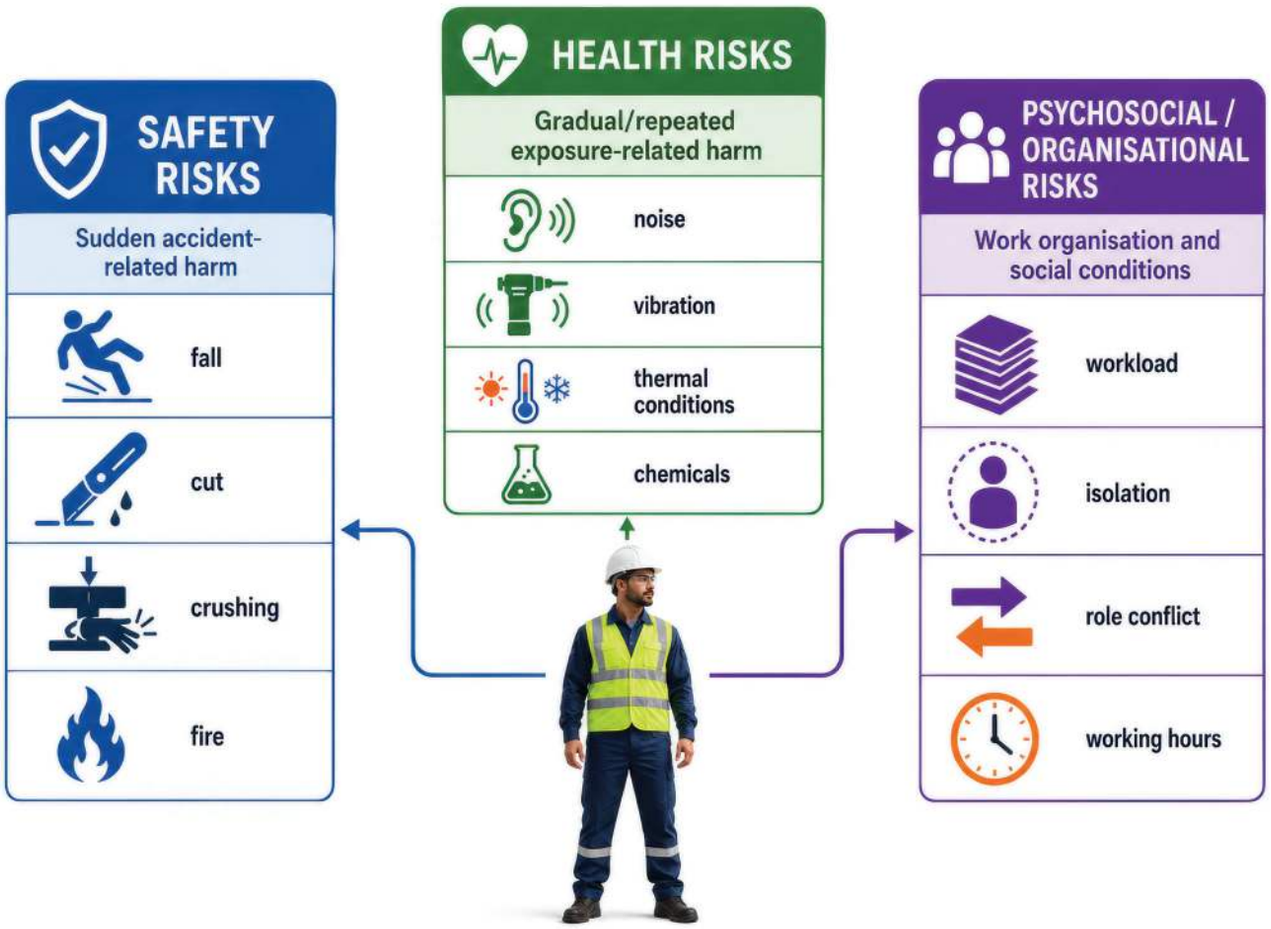


Figure 4.1 *— Workplace health and safety requires recognition of accident-related, occupational-health and psychosocial or organizational risks.*

SUBTOPIC 4.2 WHAT IS WORK-RELATED STRESS?

Work-related stress does not simply mean: "I had a difficult day." In occupational health, the term has a more specific meaning.

Definition: WORK-RELATED STRESS

Work-related stress can arise when you perceive that the demands placed on you are greater than the capacity or resources you have available to manage those demands effectively.

The resulting condition can affect:

- physical well-being
- psychological well-being
- social functioning
- performance at work

This is why work-related stress is both a **worker-health concern** and an **organizational concern.**

DEMANDS AND RESOURCES

Consider two situations

CONDITION A

A technician is given:

- a clear task
- realistic time
- suitable equipment
- clear responsibilities
- appropriate support

The demands of the task may be manageable with the resources available.

CONDITION B

The same technician is given:

- several urgent tasks at once
- unclear priorities
- inadequate equipment
- conflicting instructions
- very limited time

The demands may now exceed the resources available to manage them effectively. The worker has not changed. The **working conditions have changed.**

STRESS IS NOT ONLY ABOUT THE INDIVIDUAL

When analyzing work-related stress, avoid statements such as:

"That worker cannot handle pressure."

Instead ask:

- What demands does the work create?
- Are responsibilities clear?
- Is the workload realistic?
- Does the worker have sufficient resources and support?
- Are working hours suitable?
- Is the physical working environment appropriate?
- Are communication and relationships supporting safe work?



This shifts the focus from blaming the individual to examining the conditions in which the work is being performed.

IMPORTANT — DO NOT DIAGNOSE

As a technician, you should not diagnose a colleague as:

- "stressed"
- medically unfit
- mentally unwell
- Instead, record the **observable workplace condition.**

For example:

Avoid:

- "Ahmed is suffering from stress."

Better:

- "The task schedule includes repeated overtime, short completion deadlines and conflicting priorities." The second statement is factual and can be acted upon through the authorized workplace process.

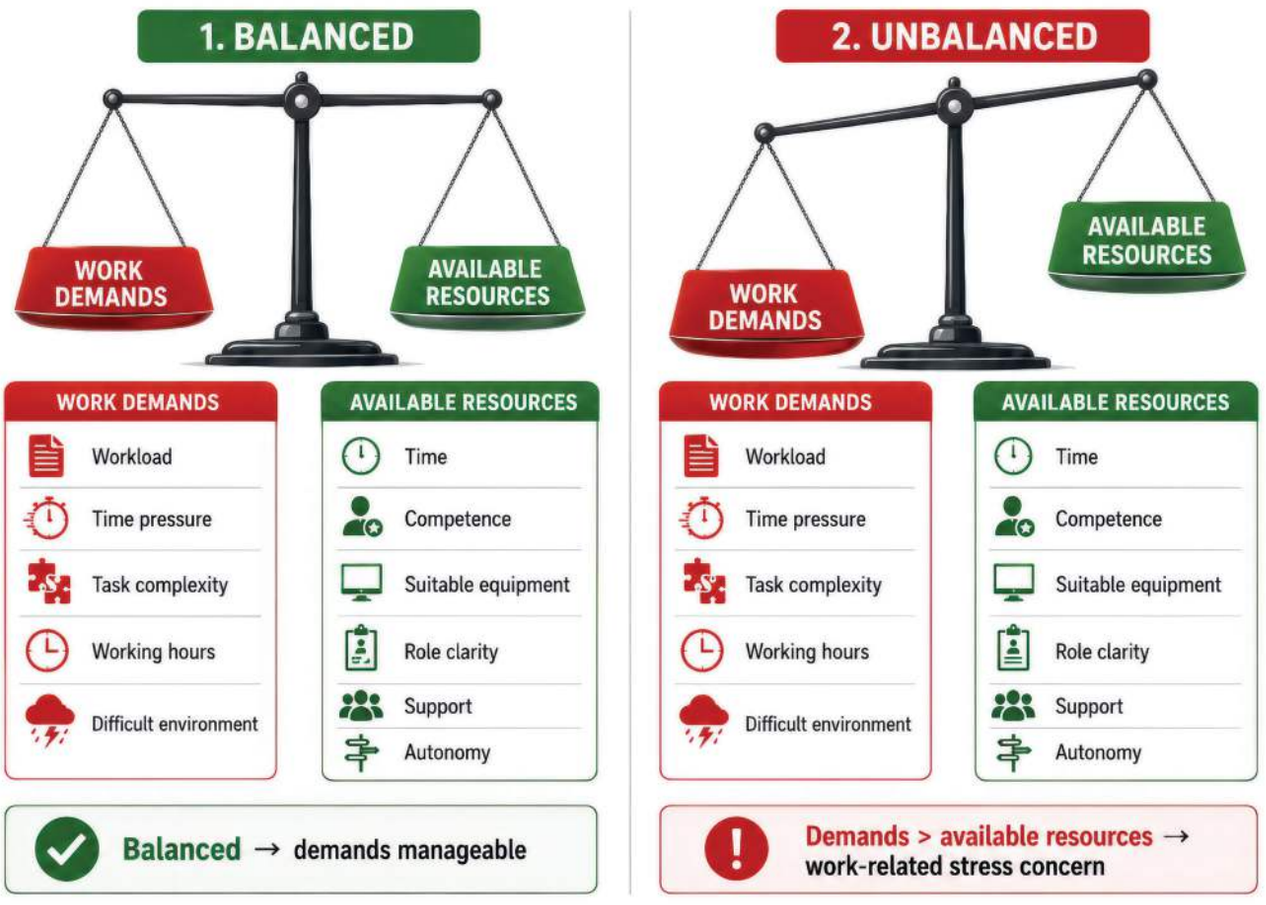


Figure 4.2 : *Work-related stress can develop when perceived work demands exceed the capacity or resources available to manage them.*

Table A1.5

Stress-inducing risk factors (source: Di Gennaro, Chiappetta and Chillemi (2023), New Course in Mechanical Technology, Volume 1, Hoepli, Table A1.5.)

Work context	
Organizational culture	Poor communication, low levels of support for problem-solving and personal development, lack of definition of organizational objectives
Role within the organization	Role ambiguity and conflict, responsibility for other people
Career development	Career uncertainty/stagnation, insufficient/excessive promotions, low pay, job insecurity, low social status attached to the work
Decision-making autonomy/control	Limited involvement in decision-making, lack of control over work (control, particularly in the form of participation, is also a broader organizational and contextual issue)
Interpersonal relationships at work	Physical or social isolation, limited contact with superiors, interpersonal conflict, lack of social support
Work-family interface	Conflicting demands between home and work, lack of support at home, dual-career issues

Job content	
Work environment and equipment	Physical working conditions, issues relating to reliability, availability, suitability, maintenance or repair of work facilities and equipment
Task planning	Monotony, short work cycles, fragmented or meaningless work, underutilization, high uncertainty
Workload/work pace	Overload or underload of work, lack of control over the pace, high levels of time pressure
Working hours	Shift work, rigid, unpredictable, excessively long working hours or hours that disrupt social rhythms

SUBTOPIC 4.3 WHAT WORKPLACE CONDITIONS CAN CONTRIBUTE TO STRESS?

The factors associated with work-related stress can be grouped into two broad categories:

1. **WORK CONTEXT:** how work is organized and managed
2. **JOB CONTENT:** the actual task and conditions in which it is performed. Understanding this distinction helps you describe the problem more accurately.

1. WORK CONTEXT: ORGANISATIONAL AND MANAGERIAL FACTORS

ORGANISATIONAL CULTURE

Stress-inducing conditions can include:

- poor communication
- limited support
- unclear objectives



Examples:

- You receive urgent instructions from two departments, but nobody explains which task has priority. The issue is not simply "pressure." There is also a **communication and work-organization problem.**

ROLE WITHIN THE ORGANISATION

Potential concerns include:

- unclear responsibilities
- role ambiguity
- conflicting responsibilities

Examples:

- Two supervisors give you different instructions for the same task.
- You should not guess which instruction to follow.
- The role or instruction conflict should be clarified through the authorized process.

CAREER DEVELOPMENT

Potential stress-inducing conditions can include:

- uncertainty
- job insecurity
- limited recognition
- lack of progression



AUTONOMY AND DECISION-MAKING

Potential concerns include:

- limited involvement in decisions
- inadequate control over work pace

This does not mean that every worker should make every decision independently. It means that an inappropriate lack of control over the way work is organized can be relevant when analyzing psychosocial conditions.

INTERPERSONAL RELATIONSHIPS

Potential concerns include:

- conflict with colleagues
- conflict with managers
- social isolation
- physical isolation

4.1 CHECK YOUR UNDERSTANDING:

- "Work-related stress does not depend on interpersonal relationships at work."

True ☐ **False** ☐

WORK–LIFE BALANCE

- Conflicting demands between work and home can also contribute to work-related stress.

2. JOB CONTENT — TASK AND ENVIRONMENTAL FACTORS

WORK ENVIRONMENT AND EQUIPMENT

Potential stress-inducing conditions include equipment or workplaces that are:

- inadequate
- unreliable
- poorly maintained

Physical conditions such as noise and unsuitable thermal conditions can also form part of the wider occupational-health environment.

TASK DESIGN

Potential concerns include work that is:

- monotonous
- fragmented
- perceived as meaningless
- not making appropriate use of the worker's skills



WORKLOAD AND PACE

Both extremes can create problems:

- chronic overload
- chronic underload
- excessive time pressure

A heavy workload is therefore not the only issue. Poorly designed work can also provide too little meaningful activity.

WORKING HOURS

Stress-inducing working-time conditions can include:

- shift work
- rigid schedules
- unpredictable schedules
- excessively long hours

PUTTING THE FACTORS TOGETHER

Work Context	Job Content
Communication	Work environment and equipment
Role clarity	Task design
Career development	Workload and pace
Autonomy	Working hours
Relationships	
Work–life balance	

WORKED EXAMPLE

A maintenance team has been working additional shifts during a shutdown

You observe:

- repeatedly extended working hours
- urgent deadlines
- several tasks competing for priority
- frequent changes to the schedule
- high background noise in the work area
- workers showing signs of fatigue after long shifts
- high temperatures in the work area



Do not record:

"The workers are stressed."

Instead, identify the conditions:

- excessive working hours
- fatigue associated with extended working hours and insufficient recovery time
- workload and time pressure
- unpredictable work organization
- occupational noise exposure
- high heat/unsuitable thermal conditions
- These are conditions that can be assessed and controlled.

TWO SOURCES OF WORK-RELATED STRESS

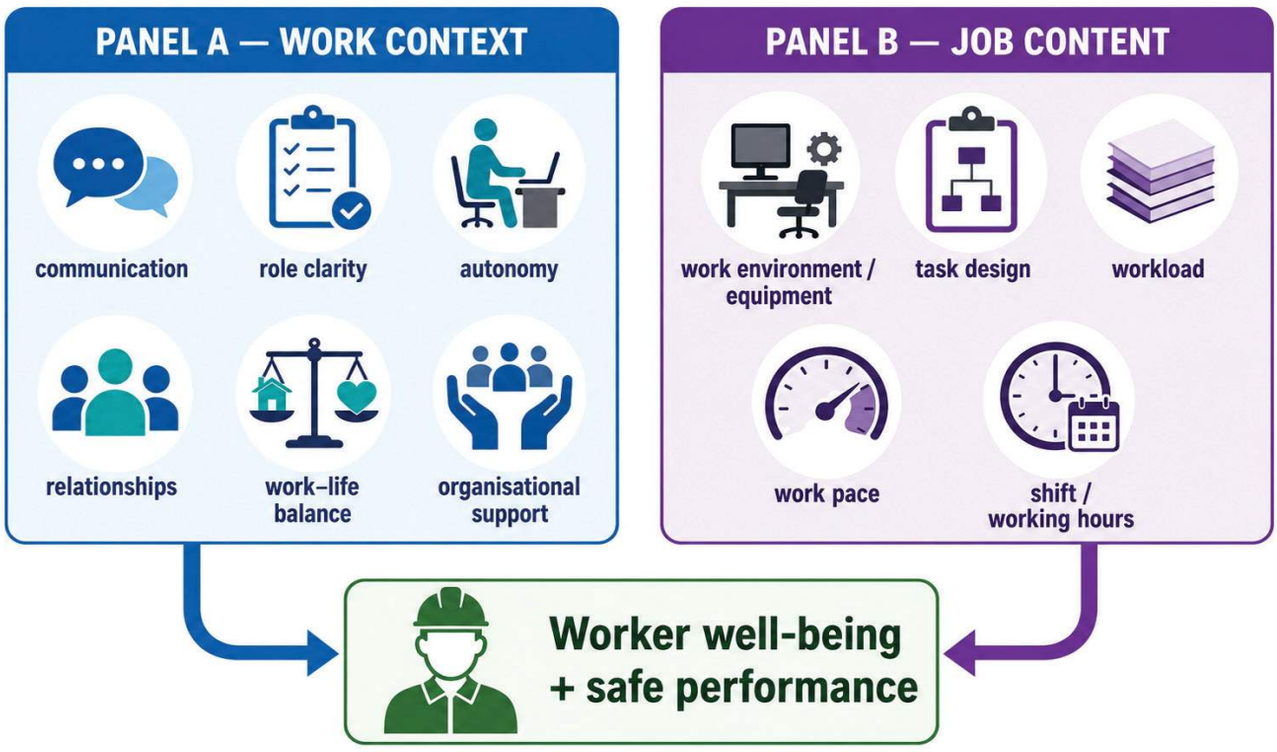


Figure 4.3 : *Work-related stress factors can arise from how work is organized and from the content and conditions of the task itself.*

SUBTOPIC 4.4 PREVENT, REPORT AND REFER

Recognising a workplace health concern is useful only if appropriate action follows. Different problems may require different responses. The correct response should be proportionate to the condition and within your authority.

PREVENTION CAN TAKE DIFFERENT FORMS

- **Occupational-health measures may be:**

1. ORGANISATIONAL

Examples can include changes to:

- work planning
- communication
- supervision
- training
- health surveillance
- workload arrangements

2. PROCEDURAL

Examples can include:

- clear work instructions
- defined reporting processes
- authorized task procedures

3. TECHNICAL

Examples can include:

engineering measures used to improve the physical work environment. For example, a physical exposure may require a technical control rather than simply telling the worker to "cope with it." You will study detailed control selection later in the unit. For now, focus on connecting the observed problem to a sensible category of action.

WHAT SHOULD YOU REPORT?

Report observable conditions such as:

- "Repeated extended shifts are shown on the current work schedule."
- "Workers in this area must raise their voices to communicate because of the background noise."
- "Two supervisors have provided conflicting priorities for the same task."
- "The maintenance team has several urgent tasks with the same completion deadline."
- Avoid unsupported statements such as:
- "Everyone is stressed."
- "The supervisor is causing mental-health problems."

Your record should describe the workplace evidence, not make a medical diagnosis or personal accusation.

WHEN MIGHT OCCUPATIONAL-HEALTH SUPPORT BE NEEDED?

An Occupational Health Doctor or other authorized occupational-health professional may support:

- assessment of occupational-health risks
- health surveillance
- appropriate medical examinations or tests
- assessment of worker fitness in relation to occupational exposure

If a concern requires medical or specialist assessment, refer; do not diagnose.

YOUR RESPONSIBILITY

As a technician:

YOU CAN:

- recognize concerning workplace conditions
- record factual information
- report the concern
- follow authorized controls
- cooperate with workplace-health arrangements

YOU SHOULD NOT:

- diagnose another person's health
- make confidential medical judgements



- change someone's fitness-for-work status
- invent exposure limits
- alter work arrangements without authority
- ignore a concern because no accident has occurred

SAFETY CRITICAL: IMMEDIATE HARM TAKES PRIORITY

- If a person appears to be in immediate danger, follow the authorized emergency and supervisory arrangements.
- Do not delay an urgent safety response simply because the issue also has an occupational-health dimension.



Figure 4.4 : *Occupational-health concerns should be recorded factually and managed through proportionate prevention, reporting and authorized referral.*



Activity 4.1 WHAT IS CREATING THE HEALTH OR STRESS CONCERN?

Purpose: Identify occupational-health and work-related stress factors and recommend an appropriate workplace response.

TIME	GROUPING	RESOURCES	EVIDENCE
20 min	pairs	Lesson notes, scenario cards and Activity 4.1 sheet	Completed classification, scenario analysis and professional reporting statement

TASK:

PART A: CLASSIFY THE RISK

Identify the main category using:

Safety risk • Health risk • Psychosocial/organizational risk

No.	Situation	Category
1	A worker may be struck by a falling object.	
2	A worker is repeatedly exposed to high workplace noise.	
3	A team receives conflicting instructions and unclear priorities.	
4	A worker repeatedly uses vibrating equipment.	
5	A team is experiencing prolonged excessive workload and time pressure.	

PART B: TRUE OR FALSE?

Work-related stress does not depend on interpersonal relationships at work.

True ☐ **False** ☐

PART C: WORK CONTEXT OR JOB CONTENT?

Classify each condition.

No.	Condition	Work context	Job content
1	Poor communication	☐	☐
2	Excessive workload	☐	☐
3	Role conflict	☐	☐
4	Unpredictable working hours	☐	☐
5	Poorly maintained equipment	☐	☐
6	Social isolation	☐	☐
7	Excessive time pressure	☐	☐
8	Limited control over work pace	☐	☐

PART D: ANALYSE THE WORKPLACE CASE

A maintenance team is completing urgent shutdown work. During the week:

- working hours have repeatedly been extended
- priorities change several times each shift
- workers report to two supervisors who sometimes give different instructions
- several urgent jobs have the same deadline
- the work area has persistent high background noise
- one worker says, "I cannot keep up with all of this."

Answer the following questions:

1. Identify two job-content factors.

Factor 1:

.....

Factor 2:

.....

2. Identify two work-context factors.

Factor 1:

.....

Factor 2:

.....

3. Identify one physical occupational-health concern.

.....

4. Why would it be inappropriate to write "the worker has a stress disorder"?

.....

5. Suggest two proportionate workplace actions.

Action 1:

.....

Action 2:

.....

6. Who might need to receive or review the concern?

.....

PART E: SAY IT PROFESSIONALLY

Rewrite this statement:

"Everyone is stressed because management has given them too much work."

- Use observable evidence rather than diagnosis or blame.

Your professional statement:

.....

.....

.....

.....

.....

CHECKPOINT Before you submit

- ☐ I distinguished safety, health and psychosocial risks.
- ☐ I identified work-context factors correctly.
- ☐ I identified job-content factors correctly.
- ☐ I considered physical workplace conditions as well as work organisation.
- ☐ I described observable conditions rather than diagnosing a person.
- ☐ My recommended action relates to the identified concern.
- ☐ I identified an appropriate reporting or referral route.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN…

Distinguish safety, health and psychosocial/organizational risks.	WITH HELP ☐	INDEPENDENT ☐
Explain what work-related stress means.	WITH HELP ☐	INDEPENDENT ☐
Identify work-context stress factors.	WITH HELP ☐	INDEPENDENT ☐
Identify job-content stress factors.	WITH HELP ☐	INDEPENDENT ☐
Recognise workload, working-hours and physical-environment concerns.	WITH HELP ☐	INDEPENDENT ☐
Recommend a proportionate preventive or workplace-control measure.	WITH HELP ☐	INDEPENDENT ☐
Identify when authorised occupational-health support may be needed.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION	EXIT TICKET
	Use one example from today's lesson.
• Which work-related stress factor was easiest for me to recognize?	• Observed workplace condition:
	Risk/concern:
• What is the difference between observing a workplace condition and diagnosing a worker?	• One question I still have…
• Which physical workplace conditions could also affect worker well-being?	• One preventive action that could be considered before work begins:

➔

TRANSFER FROM KNOWLEDGE TO PERFORMANCE

- **In Practical Session 4,** you will analyze realistic workplace health and work-related stress scenarios.

For each scenario, you will:

observe → identify the condition → classify the concern → record the evidence → recommend prevention/control → report or refer

4 SESSION — Analyze workplace health and stress scenarios and recommend suitable preventive measures, reporting routes or controls

Skill code: PC1.1 / LO3 Duration: 120 minutes/ 2 h Location: Classroom simulation, training workshop or controlled scenario-based industrial exercise

SAFETY GATE: SAFETY AND PROFESSIONAL

This activity deals with occupational-health and psychosocial workplace conditions. Treat all scenarios professionally.

You must:

- discuss workplace conditions rather than make personal judgements.
- use factual language.
- respect privacy.
- avoid asking classmates to disclose personal health information.
- avoid diagnosing physical or psychological conditions.
- work only with the information contained in the scenario.
- refer specialist or medical questions to the appropriate authorised person.



Performance task	Analyze the assigned workplace-health scenarios. **For each scenario:** **1.** identify the observable workplace condition. **2.** classify it as a physical, occupational-health or psychosocial/organisational concern. **3.** where relevant, identify whether the stress factor relates mainly to work context or job content. **4.** identify who may be affected. **5.** describe the possible impact without making a diagnosis. **6.** recommend an appropriate preventive or workplace-control measure. **7.** identify the reporting or referral route. **8.** record your reasoning clearly.
Performance standard	• analyse at least four workplace-health scenarios. • identify credible physical and psychosocial/organisational concerns. • distinguish work context from job content. • identify workload, working-hours, communication and workplace-environment factors. • recognize when more than one factor is present. • recommend proportionate preventive or workplace-control measures. • distinguish workplace reporting from occupational-health referral. • communicate concerns without diagnosing or blaming an individual. • remain within the limits of my authority.

Required evidence	• completed Occupational Health and Work-Related Stress Assessment Record. • four scenario analyses. • assessor observation. • one professional verbal report. • one prevention/control recommendation. • one reporting or referral decision.

PREPARATION

RESOURCES

- Workplace scenario cards
- Occupational-health assessment record
- Work Context / Job Content reference card
- Risk-category reference card
- Workplace images where used
- Pen or pencil

PRE-START CHECKS

- ☐ I understand that I am assessing workplace conditions, not diagnosing people.
- ☐ I can distinguish safety, health and psychosocial risks.
- ☐ I can distinguish work context from job content.
- ☐ I know that several factors can exist in one scenario.
- ☐ I know how to record evidence factually.
- ☐ I understand when a concern may require reporting or occupational-health referral.
- ☐ I know how to report or refer a workplace-health concern.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Attend the professional-conduct and task briefing. Confirm that you are assessing workplace conditions, not diagnosing individual workers.	You can explain the limits of your role and the requirement to use factual, respectful language.
2	Read or observe the complete workplace scenario and identify the conditions that may affect worker health or well-being.	Relevant workplace conditions identified without unsupported assumptions.
3	Decide whether each concern is mainly a physical occupational-health risk or a psychosocial/organizational risk.	Health and psychosocial concerns classified correctly.
4	Where work-related stress is relevant, identify whether the factor relates to work context or job content. Consider factors such as communication, role clarity, workload, work pace, working hours, isolation, equipment and the physical environment.	Work-context and job-content factors identified accurately.
5	Describe what you actually observed or what the scenario states. Avoid diagnosing, blaming or making personal judgements.	Clear factual statement based on observable workplace evidence.
6	Suggest a proportionate preventive or workplace-control measure that addresses the identified condition.	Recommended action is relevant to the workplace concern and remains within learner authority.
7	Identify who should receive the concern, such as the supervisor, responsible manager, HSE function or authorised occupational-health professional.	Appropriate reporting or referral route identified.
8	Present one significant finding using the sequence condition → concern → evidence → preventive action → reporting/referral route, then review your record for factual language.	Concise professional verbal report and completed assessment record.

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO	WE DO	YOU DO
The trainer demonstrates one case: **A maintenance team has worked several extended shifts. Several urgent tasks have the same deadline and priorities change repeatedly during each shift.** **The trainer verbalizes:** 1. **"I start with what I can actually observe."** 2. **"I can identify extended working hours, competing deadlines and changing priorities."** 3. **"I do not write that the team has a medical condition."** 4. **"Working hours and workload/pace are job-content factors."** 5. **"Changing priorities may also indicate an organisational or communication issue."** 6. **"More than one stress factor can exist at the same time."** 7. **"I record who may be affected — in this case, the maintenance team."** 8. **"I describe the potential concern without diagnosing individuals."** 9. **"I recommend a review of workload, priorities and working-time arrangements through the authorised management process."**	**The whole group analyses a second scenario:** • A worker performs a repetitive task in a noisy area. Equipment faults frequently interrupt the task, and the worker receives little support when priorities change.	**You will rotate through the remaining scenario stations or analyse assigned cases independently.** **You must contribute at least:** • two correctly identified health/stress conditions • one work-context classification • one job-content classification • one physical occupational-health observation • one preventive/workplace-control recommendation • one reporting or referral decision • one factual verbal explanation

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/ comment
Identified observable workplace-health conditions without making unsupported assumptions.	☐	☐	
Distinguished accident-related safety risks from exposure-related health risks.	☐	☐	
Recognized psychosocial and organizational workplace factors.	☐	☐	
Distinguished work-context factors from job-content factors.	☐	☐	
Identified workload and workplace concerns where present.	☐	☐	
Identified working-hours / fatigue-related concerns where present.	☐	☐	
Identified physical workplace-health factors such as noise or thermal conditions where present.	☐	☐	
Recognized when several interacting factors were present.	☐	☐	
Recommended an appropriate preventive or workplace-control measure.	☐	☐	
Selected an appropriate reporting or referral route.	☐	☐	
Avoided medical diagnosis or personal blame.	☐	☐	
Communicated findings using factual occupational-health terminology.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Identification of occupational-health concerns	Identifies physical, psychosocial and organizational concerns accurately, including interacting or less-obvious factors.	Correctly identifies the main occupational-health and stress-related conditions.	Identifies obvious issues, but misses important factors or confuses categories.	Cannot identify the relevant workplace-health concern.
Stress-factor analysis	Accurately distinguishes work context and job content and explains how several factors may interact.	Correctly classifies the main work-context and job-content factors.	Classification is partly correct, but inconsistent.	Cannot distinguish work context from job content.
Evidence and professional judgement	Uses precise observable evidence and consistently avoids unsupported diagnosis or blame.	Records factual evidence and describes the concern appropriately.	Uses some vague or person-centred statements requiring correction.	Makes unsupported medical diagnoses, personal accusations or conclusions not supported by the case.
Prevention/ workplace control	Recommends proportionate organizational, procedural or technical responses and explains why they address the observed conditions.	Recommends an appropriate preventive or workplace-control measure.	Recommendation is broadly relevant, but requires development.	Recommends ineffective, irrelevant or unauthorised action.
Reporting and referral	Clearly distinguishes supervisory/HSE reporting from occupational-health referral and explains when specialist input is required.	Identifies an appropriate reporting or referral route.	Recognizes the need to report, but needs support selecting the appropriate route.	Fails to report appropriately or attempts to make specialist decisions outside learner authority.

LESSON 5

INDUSTRIAL HAZARD IDENTIFICATION AND RECORDING

LOOK BEYOND THE OBVIOUS HAZARD



When you enter an industrial workshop, some hazards may be easy to notice. You might immediately see:

- a sharp edge
- a moving machine
- a hot surface
- an obstruction in a walkway

Other hazards may be less obvious. You may need to notice:

- electrical exposure
- excessive noise or vibration
- hazardous substance
- poor manual-handling conditions
- a missing or ineffective guard
- unsafe access to a machine danger zone
- poorly organized pedestrian and vehicle routes

Professional hazard identification means more than saying:

"That looks dangerous."

You need to describe:

1. what you observed
2. where you observed it
3. what type of hazard it is
4. how someone could be harmed
5. what consequence could result
6. whether immediate safe action is needed
7. what should be recorded and reported

In this lesson, you will develop that systematic approach.



WARM-UP

HOW MANY HAZARDS CAN YOU SEE?

Study an industrial or steel-workshop image provided by your trainer.

Look at:

- machines
- walkway
- workstations
- tools
- storage areas
- people and activities

Individually, identify five conditions that could cause harm.

Then compare your list with a partner.

Discuss

1. Did you both notice the same conditions?
2. Which hazard was easiest to identify?
3. Which condition was less obvious?
4. Can two different hazard types exist in the same location?
5. Why is simply writing "dangerous machine" not enough for a professional record?

Your first observations

1. .....
2. .....
3. .....
4. .....
5. .....

By the end of this lesson, I will be able to…

1. Analyze workplace and equipment safety requirements by identifying relevant hazard types.
2. Identify and record hazards/unsafe conditions accurately in the steel-workshop context.
3. Inspect a workplace systematically and describe the location and possible consequences of identified hazards.
4. Recognize conditions requiring immediate safe action and reporting.

KEY TERMS

• Industrial hazard • Mechanical hazard • Electrical hazard • Thermal hazard • Chemical hazard
• Physical hazard • Ergonomic hazard • Unsafe condition • Danger zone • Work equipment
• Exposure • Consequence • Inspection • Immediate safe action • Hazard record

SUBTOPIC 5.1 RECOGNIZING INDUSTRIAL HAZARD CATEGORIES

Hazards in an industrial workplace can arise from many different sources. Using hazard categories helps you inspect the workplace systematically.

For this unit, you will use six practical categories:

1. MECHANICAL
2. ELECTRICAL
3. THERMAL
4. CHEMICAL
5. PHYSICAL
6. ERGONOMIC

These categories help you organize your observations. They do not mean that every hazard fits perfectly into only one category. A single workplace condition can sometimes involve more than one hazard.

1. MECHANICAL HAZARDS

Mechanical hazards commonly arise from:

- machinery
- moving components
- sharp or pointed objects
- vehicles
- work at height
- falling objects
- crushing or trapping points

Possible events include:

- cutting
- puncture
- abrasion
- impact
- collision
- crushing
- shearing
- entrapment
- falling
- being struck by an object or vehicle

EXAMPLE

- **Observed condition:**
  - A sharp metal sheet projects into a material-handling area.
- **Hazard category:**
  - Mechanical.
- **Potential consequence:**
  - Cut, puncture, or laceration.

2. ELECTRICAL HAZARDS

- Electrical hazards concern conditions that could expose someone to electrical energy. Common hazards include:
- electric shock
- overvoltage
- electrical aspects of machinery and work equipment

At this stage, your task is to recognize and report the electrical hazard.





Examples include:

- **Observed condition:**
- An electrical condition is present that may expose a worker to electric shock.
- **Hazard category:**
- Electrical.
- Immediate action:
- Keep clear and report through the authorized process.



IMPORTANT

Do not:

- investigate inside electrical equipment
- remove covers
- touch suspect electrical parts
- attempt isolation unless you are trained and authorized

3. THERMAL HAZARDS

Thermal hazards can arise from:

- heat
- flames
- hot materials
- cold conditions
- unsuitable thermal environments

Possible consequences include:

- burns
- other heat-related injuries
- harm arising from unsuitable thermal exposure
- In a steel production environment, thermal hazards may be particularly significant because materials and processes can involve high temperatures.

4. CHEMICAL HAZARDS

Chemical hazards arise from substances that have harmful properties.

Exposure may occur through:

- inhalation
- skin or eye contact
- ingestion



Examples include substances that are:

- corrosive
- toxic
- harmful
- flammable
- oxidizing
- carcinogenic

You learned earlier that labels, pictograms, and Safety Data Sheets provide information about chemical hazards. During a workplace inspection, never identify an unknown substance by guessing. If the substance cannot be identified safely, record that the identity or hazard information requires authorized confirmation.

5. PHYSICAL HAZARDS

For this lesson, physical hazards include environmental agents such as:

- noise
- ultrasound or infrasound
- vibration
- poor lighting
- unsuitable atmospheric or environmental conditions
- electromagnetic fields
- optical radiation
- ionizing radiation
- Some of these hazards may cause harm gradually through repeated exposure.



EXAMPLE

- **Observed condition:**
- Persistent high noise is present around operating industrial equipment.
- **Hazard category:**
- Physical.
- **Potential concern:**
- Occupational noise exposure.
- Do not invent a noise measurement if one has not been taken.
- Record what you actually observed.

6. ERGONOMIC HAZARDS

- Ergonomic hazards arise when the physical demands of work may place excessive strain on the worker.

Examples include:

- lifting and carrying loads
- pushing and pulling
- repetitive handling
- biomechanical overload
- poorly arranged work activity



EXAMPLE

- **Observed condition:**
- A worker repeatedly handles heavy or awkward material manually.
- **Hazard category:**
- Ergonomic.
- **Possible consequence:**
- Musculoskeletal strain or injury.

ONE CONDITION - MORE THAN ONE HAZARD

Consider a worker manually moving hot metal components. The task might involve:



1. THERMAL HAZARD

- Contact with hot material.

2. ERGONOMIC HAZARD

- Manual handling of the load.

3. MECHANICAL HAZARD

- Impact or crushing if the component is dropped.

- **A good inspection therefore asks:**
- Have I identified all relevant hazard types, not just the most obvious one?

HAZARD-CATEGORY SUMMARY

Hazard Category	Examples to look for	Possible consequence examples
Mechanical	Sharp objects, moving equipment, crushing, entrapment, falls, vehicles, falling objects	Cuts, fractures, crushing injury, amputation
Electrical	Conditions presenting possible electrical-energy exposure	Electric shock or other electrical injury
Thermal	Heat, flames, hot/cold conditions	Burns or thermal injury
Chemical	Hazardous substances, vapors, contact hazards	Chemical burns, inhalation harm, other health effects
Physical	Noise, vibration, lighting, radiation, electromagnetic fields	Exposure-related occupational health effects
Ergonomic	Manual handling, pushing/pulling, repetitive work, biomechanical overload	Musculoskeletal strain or injury

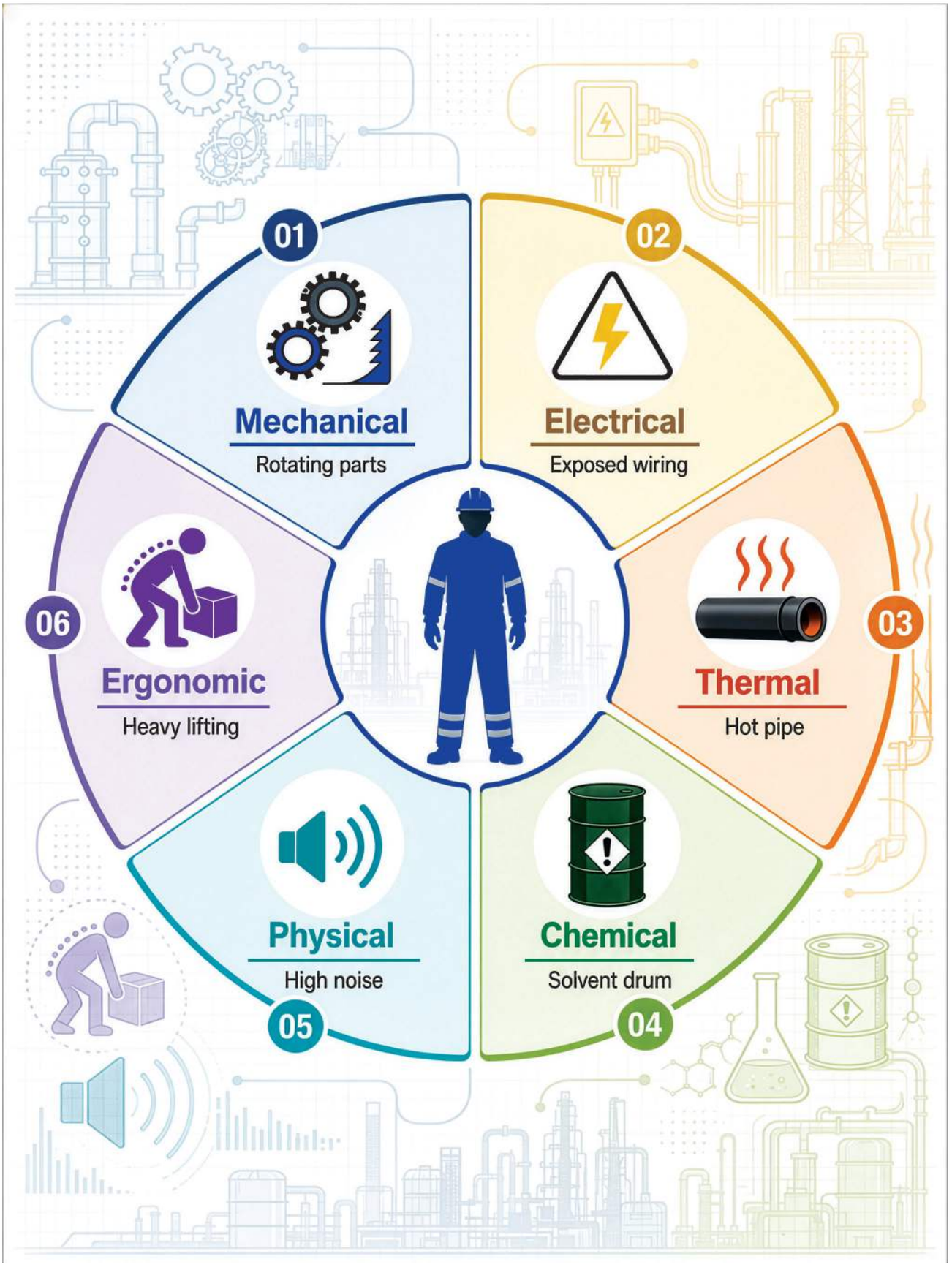


Figure 5.1: *Industrial hazard categories help you inspect a workplace systematically rather than relying only on the most obvious dangers.*

SUBTOPIC 5. 2 FROM WORKPLACE CONDITION TO POTENTIAL CONSEQUENCE?



EXAMPLE 1: SHARP MATERIAL

Observed condition:	Sharp sheet-metal edge projecting into a handling area.
Hazard	Sharp edge.
Exposure/event:	Worker's hand or body contacts the edge.
Potential consequence:	Cut or laceration



EXAMPLE 2: MATERIAL IN A WALKWAY

Observed condition:	Loose material is present in a pedestrian route.
Hazard	Obstruction in the route.
Exposure/event:	A person strikes or steps on the material.
Potential consequence:	Trip, fall, impact injury, or fracture.



EXAMPLE 3: MACHINE DANGER ZONE

Definition: DANGER ZONE

A danger zone is an area within or close to work equipment where a person's presence can create a risk to health or safety.

A guard creates a physical barrier between the person and the hazardous zone.

During your inspection, ask:

- Is access to a hazardous area prevented?
- Is a guard or barrier visibly present where expected?
- Is the person being kept outside the danger zone?
- Does anything appear damaged, displaced, or missing?

You are inspecting the condition. You are not testing the safety device by entering the danger zone.

EXAMPLE 4: MANUAL HANDLING

Observed condition:	Material must be moved manually repeatedly.
Hazard	Biomechanical/manual-handling demand.
Exposure/event:	Repeated lifting, carrying, pushing, or pulling.
Potential consequence:	Musculoskeletal strain or injury.



DO NOT CONFUSE THE FOLLOWING TERMS

Term	Example
Observed condition	Sharp sheet projects into walkway
Hazard	Sharp metal edge
Exposure/event	Worker contacts edge
Potential consequence	Laceration

Avoid writing:

- "Hazard: injury."
- An injury is the possible consequence, not the hazard.



Figure 5.2: *A professional hazard record links the observed workplace condition to the hazard, exposure event and potential consequence.*

SUBTOPIC 5. 3 INSPECT THE WHOLE WORKPLACE — NOT JUST THE MACHINE

A workplace is not only the machine or workbench in front of you. Your inspection should consider the wider working area. Workplace requirements include floors, routes, emergency access and work equipment.

LOOK DOWN: FLOORS AND WALKWAYS:

Workplace floors should be:

- stable
- fixed
- non-slip
- free from dangerous projections, holes or other unsafe surface conditions

During an inspection, look for:

- obstructions
- material left in pedestrian routes
- unstable surfaces
- conditions that could cause slipping or tripping

LOOK AROUND: PEDESTRIANS AND VEHICLES

- Traffic routes should be clearly organized. Where vehicles and pedestrians use nearby routes, workers need sufficient separation to move safely. During an inspection, ask:
- Is the route clearly identifiable?
- Is material obstructing the route?
- Could a worker enter a vehicle movement area unexpectedly?
- Has access become restricted?

LOOK TOWARD ACCESS AND EXIT ROUTES

Emergency routes and exits need to remain clear. For hazard-identification purposes, you should therefore **recognize:**

- an obstructed route
- material stored where access is required
- conditions that could delay movement to safety
- In this lesson, you only need to identify and report the unsafe condition. You do not need to explain the emergency procedures in detail.

LOOK AT THE EQUIPMENT

Work equipment includes:

- machines
- apparatus
- equipment and components used in production
- tools
- plant

Use of work equipment includes activities such as:

- operation
- transport
- maintenance
- cleaning
- assembly
- dismantling

This means hazards can appear during more than normal production. A machine that appears controlled during normal operation may present different conditions during cleaning or maintenance.

LOOK AT THE MATERIALS AND SUBSTANCES

Ask:

- Are materials stored securely?
- Could material fall, move, or obstruct access?
- Is a chemical clearly identified?
- Is the relevant hazard information available?
- Is hot material located where people could contact it?
- Do not move a substance or material simply to complete your inspection.

LOOK AT THE TASK

- Some hazards arise from **what the worker is doing**, rather than from a defective object. Examples include:
- working at height
- manual handling
- repetitive movement
- working near moving equipment

A SYSTEMATIC INSPECTION PATTERN

Rather than walking around randomly, use this pattern:

1. AREA
- Floors • routes • access • environment

2. EQUIPMENT
- Machines • tools • danger zones • visible protective arrangements

3. MATERIALS
- Storage • sharp objects • hot materials • substances

4. ACTIVITY
- Movement • manual handling • work at height • interaction with equipment

5. PEOPLE
- Who could be exposed and where?
- This makes it less likely that you will overlook a hazard.



Figure 5.3: *A systematic workplace inspection should consider the work area, equipment, materials, activities and people who may be exposed; hazards may be present in more than one of these aspects.*



Figure 5.4: Examples *of hazards that may be identified during a systematic workplace inspection.*

SUBTOPIC 5. 4 RECORD THE HAZARD AND TAKE THE CORRECT IMMEDIATE ACTION

A hazard that is noticed, but not communicated can remain a hazard. Your inspection record needs to allow another authorized person to understand:

- what you saw
- where it was
- what hazard category it belongs to
- what could happen
- what immediate action was taken or is required



WHAT SHOULD YOUR RECORD CONTAIN:

1. LOCATION

- Where exactly did you observe the condition?

Too general: "Workshop."

Be specific: "Material-handling route beside the sheet-storage rack."

2. OBSERVED CONDITION

What did you actually see?

Too general: "Very dangerous."

Be specific: "Sharp sheet-metal edge projects approximately into the pedestrian side of the handling route."

3. HAZARD TYPE

Choose the relevant category:

- mechanical
- chemical
- electrical
- physical
- thermal
- ergonomic

If more than one category applies, record the main hazard and note the additional hazard where appropriate.

4. POTENTIAL CONSEQUENCE

- What could reasonably happen if exposure occurs?

Examples:

- cut
- electrical injury
- fracture
- chemical exposure
- crushing injury
- musculoskeletal strain
- burn

5. IMMEDIATE SAFE ACTION

Ask:

What should happen now to prevent exposure while the condition is reviewed?



Examples may include:

- remain clear of the area
- warn people in the immediate vicinity
- stop the task where authorized safety arrangements require it
- prevent unauthorized approach
- inform the supervisor or responsible person
- follow the site's reporting process

PRIORITY FOR ACTION: BEFORE FORMAL RISK SCORING

- You will learn how to assess risk formally in Lesson 6. For now, do not assign a numerical risk score. Instead, focus on identifying conditions that may require immediate attention or action.

IMMEDIATE SAFE ACTION IS NEEDED WHEN:

- exposure is occurring or could occur immediately
- a person may enter an obvious danger zone
- the condition could result in serious harm
- a required safety arrangement appears to be absent or ineffective
- continued work would expose people unnecessarily

In these situations:

Keep clear	warn if necessary	stop within your authority	report immediately.

For other conditions:

Record accurately	report through the authorized route	allow the responsible person to determine the formal priority.

DO NOT FIX WHAT YOU ARE NOT AUTHORISED TO FIX

During inspection, do not:

- repair machinery
- reconnect electrical equipment
- remove material from a danger zone
- handle an unidentified chemical
- adjust a guard
- bypass a protective device
- enter a restricted area



Your responsibility is:

OBSERVE	IDENTIFY	RECORD	TAKE IMMEDIATE SAFE ACTION	REPORT

INDUSTRIAL EXAMPLE:

1. Observation

- Several steel offcuts are stored partly across a marked pedestrian route.

2. Location

- Pedestrian route beside fabrication workstation 2.

3. Hazard type

- Mechanical

4. Potential event

- A person could strike or trip over the material.

5. Potential consequence

- Fall, impact injury, or fracture.

6. Immediate safe action

- Keep pedestrians away from the affected section and report the obstruction through the authorized workplace process.

7. What you do not do automatically

- Do not lift or relocate the steel yourself unless specifically authorized and trained to do so.

Activity 5.5 IDENTIFY, CLASSIFY AND RECORD INDUSTRIAL HAZARDS

Purpose: Practice identifying relevant industrial hazards and accurately recording unsafe conditions in preparation for the practical steel-workshop inspection.

TIME	GROUPING	RESOURCES	EVIDENCE
30 min	Individual ➡ pairs	Industrial scenario images, Lesson 5 hazard-category card and Activity 5.1 sheet	Completed hazard classification, unsafe-condition analysis, and professional hazard records

TASK:

PART A: MATCH THE HAZARD CATEGORY

- **Choose from**

Mechanical • Electrical • Thermal • Chemical • Physical • Ergonomic

No.	Observed condition	Main hazard category
1	Sharp steel edge exposed in a handling area	
2	Repeated manual lifting and carrying of material	
3	Persistent high noise around production equipment	
4	A condition that may expose a worker to electric shock	
5	Contact possible with hot material	
6	Corrosive substance being handled	
7	Repeated vibration exposure from equipment	
8	Material positioned where it may fall onto a worker	

PART B: WHAT COULD HAPPEN?

- **Complete the chain.**

1. Sharp sheet-metal edge

Hazard:

.....

Exposure/event:

.....

Potential consequence:

.....

2. Material obstructing a pedestrian route

Hazard:

.....

Exposure/event:

.....

Potential consequence:

.....

3. Repeated manual handling

Hazard:

.....

Exposure/event:

.....

Potential consequence:

.....

PART C: SPOT THE UNSAFE CONDITION

1. CASE 1:

Materials have been stored partly across a pedestrian route.

Observed condition:

.....

Hazard category:

.....

Potential consequence:

.....

2. CASE 2:

A machine has a hazardous operating area. A physical guard that normally prevents access appears to be displaced.

Observed condition:

.....

Hazard category:

.....

Immediate safe action:

.....

CASE 3:

A route intended to remain clear for emergency movement contains stored equipment.

Observed condition:

.....

Why does it require action?

.....

Immediate safe action/report:

.....

PART D: CHECK THE WORKPLACE REQUIREMENT

The lesson describes workplace barriers, guards, and protective arrangements.

- **Answer:**

1. Why can a barrier or guard be important during hazard identification?

.....

.....

2. If a guard appears missing or ineffective, should you enter the danger zone to investigate?

Yes ☐ **No** ☐

- **Explain:**

.....

.....

3. Collective protective arrangements are intended to protect people from workplace health and safety risks.

True ☐ **False** ☐

PART E: WRITE IT PROFESSIONALLY

- **Rewrite each weak statement.**

1. Weak statement 1

"There is dangerous stuff on the floor."

Professional record:

.....

.....

.....

2. Weak statement 2

"The machine is unsafe."

Professional record:

.....

.....

.....

Weak statement 3

"Someone will get hurt."

Professional record:

.....

.....

.....

PART F: WHICH NEEDS ATTENTION FIRST?

- **Consider these two observations.**
- **Observation A**
  - A labelled container is correctly closed and stored in its designated location, but a small amount of the substance has spilled on the floor nearby.
- **Observation B**
  - A worker is standing close to an accessible hazardous machine area where the expected physical separation is absent.
- **Which condition requires the more immediate safe response?**

.....

.....

.....

- **Why?**

.....

.....

.....

- **What should you do?**

.....

.....

.....

✓

CHECKPOINT Before you submit

- ☐ I identified the correct hazard category.
- ☐ I distinguished the observed condition from the potential consequence.
- ☐ I identified the location clearly.
- ☐ I linked the hazard to a reasonable exposure or event.
- ☐ I identified an appropriate immediate safe action.
- ☐ I did not propose an action outside my authority.
- ☐ My records are factual and specific.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN…

Identify mechanical hazards.	WITH HELP ☐	INDEPENDENT ☐
Identify electrical hazards at recognition level.	WITH HELP ☐	INDEPENDENT ☐
Identify thermal and chemical hazards.	WITH HELP ☐	INDEPENDENT ☐
Identify physical and ergonomic hazards.	WITH HELP ☐	INDEPENDENT ☐
Inspect an area systematically.	WITH HELP ☐	INDEPENDENT ☐
Distinguish a hazard, exposure/event and consequence.	WITH HELP ☐	INDEPENDENT ☐
Record a hazard type and location accurately.	WITH HELP ☐	INDEPENDENT ☐
Recognize when immediate safe action is needed.	WITH HELP ☐	INDEPENDENT ☐
Report a hazard without attempting unauthorized repair.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION	EXIT TICKET
• Which hazard category was easiest for me to identify?	Choose one industrial hazard and complete the hazard record below: **Location:**
• Which hazard category do I need more practice recognizing?	**Risk/concern:**
• Why is a systematic inspection better than looking only for obvious hazards?	**Observed condition:**
• What makes a hazard record useful to another person?	**Potential consequence:**
	• **Immediate safe action**

➔

TRANSFER FROM KNOWLEDGE TO PERFORMANCE

- In Practical Session 5, you will complete a structured hazard inspection in an authorized steel workshop or simulated industrial environment. Your working sequence will be:

SCAN ➡ OBSERVE ➡ LOCATE ➡ CLASSIFY ➡ EXPLAIN ➡ RECORD ➡ ACT / REPORT

5 SESSION Complete structured hazard identification and recording in a simulated steel-workshop environment

Skill code: PC1.1 / TPC1.1 Duration: 240 minutes/ 4 h

Location: Authorized steel workshop, controlled training workshop, or realistic simulated industrial environment

SAFETY GATE: INSPECT WITHOUT CREATING EXPOSURE

This session requires you to inspect a real or simulated industrial environment. Hazard identification must never create a new hazard.

Before beginning, confirm:

- the inspection boundary
- required PPE
- restricted areas
- pedestrian routes
- the stop/report instruction for an immediate danger
- equipment that is observation-only
- photography rules
- the reporting route

Performance task	Inspect the assigned steel-workshop zones systematically. Identify and record relevant hazards and unsafe conditions. **For every finding, record:** **1.** location. **2.** observed condition. **3.** hazard category. **4.** who may be exposed. **5.** how exposure or the harmful event could occur. **6.** potential consequence. **7.** immediate safe action/reporting requirement.
Performance standard	**By the end of the session, you should be able to:** • complete a systematic inspection of all assigned areas. • identify at least eight credible hazards or unsafe conditions, where present. • demonstrate recognition across at least four different hazard categories, where the inspection area provides them. • distinguish the observed condition from the possible injury. • identify the exact location of each finding. • identify who could be exposed. • describe a reasonable potential consequence. • recognize conditions requiring immediate safe action. • complete the hazard record accurately. • report one significant finding verbally. • remain within the limits of your authority.

Required evidence	• completed Steel-Workshop Hazard Identification Record. • inspection-zone check sheets where provided. • direct-observation evidence. • one verbal hazard report. • one priority/immediate-action explanation.

PREPARATION

RESOURCES

Wear the PPE required by the authorized training location.

- Safety footwear
- Eye protection
- High-visibility clothing where required
- Head protection where required
- Hearing protection where required by the site
- Other authorized PPE:

TOOLS & RESOURCES

- Clipboard / digital inspection device
- Pen or pencil
- Steel-Workshop Hazard Identification Record
- Hazard-category reference card
- Workshop plan / inspection-route sheet
- Camera only where authorized
- Other:

PRE-START CHECKS

- ☐ I know the authorized inspection route.
- ☐ I know which areas are restricted.
- ☐ I know which equipment is observation-only.
- ☐ I understand that I must not test a hazard by exposing myself to it.
- ☐ I know the six hazard categories used in this lesson.
- ☐ I know how to report immediate danger.

WORKFLOW

Step	Learner action	Evidence/checkpoint
1	Attend the workshop briefing and confirm PPE, boundaries, restricted areas and reporting arrangements.	You can explain where you may inspect and what actions are prohibited.
2	Observe the assigned zone before approaching individual items. Look at the area, equipment, materials, activities, and people.	Initial overview completed without entering hazardous areas.
3	Locate credible hazards or unsafe conditions.	Relevant observations identified without touching or operating equipment.
4	Classify each finding as mechanical, electrical, thermal, chemical, physical, or ergonomic.	Appropriate hazard category recorded.
5	Identify who could be exposed, how the event could occur, and the possible consequence.	Hazard ➡ exposure/event ➡ consequence link is clear.
6	Decide whether immediate safe action or urgent reporting is required. Do not calculate a formal risk score.	Immediate action/reporting requirement recorded.
7	Complete all required fields using factual and specific language.	Traceable hazard record produced.
8	Present one significant finding using the authorized reporting route.	Clear verbal report supported by the written record.
9	Re-check the area and your record for overlooked hazard categories, vague wording or unsupported assumptions.	Record corrected before submission.
10	Complete the self-check and submit your evidence.	Complete authentic inspection evidence.

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO	WE DO	YOU DO
The trainer demonstrates one controlled steel-workshop example: Several steel offcuts extend partly into a marked pedestrian route beside a fabrication workstation. **The trainer verbalizes:** **1.** "I begin by observing the condition from within the authorized inspection route." **2.** "I do not move the steel simply because I have noticed the problem." **3.** "I record the exact location: pedestrian route beside fabrication workstation 2." **4.** "My observed condition is factual: several steel offcuts extend into the marked pedestrian route." **5.** "The main hazard category is mechanical." **6.** "Pedestrians and workers moving through this area may be exposed." **7.** "Exposure could occur if someone strikes, steps on or trips over the material." **8.** "A reasonable potential consequence is a fall, impact injury or fracture." **9.** "Because people *could encounter the obstruction while using the route, immediate safe action may be required.*"	**The whole group analyses several safely staged examples of unsafe conditions:** • a sharp sheet-metal edge projecting into a handling area • an accessible machine danger zone where the expected guard appears displaced • a possible electrical exposure condition visible from the authorized route • hot material positioned close to an access area • a hazardous substance with an unreadable or incomplete label • persistent high workplace noise • repeated manual lifting of awkward material	**You will complete the authorized steel-workshop inspection using the structured hazard record.** **You must contribute evidence of:** • at least eight credible hazard or unsafe condition observations, where present • hazards from at least four categories, where the inspection area provides them • accurate locations for each recorded finding • appropriate hazard classifications • potential consequences • immediate safe-action or reporting decisions • one significant verbal hazard report You must not invent hazards simply to meet a numerical target.

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/ comment
Followed the authorized inspection route, PPE requirements and workshop instructions.	☐	☐	
Inspected hazards without touching, operating, moving or modifying equipment or materials.	☐	☐	
Used a systematic inspection approach covering area, equipment, materials, activities and people.	☐	☐	
Identified credible mechanical hazards where present.	☐	☐	
Recognized possible electrical hazards without attempting unsafe investigation.	☐	☐	
Identified thermal, chemical, physical and ergonomic hazards where present.	☐	☐	
Distinguished the observed condition from the hazard and possible consequence.	☐	☐	
Identified the location of each hazard accurately.	☐	☐	
Identified who may be exposed and how the harmful event could occur.	☐	☐	
Identified reasonable potential consequences.	☐	☐	
Recognized when immediate safe action or urgent reporting was required.	☐	☐	
Completed the hazard record using clear and factual technical language.	☐	☐	
Communicated one significant finding clearly through the authorized reporting process.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Systematic workplace inspection	Uses a highly systematic scan and identifies obvious and less-obvious conditions across the work area, equipment, materials, activities and people.	Completes the assigned inspection systematically and identifies the required range of credible conditions.	Focuses mainly on obvious hazards or overlooks parts of the assigned inspection area.	Inspects randomly, misses significant conditions or enters areas outside the authorized route.
Hazard identification and classification	Accurately identifies multiple hazard types and recognizes where one condition creates more than one hazard.	Correctly identifies and classifies the main mechanical, electrical, thermal, chemical, physical or ergonomic hazards present.	Identifies hazards, but makes some category errors or confuses hazard with consequence.	Cannot identify the main hazard or repeatedly uses incorrect classifications.
Exposure and consequence reasoning	Clearly links the condition, exposure/event and potential consequence and recognizes multiple credible exposure routes where relevant.	Correctly identifies who may be exposed, how the event could occur and a reasonable potential consequence.	Reasoning is partly correct, but vague or incomplete.	Cannot explain how the observed condition could result in harm.
Immediate safe action	Correctly distinguishes urgent conditions from routine findings and explains a proportionate immediate response within the limits of authority.	Identifies appropriate immediate safe action or reporting requirements.	Recognizes that action is needed, but requires guidance on what is proportionate or authorized.	Ignores an obvious urgent condition or proposes unsafe/unauthorized intervention.
Hazard records and communication	Produces concise, traceable and technically precise records and gives a clear action-oriented verbal report.	Records all required information clearly and communicates one significant finding appropriately.	Records contain omissions, vague wording or minor inconsistencies.	Records are incomplete, inaccurate or contain unsupported assumptions.

LESSON 6

Risk Assessment and Preventive Measures

FROM FINDING A HAZARD TO DECIDING WHAT TO DO

In Lesson 5, you learned how to inspect an industrial workplace and record hazards accurately. Now you need to answer two further questions:

- How significant is the risk?
- What should be done to control it?

Not every hazard creates the same level of risk. The level of risk depends on the conditions in which the hazard is encountered, particularly:

- how likely exposure or a harmful event is
- how serious the possible consequence could be

Once the risk has been assessed, the next step is to select the most effective feasible preventive or protective measure. You should not automatically begin with PPE. Where possible, control the hazard at its source.

WARM-UP

SAME HAZARD, DIFFERENT PRIORITY

Consider the same sharp metal edge under two conditions.

CONDITION A

The sharp edge is:

- inside a controlled storage rack
- physically separated from the pedestrian route
- unlikely to be contacted during normal work

CONDITION B

The same type of sharp edge:

- projects into a frequently used handling route
- is at hand height
- can easily be contacted by workers passing the area

Discuss:

1. Is the hazard the same?



2. Is exposure equally likely?
3. Could the severity of a cut be similar?
4. Which condition should receive greater attention?
5. What control would act most directly on the problem?

Record your starting decision:

.....

.....

BY THE END OF THIS LESSON, I WILL BE ABLE TO:



1. analyze workplace and equipment safety requirements.
2. apply preventive safety measures in simulated scenarios.
3. document the relationship between identified risk and selected control.
4. explain how likelihood and severity influence risk.
5. apply the hierarchy of controls.

KEY TERMS

• Risk assessment • Likelihood • Probability • Severity • Risk priority • Initial risk • Residual risk • Prevention • Protection • Elimination • Substitution • Engineering control • Administrative control • PPE • Collective protection • Corrective action • Improvement action • Guard • Protective device

SUBTOPIC 6.1 ASSESS RISK USING LIKELIHOOD AND SEVERITY

You already know that a hazard is something with the potential to cause harm. Risk considers whether that harm could actually occur under the conditions present.

The following two factors are central:

1. LIKELIHOOD

Definition - LIKELIHOOD

Likelihood describes how probable it is that exposure to a hazard or a harmful event will occur under the conditions being assessed.

Ask:

How likely is the harmful event to occur?

Consider evidence such as:

- how often people enter the area
- how frequently the task is performed

- how close workers are to the hazard
- whether access is prevented
- whether an existing control is present
- whether the work is routine or unusual
- whether the condition makes exposure easy or difficult



Do not choose a likelihood simply because it "feels right."

Use the evidence in the scenario and the authorized assessment method.

2. SEVERITY

Definition - SEVERITY

Severity describes how serious the potential harm could be if the hazardous event occurs.

Ask:

If this event happens, how serious could the consequence reasonably be?

Possible consequences may range from minor injury to:

- serious laceration
- fracture
- crushing injury
- permanent impairment
- serious occupational ill health
- fatal injury

The consequence must be credible for the hazard being assessed. Do not select the worst imaginable outcome if there is no reasonable connection to the condition.

The relationship can be represented simply as

LIKELIHOOD + SEVERITY → RISK PRIORITY

If workers are more likely to be exposed to a hazard, and the possible harm is more serious, the condition should be given higher priority for action.

USE THE AUTHORISED RISK-ASSESSMENT METHOD

Your workplace or training organization may use:

- descriptive ratings
- numerical ratings
- a risk matrix
- another approved method

IMPORTANT

Do not invent your own rating scale. If the authorized workplace matrix differs from one you have seen elsewhere, use the authorized workplace method.

INITIAL RISK

Definition - INITIAL RISK

The initial risk is the level or priority determined from the conditions and controls that exist when you carry out the assessment. This means you should record existing controls before deciding what additional action is needed.

EXAMPLE

Condition:

A machine has moving hazardous components.

Existing control:

A suitable fixed guard prevents normal access to the danger zone. The hazard associated with the moving machinery still exists. However, the guard changes the conditions of exposure. Compare this with the same machine where the required guard is missing or displaced. The hazard may be similar, but the likelihood of exposure is very different.

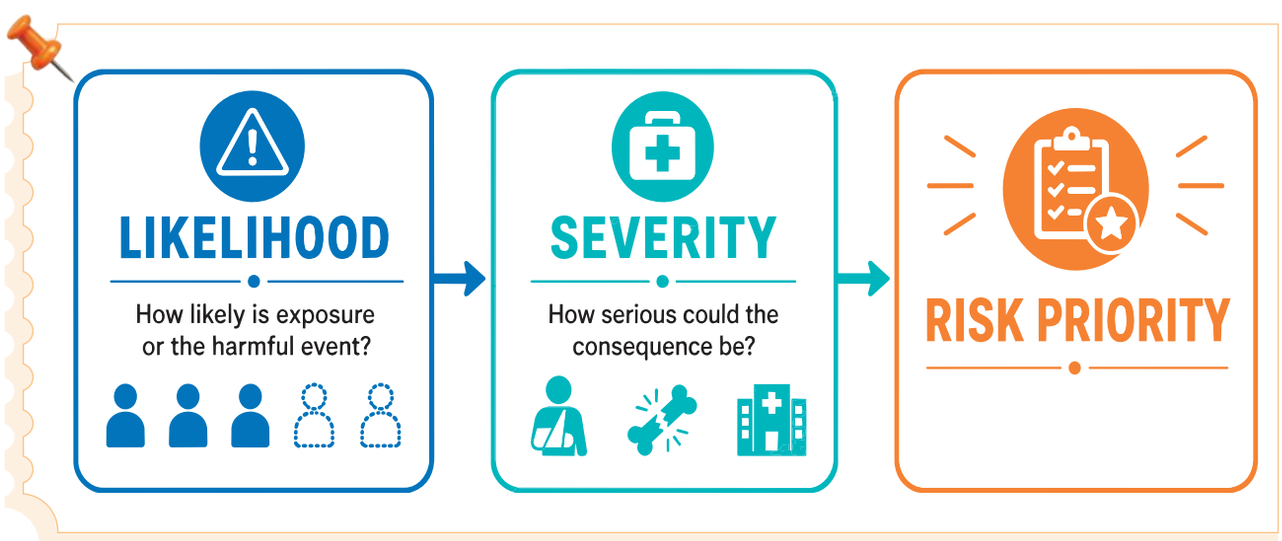


Figure 6.1: Risk assessment considers both the likelihood of a harmful event and the severity of the possible consequence.

SUBTOPIC 6.2 SELECT THE STRONGEST FEASIBLE CONTROL

Risk assessment is not complete simply because you have assigned a priority. The purpose is to decide what action is needed to eliminate or reduce the risk.

A useful principle is: **Control the hazard as close to its source as possible.**

PREVENTION AND PROTECTION

PREVENTION:

Prevention aims to stop the harmful event from occurring or make it less likely. Examples include:

- removing the hazard
- replacing a hazardous substance
- preventing access
- reducing exposure at source

PROTECTION:

Protection aims to limit exposure or reduce the consequence if a hazardous event occurs.

Examples include:

- barriers
- protective systems
- PPE

Prevention and protection often work together.

THE HIERARCHY OF CONTROLS

When selecting a control, begin at the highest feasible level.



1. ELIMINATION

Remove the hazard.

Examples:

- remove an unnecessary hazardous activity
- remove a dangerous obstruction
- design the task so entry into a danger zone is unnecessary

Elimination is strongest because the source of danger is removed.

2. SUBSTITUTION

Replace the hazard with a less hazardous alternative.

Examples:

Replace a hazardous material or process with a safer alternative where this is technically suitable.

3. ENGINEERING CONTROLS

Separate people from the hazard through physical or designed measures. Examples include:

- machine guards
- interlocking systems
- barriers
- local exhaust ventilation
- enclosures
- protective devices

Engineering controls act through the design of the workplace or equipment rather than relying only on a person's behavior.

4. ADMINISTRATIVE CONTROLS

Change how the work is organized or performed. Examples include:

- safe-work procedures
- information
- training
- supervision
- restricted access
- warning arrangements

These controls depend on people understanding and following the required process.

5. PERSONAL PROTECTIVE EQUIPMENT

Protect the individual from remaining exposure. Examples include:

- gloves
- safety glasses
- hearing protection
- protective clothing

PPE may be necessary, but the hazard normally remains present.

For this reason:

Do not choose PPE automatically when a stronger feasible control can act on the source. You will study PPE selection in detail in Lesson 7.

COLLECTIVE PROTECTION BEFORE INDIVIDUAL PROTECTION

Some controls protect everyone exposed to the area, rather than only the person wearing equipment. Examples include:

- barriers and parapets
- safety nets
- local exhaust ventilation
- extraction hoods
- safety cabinets

These are collective protective measures. Where feasible, controlling exposure at the source or protecting the whole group takes priority over relying only on individual PPE.

EXAMPLE — WELDING FUME

Consider two possible responses:

OPTION A

Tell every worker to rely only on respiratory PPE.

OPTION B

Use suitable local exhaust ventilation at the source, together with any additional controls required.

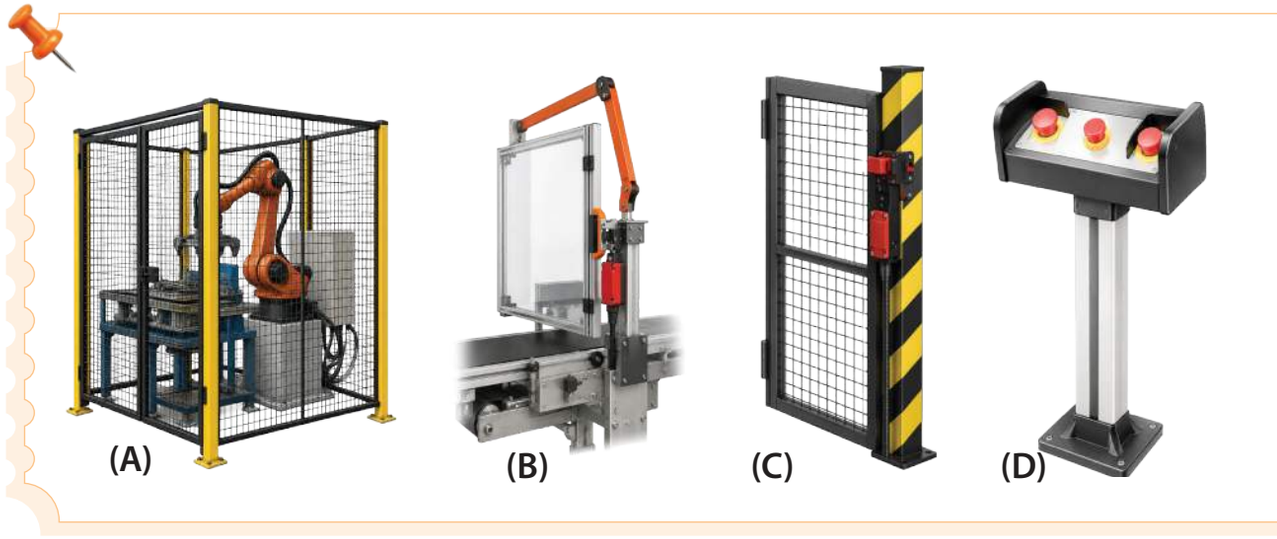
Question: Which acts closer to the source?

Answer: Option B.

PPE may still be required, but it should not automatically replace a feasible source-based control.



Figure 6.2: Control selection should begin with the highest feasible level and should not rely automatically on PPE.



Examples of work-equipment engineering controls: (a) fixed guard, (b) adjustable guard with interlock, (c) interlocked guard and (d) two-hand control device. These measures help prevent or control access to hazardous machine functions and danger zones.

SUBTOPIC 6.3 APPLY CONTROLS TO WORKPLACES AND WORK EQUIPMENT

The hierarchy becomes useful when you apply it to actual workplace conditions.

WORKPLACE CONTROLS

Workplace conditions should support safe movement and work. Examples include:

- stable, non-slip floors
- routes free from dangerous obstructions
- clear pedestrian routes
- suitable separation between pedestrians and vehicles
- emergency routes kept clear



If material repeatedly blocks a route, simply telling workers, "Be careful." is weaker than correcting the condition that creates the obstruction.

COLLECTIVE CONTROLS EXAMPLES:

• BARRIERS AND PARAPETS:

Used to prevent access or falls.

• LOCAL EXHAUST VENTILATION:

Captures harmful contaminants such as fumes close to the source.

• SAFETY CABINETS:

Provide controlled storage for hazardous or flammable substances.

These measures protect more than one individual and can reduce exposure at the source.

• WORK-EQUIPMENT CONTROLS:

Work equipment can create danger zones. A guard creates a physical barrier between the worker and the hazardous area.

Common types of work-equipment controls include:

Guard	Basic function
Fixed guard	Can normally be removed only using tools or by destroying its fastenings
Movable guard	Can be opened without tools
Adjustable guard	Can be adjusted for the operation and remains set during that operation
Interlocked guard	Prevents hazardous functions from operating when the guard is open and sends a stop command if opened during hazardous operation
Interlocked guard with guard locking	Keeps access locked until the hazardous condition has disappeared

· PROTECTIVE DEVICES:

A protective device protects without necessarily being a physical guard. Examples include:

TWO-HAND CONTROL

Requires simultaneous use of two controls so that the operator's hands remain outside the danger zone during activation.

· LIGHT CURTAIN

Detects a person or object entering its sensing field and initiates a stop.

· SAFETY MAT

Detects pressure when someone steps on it and triggers a stop.

· GUARD OR PROTECTIVE DEVICE?

Do not confuse them. Both can form part of an engineering-control strategy.

Guard = physical barrier

Protective device = protective function other than a guard

ELECTRICAL HAZARDS IN A RISK-ASSESSMENT SCENARIO

If an electrical hazard appears in your scenario, your assessment may identify the need for:

- authorized electrical protection
- safe isolation through the approved workplace process
- restricted access
- specialist review

Figure 6.3: Effective controls may act on the workplace, the source of exposure or the equipment danger zone.

SUBTOPIC 6.4 RECORD CORRECTIVE ACTION AND REASSESS THE RISK

A control recommendation needs to be documented clearly. Another person should be able to understand:

- what hazard was assessed
- what controls already existed
- what risk priority was identified
- what additional action is required
- who is responsible for the action
- whether the risk has been reduced after the action is completed



Definition - CORRECTIVE ACTION

A corrective action addresses a condition where the existing risk is not acceptable, and action is required to remove or control it.

Example

A required machine guard is missing. The missing guarding condition requires correction before the equipment is used under the authorized process.

Definition - IMPROVEMENT ACTION

An improvement action further reduces a risk that is already controlled to an acceptable level.

Example

A controlled work area already has adequate separation, but the organization identifies a practical way of improving visibility or access control further.

THREE FORMS OF SAFETY MEASURE

Measures may be:

1. ORGANISATIONAL

Examples:

- information
- training
- supervision
- health surveillance

2. PROCEDURAL

Example:

- a safe-work instruction

3. TECHNICAL

Examples:

- guarding
- barriers
- extraction
- engineered protection

These categories can be used alongside the hierarchy of controls.



Definition - RESIDUAL RISK

Residual risk is the risk that remains after feasible controls have been applied.

Even after good controls are introduced, some risk may remain. This remaining risk may require:

- additional precautions
- training
- monitoring
- PPE
- safe procedures
- review

DO NOT REDUCE THE RATING JUST BECAUSE A CONTROL HAS BEEN SUGGESTED

A proposed control and an implemented control are not the same thing.

Use this sequence:

ASSESS EXISTING CONDITION

↓

SELECT ADDITIONAL CONTROL

↓

IMPLEMENT THROUGH AUTHORISED PROCESS

↓

CHECK THAT THE CONTROL IS IN PLACE

↓

REASSESS THE RESIDUAL RISK

If the control has not been implemented, record it as an action required. Do not pretend that the risk has already been reduced.

INDUSTRIAL EXAMPLE

1. Observation

A machine danger zone is accessible because a required guard appears to be absent.

2. Hazard

Moving machinery/access to hazardous components.

3. Existing controls

- frequency of access
- worker proximity
- task frequency
- whether exposure can occur easily

4. Severity

Consider the credible consequence associated with contact with the moving components.

5. Initial risk priority

Determine this using the authorized matrix or method.

6. Additional control

A suitable engineering control preventing access to the danger zone should be considered before relying on administrative warnings or PPE.

7. Corrective action

Do not operate the equipment under unsafe conditions. Report it for authorized corrective action.

8. Residual risk

Reassess only after the required control has been implemented and confirmed through the authorized process.

Activity 6.1 ASSESS THE RISK AND SELECT THE CONTROL

TIME	GROUPING	RESOURCES	EVIDENCE
30 min	Individual → pairs	Lesson notes, authorized risk-assessment method, hierarchy-of-controls card and Activity 6.1 sheet	Completed risk/control analysis and justified corrective-action statement

TASK:

PART A: LIKELIHOOD OR SEVERITY?

- Write L for likelihood or S for severity.

Question considered during assessment	L / S
How often could workers enter the hazardous area?	
Could the event result in a serious crushing injury?	
Is exposure possible every time the task is performed?	
Could the consequence be a minor cut or a serious laceration?	
Does an existing barrier make contact less likely?	
How serious could the harm be if contact occurs?	

PART B: PUT THE CONTROLS IN ORDER?

- Number the following controls from **1 (highest in the hierarchy) to 5 (lowest in the hierarchy).**

Control	Order
PPE	
Engineering control	
Elimination	
Administrative control	
Substitution	

PART C: WHICH CONTROL IS STRONGER?

CASE 1: WELDING FUME

Two options are proposed:

A: Require workers to rely only on respiratory PPE.

B: Install suitable local exhaust ventilation close to the fume source and assess whether additional PPE is still required

Which option acts closer to the source?

.....

Which hierarchy level does it represent?

.....

.....

CASE 2: MACHINE DANGER ZONE

A machine danger zone can be reached during normal operation.

Suggested actions:

- place a warning sign
- install a suitable guard preventing access
- provide gloves
- tell workers to be careful

Which control should normally be considered before the others?

.....

Why?

.....

.....

PART D: UNDERSTANDING SAFETY CONTROLS

1. A fixed guard normally requires tools, or destruction of its fastenings, before it can be removed.
   True ☐ False ☐
2. Collective protective measures are designed to protect people from workplace health and safety risks.
   True ☐ False ☐
3. Which is the stronger general approach where feasible?
   ☐ Rely only on individual PPE.
   ☐ Control the risk at source using collective or engineering measures.

PART E: CORRECTIVE OR IMPROVEMENT ACTION?

- Write C for Corrective or I for Improvement.

Situation	C / I
A required guard is missing, and the danger zone is accessible.	
A controlled route already meets requirements, but an additional visual aid could make it clearer.	
A hazardous substance is stored in an unacceptable location and needs to be moved through the authorized process.	
Existing controls are acceptable, but exposure can feasibly be reduced further.	

PART F: MINI RISK ASSESSMENT

Scenario: Steel offcuts obstruct part of a frequently used pedestrian route.

Complete the assessment using the authorized risk-assessment method provided by your trainer.

1. **Hazard:**

.....

2. **Who may be exposed?**

.....

3. **Existing controls:**

.....

4. **Likelihood rating/description:**

.....

5. **Evidence supporting likelihood:**

.....

6. **Severity rating/description:**

.....

7. **Credible consequence:**

.....

8. **Initial risk priority:**

.....

9. **Highest feasible control you would consider:**

.....

10. **Why is this control appropriate?**

.....

11. **Corrective action required:**

.....

12. **Residual risk:**

☐ Reassessed after control implementation

☐ Not yet reassessed — control still pending

✓

CHECKPOINT Before you submit

☐ I used the authorized risk-assessment method.
☐ My likelihood judgement is supported by evidence.
☐ My severity judgement matches a credible consequence.
☐ I identified the existing controls before selecting additional action.
☐ I considered higher-level controls before PPE.
☐ My control addresses the hazard or exposure route.
☐ I distinguished corrective action from improvement action.
☐ I did not reduce residual risk before the control was implemented.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN…

Explain how likelihood and severity influence risk.	WITH HELP ☐	INDEPENDENT ☐
Use the authorized workplace or training method to determine risk priority.	WITH HELP ☐	INDEPENDENT ☐
Distinguish an initial risk from the residual risk remaining after controls.	WITH HELP ☐	INDEPENDENT ☐
Explain the difference between prevention and protection.	WITH HELP ☐	INDEPENDENT ☐
Apply the hierarchy of controls.	WITH HELP ☐	INDEPENDENT ☐
Prioritize source-based and collective controls over reliance on PPE where feasible.	WITH HELP ☐	INDEPENDENT ☐
Recognize organizational, procedural, and technical safety measures.	WITH HELP ☐	INDEPENDENT ☐
Apply workplace and work-equipment safety requirements when selecting controls.	WITH HELP ☐	INDEPENDENT ☐
Distinguish corrective action from an improvement action.	WITH HELP ☐	INDEPENDENT ☐
Document why a selected control is appropriate for the identified risk.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION

- Which is more difficult for me to judge: likelihood or severity? Why?
- Why should PPE not automatically be the first control selected?
- Why must residual risk be reassessed after a control is implemented?

EXIT TICKET

- Choose one hazard from Lesson 5 and use it to complete the risk assessment below:
- Hazard:

.....

- Likelihood evidence:

.....

- Possible severity:

.....

- Selected control:

.....

- Hierarchy level:

.....

- Why is the control appropriate?

.....

➔

TRANSFER FROM KNOWLEDGE TO PERFORMANCE

In Practical Session 6, you will use the hazards identified in a simulated industrial area to complete a structured risk assessment and corrective action record.

You will:

IDENTIFY → CHECK EXISTING CONTROLS → ASSESS LIKELIHOOD → ASSESS SEVERITY → PRIORITISE → SELECT CONTROL → RECORD ACTION → REASSESS

6 SESSION

Complete a risk assessment and corrective-action record using likelihood, severity, risk priority and appropriate controls

Skill code: PC1.2 • Duration: 180 minutes / 3 h • Location: Classroom simulation, controlled training workshop or authorized simulated industrial area

SAFETY GATE ASSESS WITHOUT CREATING EXPOSURE

Risk assessment is not a reason to approach a hazard more closely. You must not:

- enter a danger zone
- operate machinery
- remove or test guards
- open electrical equipment
- touch suspect electrical parts
- handle unidentified chemicals
- move unstable or hot materials
- bypass a protective device
- implement a technical control unless specifically authorized

Use the information provided and the observations you can make from an authorized safe position.

Performance task	Complete a risk assessment for hazards identified in the simulated industrial area. For each finding: 1. confirm the hazard and possible consequence. 2. identify who may be exposed. 3. record existing controls. 4. assess likelihood using the authorized method. 5. assess severity using the authorized method. 6. determine initial risk priority. 7. select additional preventive/protective measures using the hierarchy. 8. identify whether the action is corrective or improvement-oriented. 9. identify who must receive or own the action. 10. reassess residual risk only after the simulated control is confirmed as implemented.
Performance standard	• complete at least four full risk assessments. • justify likelihood using observed evidence. • justify severity using a credible consequence. • determine risk priority using the authorized method. • distinguish existing controls from proposed controls. • select controls using the hierarchy.

Performance standard	• choose a source-based or collective control where feasible rather than relying automatically on PPE. • analyse at least one workplace-safety condition. • analyse at least one work-equipment condition. • distinguish corrective action from improvement action. • identify an appropriate action owner/reporting route. • reassess residual risk appropriately. • remain within your authority.
Required evidence	• completed Risk Assessment and Corrective-Action Record • at least four assessed hazards • documented likelihood and severity reasoning • control-selection justification • corrective/improvement classification • residual-risk review where simulated controls have been implemented • one verbal explanation of the highest-priority finding

PREPARATION

RESOURCES

- ☐ Session 5 hazard records where used
- ☐ Authorised workplace/training risk-assessment method
- ☐ Hierarchy-of-controls reference card
- ☐ Risk Assessment and Corrective-Action Record
- ☐ Scenario cards / authorised inspection area
- ☐ Pen or pencil
- ☐ Workplace/equipment-control reference sheet

PRE-START CHECKS

- ☐ I understand the authorised likelihood method.
- ☐ I understand the authorised severity method.
- ☐ I know how the authorised method determines risk priority.
- ☐ I understand the hierarchy of controls.
- ☐ I know the difference between existing and additional controls.
- ☐ I understand the difference between corrective and improvement action.
- ☐ I know that residual risk is reassessed after controls are implemented.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Review the hazard record or scenario and confirm the hazard, exposed persons, and credible consequence.	Complete risk statement.
2	Record the controls already in place at the time of assessment.	Existing controls separated from proposed action.
3	Assess likelihood using the authorised method and justify the rating from observable evidence.	Likelihood rating/description + evidence.
4	Assess severity using the authorised method and justify it using the credible potential consequence.	Severity rating/description + reasoning.
5	Determine the initial risk priority using the authorised method.	Initial risk level/priority recorded.
6	Work down the hierarchy from elimination and select the highest feasible control.	Hierarchy level and control justification recorded.
7	State whether the measure is corrective or improvement-oriented and identify the action owner/reporting route.	Clear corrective-action record.
8	Where the simulated control has been implemented and confirmed, reassess the residual risk. Otherwise record the action as pending.	Valid residual-risk status.
9	Present the highest-priority finding and selected control.	Concise verbal justification.
10	Check that ratings, controls, and actions are internally consistent.	Completed record ready for submission.

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO	WE DO	YOU DO
The trainer demonstrates one controlled industrial scenario: A machine danger zone is accessible because the expected physical guard is absent in the simulated condition. Warning signage is present, but workers may approach the machine during normal work. The trainer verbalises: **1.** "I begin with the hazard already identified: access to moving hazardous machine components." **2.** "I identify who may be exposed — operators, maintenance personnel or anyone who can approach this part of the equipment." **3.** "I record the controls that already exist. In this example, a warning sign is present, but there is no effective physical separation at the access point." **4.** "I assess likelihood using the authorized method. I consider how often people approach the machine, how easily the danger zone can be reached and whether existing controls prevent exposure." **5.** "I do not choose the likelihood rating simply because the machine looks dangerous. I justify it using the evidence."	The whole group assesses several controlled scenarios, such as: • steel material obstructing a frequently used pedestrian route • welding fume generated without effective source extraction • repeated manual handling of awkward steel components • an accessible machine danger zone; • a hazardous substance stored incorrectly • a physical workplace hazard requiring a collective control • a possible electrical hazard requiring authorized specialist protection or safe isolation	You will complete structured risk assessments for assigned simulated industrial hazards or selected findings from Session 5. **You must provide evidence of:** • at least four complete risk assessments • likelihood judgements supported by observable evidence • severity judgements linked to credible consequences • initial risk priorities determined using the authorised method • existing controls recorded separately from proposed controls • control selection using the hierarchy of controls • at least one source-based or collective control where feasible • at least one corrective/ improvement classification • an appropriate action owner or reporting route • residual-risk status recorded correctly • one verbal justification of the highest-priority finding

6. "I assess severity by considering the credible harm if contact with the moving machinery occurs."

7. "The possible consequence may include serious crushing, cutting or other major mechanical injury."

8. "I use the authorised risk-assessment method to determine the initial risk priority."

9. "Now I select additional controls using the hierarchy."

10. "I do not begin automatically with PPE or a warning sign."

11. "Because access to the danger zone is the problem, I consider an engineering control that physically prevents or controls access."

12. "A suitable guard or protective system acts more directly on the exposure route than simply telling workers to be careful."

13. "I record the required action as a corrective action because the simulated condition requires an unacceptable safety deficiency to be addressed."

14. "I identify the authorized person or function responsible for the corrective action."

15. "I do not personally install or modify the guard unless I am trained and authorized."

16. "I do not reduce the risk rating merely because I have recommended a control."

17. "Residual risk is reassessed only after the control has been implemented and its presence or effectiveness has been confirmed through the authorised process."

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/comment
Followed the authorised assessment process and remained within all safety boundaries.	☐	☐	
Correctly identified the hazard, exposed persons and credible potential consequence.	☐	☐	
Recorded existing controls before proposing additional measures.	☐	☐	
Assessed likelihood using the authorized method.	☐	☐	
Justified the likelihood judgement using relevant workplace or scenario evidence.	☐	☐	
Assessed severity using a credible consequence rather than an unsupported worst-case assumption.	☐	☐	
Determined initial risk priority using the authorized risk-assessment method.	☐	☐	
Applied the hierarchy of controls rather than beginning automatically with PPE.	☐	☐	
Considered elimination or substitution where feasible.	☐	☐	
Selected an appropriate engineering or collective control where feasible.	☐	☐	
Identified relevant administrative controls where required.	☐	☐	
Recognized PPE as a possible residual control rather than automatically the primary control.	☐	☐	
Distinguished corrective action from improvement-oriented action.	☐	☐	
Identified an appropriate action owner or reporting route.	☐	☐	
Reassessed residual risk only after the simulated control was confirmed as implemented.	☐	☐	
Communicated the relationship between the identified risk and selected control clearly.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
Risk-assessment reasoning	Uses clear evidence to distinguish exposure conditions, likelihood, and credible severity and explains uncertainty appropriately.	Correctly assesses likelihood and severity using the authorized method.	Ratings are partly correct, but some reasoning is unsupported or inconsistent.	Cannot apply the authorized risk-assessment method or produces unsafe prioritization.
Existing-control analysis	Identifies the function and limitations of existing controls and explains how they influence exposure.	Correctly records relevant existing controls before determining additional action.	Misses some controls or confuses existing controls with proposed measures.	Does not identify existing controls or assumes proposed controls already exist.
Control selection	Selects the highest feasible control, explains why it acts on the source/exposure route and identifies appropriate supporting controls.	Applies the hierarchy and selects an appropriate preventive/protective measure.	Relies mainly on administrative controls or PPE despite a stronger feasible option.	Selects an ineffective, irrelevant, or unsafe control.
Corrective action and residual risk	Clearly distinguishes corrective and improvement actions, identifies ownership and reassesses residual risk only after verified implementation.	Records appropriate action and residual-risk status correctly.	Understands the need for action, but needs support with action classification, ownership or reassessment.	Treats a recommendation as implemented or reduces residual risk without evidence.
Professional communication	Produces precise, traceable and technically justified records linking risk evidence directly to selected controls.	Completes all required fields and clearly explains one risk/control decision.	Records contain omissions or vague justification.	Records are incomplete, unsupported, or communicate unsafe decisions.

7 LESSON

PERSONAL PROTECTIVE EQUIPMENT AND EQUIPMENT-SAFETY INSPECTION

PPE MUST MATCH THE RISK — AND IT DOES NOT REPLACE A SAFE WORKPLACE

In Lesson 6, you learned that PPE is the lowest level in the hierarchy of controls. This does not mean that PPE is unimportant.

It means that PPE should protect you from remaining exposure after stronger feasible controls have been considered. The wrong PPE can create a false sense of protection. For PPE to be useful, you need to check:

- what residual risk remains
- which part of your body requires protection
- whether the PPE is designed for that type of hazard
- whether its performance is suitable
- whether it fits and can be used correctly
- whether it is compatible with other PPE
- whether its condition is acceptable

You also need to remember an important rule:

PPE does not make an unsafe machine, blocked route, or missing guard acceptable.

Safe work requires the correct combination of:

safe workplace + safe equipment + suitable controls + appropriate PPE



WARM-UP

SAME GLOVES, DIFFERENT TASKS

You will be shown two work situations:

TASK A

A worker handles clean cardboard boxes.

TASK B

A worker handles sheet steel with sharp edges. The same pair of general-purpose gloves is available for both tasks.



Discuss:

1. Is "wear gloves" enough information to select PPE?
2. What hazard needs to be controlled in Task B?
3. Which part of the body is exposed?
4. What performance information would you want to check?

5. Could a glove that is suitable for one task be unsuitable for another?

6. What else should you inspect before using the gloves?

Record your starting idea:

.....

.....

.....

BY THE END OF THIS LESSON, I WILL BE ABLE TO:

1. identify appropriate PPE.

2. analyze workplace and equipment safety requirements.

3. apply preventive safety measures in a simulated task.

4. distinguish PPE from ordinary clothing that is not designed for protection.

KEY TERMS

PPE • Residual risk • Category I, II and III • CE marking • Performance level • Pictogram • EN 388 • Compatibility • Fit • Inspection • Guard • Protective device • Danger zone • Collective protection • Defective equipment

SUBTOPIC 7.1 USING PPE AS THE FINAL PROTECTIVE BARRIER



Definition - PERSONAL PROTECTIVE EQUIPMENT

Personal Protective Equipment (PPE) is equipment intended to be worn or held by you to protect against one or more workplace risks that could endanger your safety or health. Accessories or components specifically designed to provide that protection can also form part of PPE.

Examples can include:

- eye and face protection
- safety helmets
- respiratory protection
- protective gloves
- safety footwear
- fall-protection equipment
- hearing protection
- protective clothing

PPE IS NOT THE FIRST ANSWER TO EVERY HAZARD

PPE protects the **individual wearer**. It does not normally remove the hazard itself. For example:

MACHINE DANGER ZONE

Wearing gloves does not make access to moving machinery acceptable.

WELDING FUME

A respirator may be required for residual exposure, but it should not automatically replace effective source extraction where that is feasible.

OBSTRUCTED WALKWAY

Safety footwear does not make material stored across a pedestrian route acceptable. This is why PPE should be used where risks cannot be adequately controlled through stronger technical, collective or organisational measures.

PPE OR ORDINARY CLOTHING?

Not everything you wear at work is PPE. Ordinary workwear or uniforms that are not specifically designed to protect your health or safety are not automatically PPE.

Ask:

Was this item specifically designed to provide protection against an identified risk?

If not, do not assume that it provides protective performance.

THREE PPE CATEGORIES:

CATEGORY I — SIMPLE DESIGN / LOW RISK

Category I PPE is intended for relatively minor hazards where you can normally recognise the onset of the harmful effect in time.

Examples of the types of risks covered include:

- superficial mechanical effects
- minor reversible effects from cleaning products
- contact with hot objects below the stated temperature threshold
- ordinary atmospheric conditions
- superficial mechanical injury
- sunlight

CATEGORY II — INTERMEDIATE RISK

Category II covers PPE that does not fall into Category I or Category III. It includes many common industrial applications.

Examples:

- face shields
- protective gloves
- hearing protection
- safety helmets
- safety footwear
- safety goggles

Category II PPE must carry the relevant conformity marking and undergo the required type examination.

CATEGORY III — HIGHEST RISK

Category III PPE is intended to protect against risks involving:

- death
- serious and permanent injury

Examples include:

- full-face respiratory protection for highly hazardous atmospheres
- fall-arrest harnesses
- firefighter protective equipment

Specific training may be required for certain Category III PPE.

CATEGORY DOES NOT REPLACE RISK-BASED SELECTION

Do not select PPE simply by saying:

"Category III must always be best."

The PPE must be **appropriate to the actual hazard and task.** A higher category does not make an unsuitable item suitable for a different risk.

Category I



Category II



Category III



Figure 7.1: Examples of personal protective equipment across Categories I, II, and III, illustrating protection for minor risks, intermediate industrial risks, and risks that may result in death or serious and permanent injury.

SUBTOPIC 7.2 SELECT PPE FROM THE RISK — NOT FROM HABIT

After the risk assessment has identified residual exposure, PPE selection should follow the risk. Use this sequence:
RESIDUAL RISK ➡ BODY PART EXPOSED ➡ REQUIRED PROTECTION ➡ PERFORMANCE ➡ FIT / COMFORT ➡ COMPATIBILITY.

1. WHAT RISK REMAINS?

Start with the risk assessment from Lesson 6. And think about **what exposure remains after higher-level controls have been considered.**

Examples may involve:

- mechanical contact
- chemical contact
- heat or flame
- noise
- particles or foreign bodies

2. WHICH PART OF THE BODY IS EXPOSED?

Possible body areas include:

- head
- eyes
- face
- hearing
- respiratory system
- hands
- body
- feet

The PPE must protect the part that is actually exposed.

3. WHAT PERFORMANCE IS REQUIRED?

Two items that look similar may provide different levels or types of protection. PPE markings and pictograms help communicate what the item is designed to protect against. For protective gloves, examples include protection against:

- mechanical risks
- chemical risks
- heat and flame
- impact-related cutting risks

4. MECHANICAL-GLOVE PERFORMANCE

For gloves intended to protect against mechanical risks, EN 388 performance testing includes resistance to:

- abrasion
- blade cutting
- tearing
- puncture

The performance information is shown on the glove or its packaging. This means that you should not select a glove only because it is thick or it looks strong. Check the protective performance information relevant to the task.

EXAMPLE — SHARP STEEL:

Hazard: Sharp sheet-metal edge.

Body part exposed: Hands.

Required protection: Protection appropriate to cutting/puncture hazards.

Information to check: Relevant glove pictogram and performance information.

Additional question: Could wearing the glove create or increase another risk in the task?

Key point:

You should not choose PPE based only on hazard. You also need to think about what the worker is actually doing, because the PPE itself could create another problem.

5. ERGONOMICS AND COMFORT

Suitable PPE should allow the task to be performed safely. Selection should consider:

- ergonomics
- worker comfort

PPE that interferes unnecessarily with safe movement or work performance may not be appropriate for the task.

6. COMPATIBILITY

Sometimes you need to wear more than one item. Examples include:

- helmet + eye protection
- eye protection + face shield
- hearing protection + helmet
- respirator + eye/face protection

The items must be compatible when worn together. One item should not prevent another from providing its intended protection.



1



RESIDUAL RISK

Identify the hazard that remains after control measures are in place and assess the level of risk

2



BODY PART EXPOSED

Determine which body part(s) are exposed to the residual risk.

3



PPE TYPE

Select the type of PPE that is suitable for the identified risk and body part

4



PERFORMANCE / MARKING

Ensure the PPE meets the required performance standards and has the correct marking.

5



FIT & COMFORT

Ensure the PPE fits properly and is comfortable enough to wear for the required duration

6



COMPATIBILITY

Ensure the PPE works effectively together and is compatible with other required PPE.

WORKED EXAMPLE



Sharp sheet edge



Hands



Protective gloves



Mechanical performance



Correct fit



Compatible with other required PPE

Figure 7.2: PPE selection should begin with the residual risk and finish with confirmation that the selected items can be worn and used together correctly.

#	Performance level
a	Abrasion resistance (Cycles)
b	Blade cut resistance (Coup test/Index)
c	Tear resistance (Newtons)
d	Puncture resistance (Newton)
e	Shear strength EN ISO (Newton)
f	EN impact protection (Pass/Fail)

Figure 7.3 — EN 388 Mechanical protective-glove markings indicate performance against defined hazards, including abrasion, cutting, tearing, puncture, ISO cut resistance and, where applicable, impact protection.

SUBTOPIC 7.3 FIT, INSPECT, USE AND CARE FOR PPE CORRECTLY

Selecting the correct PPE type is not enough. The item must also be **fit for use.** Different PPE items require different fitting and inspection methods:

Always use:

- the manufacturer's instructions
- the authorized workplace procedure
- the training provided for that PPE

BEFORE YOU USE PPE

Use a simple sequence:

IDENTIFY ➡ CHECK ➡ FIT ➡ COMPATIBILITY ➡ USE

1. IDENTIFY

Confirm that you have the **correct PPE item for the identified risk.**

Check:

- type
- intended protection
- relevant markings or performance information

Do not substitute a visually similar item without confirmation.

2. CHECK THE CONDITION

Before use, inspect the item as required by the authorized procedure. Look for any obvious condition that could affect its protective function. If the PPE appears damaged, contaminated, incomplete or otherwise unsuitable:

Do not simply continue because it is the only item available. Follow the authorized reject/replacement or reporting process.

3. FIT IT CORRECTLY

PPE should be worn or positioned in accordance with the instructions provided for the item. A correct item that is fitted incorrectly may not provide the intended protection.

Do not:

- modify PPE to make it fit
- cut, drill or alter components
- remove protective parts
- improvise attachments

4. CHECK COMPATIBILITY

If you wear more than one item, check that they can be worn together without interfering with their protective function.

5. USE IT FOR THE INTENDED TASK

Do not use PPE for a purpose for which it was not designed.

AFTER USE

Follow the authorized arrangements for:

- storage
- care
- maintenance
- replacement
- reporting defects

The exact process depends on the PPE supplied. The important principle is:

PPE must remain in a condition that allows it to provide its intended protection.

ACCEPT OR REJECT?

During an inspection, your decision should be based on the information available.

ACCEPT FOR THE TRAINING TASK

The item:

- matches the identified protection requirement
- has the required information available
- is in an acceptable visible condition
- can be fitted according to instructions
- is compatible with the other PPE required

REJECT / REFER

The item:



- is the wrong type
- does not provide the required protection
- has relevant information missing or unreadable
- has an obvious defect affecting use
- cannot be fitted correctly
- conflicts with another required PPE item
- requires specialist inspection that has not been completed

IMPORTANT — DO NOT CLAIM MORE THAN YOU HAVE CHECKED

Your inspection does not replace any specialist examination, testing or certification required for particular PPE. Record only what you are authorized and competent to confirm.

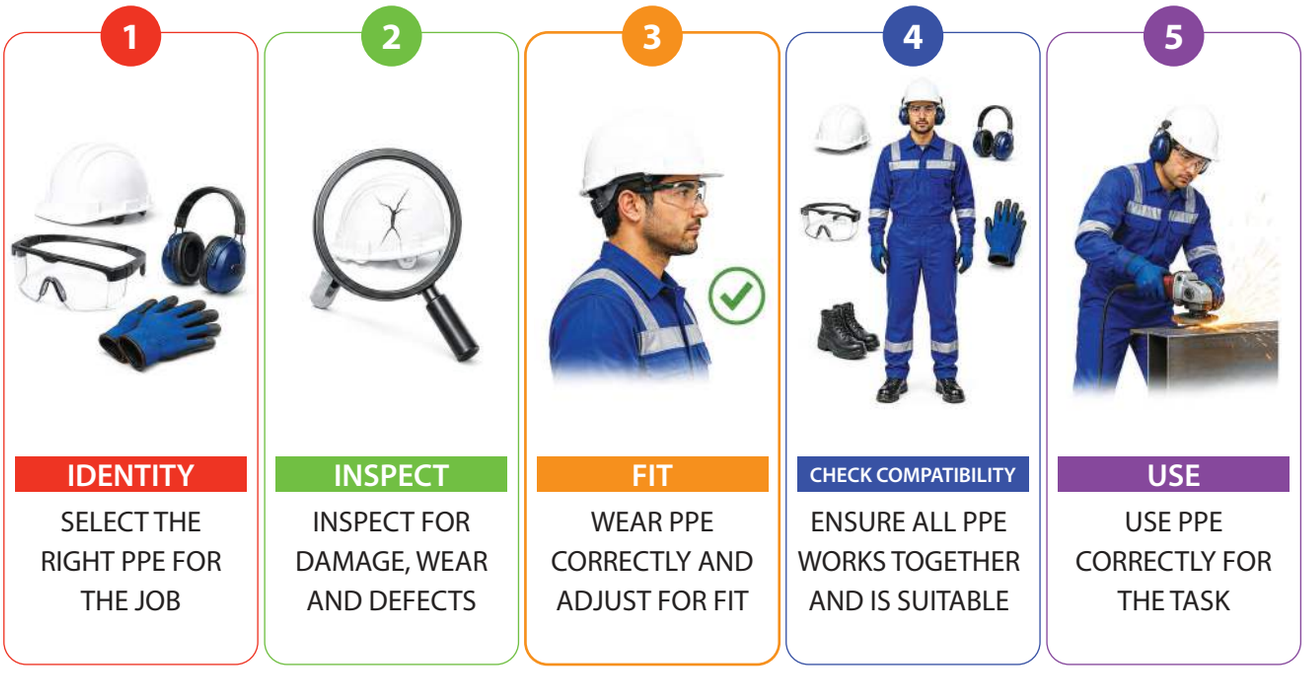


Figure 7.4: Correct PPE selection must be supported by appropriate pre-use checks, fitting, compatibility and authorised care arrangements.

SUBTOPIC 7.4 PPE DOES NOT REPLACE SAFE WORKPLACES OR SAFE EQUIPMENT

PPE is only one part of the safety system. Accident prevention also depends on:

correct PPE + safe workplace + safe work equipment

You should therefore check the surrounding work conditions before relying on PPE.

CHECK THE WORKPLACE

Examples of workplace conditions:

• FLOORS AND ACCESS ROUTES

Look for routes that are:

- stable
- clear
- suitable for safe movement

A blocked pedestrian route should be corrected.

Do not say: "Workers have safety boots, so the obstruction is acceptable."

• CHECK THE EQUIPMENT DANGER ZONE

A **danger zone** is an area in or close to work equipment where a person's presence creates a safety or health risk.

Think about:

- Is access physically controlled?
- Is the expected guard present?
- Is the guard visibly in position?
- Is a protective device provided where required?
- Is an obvious equipment defect present?

• GUARDS

A **guard** is a physical barrier between you and a hazardous machine area. Types introduced earlier include:

- fixed guards
- movable guards
- adjustable guards
- interlocked guards
- interlocked guards with guard locking

• PROTECTIVE DEVICES

A protective device provides protection by another means. Examples include:

- **TWO-HAND CONTROL**

Requires both controls to be operated together, helping keep the hands outside the danger zone.

- **LIGHT CURTAIN**

Detects entry into its sensing field and initiates a machine stop.

- **SAFETY MAT**

Detects pressure when someone enters a protected area and triggers a stop.



Key Idea:

GUARD ≠ PROTECTIVE DEVICE

Remember:

Guard = physical barrier

Protective device = another protective means

Both can help prevent access or exposure to a danger zone.

DO NOT USE PPE TO COMPENSATE FOR A MISSING CONTROL

Consider this case: A machine has an accessible moving-part danger zone because the required guard is missing.

Someone suggests: "Just give the operator thicker gloves."

Is this acceptable? The answer is **No**.

The problem is uncontrolled access to the danger zone. PPE does not replace the required equipment-safety control.

The appropriate response is to:

identify ➡ prevent unsafe use within your authority ➡ report for corrective action

DEFECTIVE-EQUIPMENT REPORTING

If you observe:

- a missing guard
- a displaced guard
- an obvious defective protective device
- damaged equipment
- an unsafe access condition

do not carry out an unauthorised repair. Record the following:

1. what you observed
2. where it is
3. why it affects safe use
4. what action/reporting route is required



Figure 7.5: PPE protects against residual exposure; it must not be used as a substitute for required workplace or equipment-safety controls.

Activity 7.1 SELECT THE PPE - THEN CHECK THE WHOLE SAFETY SYSTEM

Purpose: Apply PPE-selection principles, interpret protective information and recognise when workplace or equipment controls are still required.

TIME	GROUPING	RESOURCES	EVIDENCE
20 min	Individual ➡ pairs	PPE images/samples, glove-marking cards, risk scenarios and Activity 7.1 sheet	Completed PPE-selection table, marking interpretation and safety-control decisions

TASK:

PART A: PPE OR NOT PPE?

Decide whether each item is PPE **for the situation described**:

Item	PPE?	Reason
Ordinary company polo shirt with no protective function	☐Yes ☐No	
Protective gloves designed for a defined mechanical risk	☐Yes ☐No	
Safety goggles designed to protect the eyes	☐Yes ☐No	
Ordinary uniform worn only for company identification	☐Yes ☐No	
Safety footwear intended to protect the feet	☐Yes ☐No	

PART B: CHECK THE PPE CATEGORIES

1. **Category 1 PPE is designed primarily for:**

☐ Minor/low-level risks
☐ Risks of death only
☐ Electrical equipment only
☐ Every industrial risk

2. **PPE that does not belong to Category 1 or Category 3 generally belongs to:**

☐ Category 0
☐ Category 2
☐ Category 4
☐ No category

3. **Give one example of a type of high-risk PPE associated with Category 3.**

.....

.....

PART C: READ THE GLOVE INFORMATION

A protective glove carries a marking for mechanical protection.

1. **What types of mechanical performance may need to be considered?**
Tick all that apply.

☐ Abrasion resistance
☐ Blade-cut resistance

☐ Tear resistance
☐ Puncture resistance

2. Under EN 388, blade-cut resistance performance relates to:

☐ Performance against laceration/cutting
☐ Biological protection
☐ Thermal protection
☐ Hearing protection

3. True or false?

Abrasion-resistance testing applies to gloves intended to protect against mechanical risks

True ☐ False ☐

4. Why should you check the glove performance information instead of selecting a glove only by appearance?

.....

.....

PART D: SELECT PPE FROM THE RISK

Complete the table.

Residual risk/task	Body part exposed	PPE type to consider	Information you would check
Sharp metal handling			
Flying particles			
High workplace noise where hearing PPE is required by the authorized assessment			
Residual contact risk to the feet from workplace material			

Do not assume that one item is suitable for every version of the task.

PART E: FIT AND CONDITION

a. A pair of gloves is the correct protection type for the task, but one glove is visibly damaged.

Answer the following:

1. Is "correct PPE type" enough to accept it for use?

☐ Yes ☐ No

2. What should you do?

.....

3. Why?

.....

.....

b. A worker needs to wear both eye protection and another item of face/head PPE.

What additional factors must be checked?

.....

PART F: PPE OR HIGHER-LEVEL CONTROL?

For each situation, decide whether PPE alone is an adequate response.

CASE 1. WELDING FUME EXPOSURE

☐ PPE alone is adequate
☐ Equipment-safety corrective action is required

Explain:

.....

.....

CASE 2. WELDING FUME EXPOSURE.

Welding fume is generated, and suitable source extraction can be provided. Which should normally be considered before relying only on respiratory PPE?

.....

.....

CASE 3. Steel is stored across a pedestrian route. Which is the better response?

☐ Tell workers to wear safety footwear and step around it.
☐ Correct the obstructed-route condition through the authorised process.

Explain:

.....

PART G: GUARD OR PROTECTIVE DEVICE?

Match the item:

Item	Guard	Protective device
• Fixed physical barrier	☐	☐
• Light curtain	☐	☐
• Two-hand control	☐	☐
• Interlocked physical guard	☐	☐
• Safety mat	☐	☐

PART H: SAY IT PROFESSIONALLY

Read the following statement and rewrite it as a professional statement:
"The machine is okay because the operator has gloves."

.....

.....

CHECKPOINT Before you submit

- ☐ I selected PPE from the residual risk.
- ☐ I identified the body part requiring protection.
- ☐ I considered relevant performance information.
- ☐ I distinguished PPE from ordinary work clothing.
- ☐ I considered fit and compatibility.
- ☐ I rejected visibly unsuitable PPE rather than relying only on its type.
- ☐ I distinguished a guard from a protective device.
- ☐ I recognized that PPE cannot replace required equipment or workplace controls.

Check, reflect and transfer

BY THE END OF THIS LESSON, I CAN:

Explain what PPE is and why it is considered a residual or last-resort protective measure.	WITH HELP ☐	INDEPENDENT ☐
Distinguish PPE from ordinary clothing that is not designed for protection.	WITH HELP ☐	INDEPENDENT ☐
Recognize the three broad PPE risk categories.	WITH HELP ☐	INDEPENDENT ☐
Select PPE according to the hazard, exposed body part and required performance.	WITH HELP ☐	INDEPENDENT ☐
Interpret basic PPE markings and protective-glove performance information.	WITH HELP ☐	INDEPENDENT ☐
Consider ergonomics, comfort and compatibility when several PPE items are worn together.	WITH HELP ☐	INDEPENDENT ☐
Check the correct fit, use and visible condition of PPE using the authorized instructions.	WITH HELP ☐	INDEPENDENT ☐
Identify when PPE should be rejected, replaced or referred for authorized review.	WITH HELP ☐	INDEPENDENT ☐
Recognise workplace and equipment-safety conditions that PPE cannot replace.	WITH HELP ☐	INDEPENDENT ☐

REFLECTION	EXIT TICKET
• What information should I check before selecting PPE?	• Choose one residual risk. • Residual risk:
• Why can two visually similar gloves provide different protection?	• Body part requiring protection:
• Why should PPE never be used to justify working with a missing machine guard?	• PPE selected:
	• Performance/marking to check:
	• Fit/compatibility check:

TRANSFER FROM KNOWLEDGE TO PERFORMANCE

In Practical Session 7, you will use authorized PPE samples and controlled workplace/equipment examples.

You will:

REVIEW THE RISK ➡ SELECT ➡ CHECK MARKINGS ➡ INSPECT ➡ FIT ➡ CHECK COMPATIBILITY ➡ USE / REJECT ➡ CHECK WORKPLACE & EQUIPMENT ➡ RECORD / REPORT:

7 SESSION — Select, fit and inspect PPE and complete workplace and equipment-safety checks

Skill code: PC1.2 / TPC1.2 • Duration: 240 minutes/ 4 h • Location: Training workshop, PPE practical area or authorised simulated industrial environment

SAFETY GATE **SIMULATION ONLY — NO DELIBERATE HAZARD EXPOSURE**

This practical session uses controlled tasks and training PPE. You are demonstrating that you can make the correct PPE decision **before exposure occurs.** Do not deliberately expose yourself to a hazard to test the PPE.

You must not:

- approach moving machinery to test protective clothing.
- touch a sharp edge to test gloves.
- enter a noisy hazardous environment simply to test hearing protection.
- expose PPE to chemicals.
- test respiratory PPE in a hazardous atmosphere.
- test fall-protection PPE by creating a fall.
- enter a machine danger zone.
- bypass guards or protective devices.

Performance task	For each assigned simulated task: **1.** review the hazard and residual-risk information. **2.** identify the body part requiring protection. **3.** select the appropriate PPE type from the authorized choices. **4.** check the PPE's markings and performance information. **5.** inspect its visible condition. **6.** fit it according to the provided instructions. **7.** check comfort and compatibility with other required PPE. **8.** accept or reject the item for the simulated task. **9.** complete a short workplace/equipment-safety check. **10.** record and explain your decision.

Performance standard	• complete at least four PPE-selection and inspection tasks. • link each PPE selection to an identified residual risk. • identify the exposed body part correctly. • select an appropriate PPE type. • interpret relevant markings or performance information provided. • check visible condition before use. • fit the supplied PPE according to the authorised instructions. • identify obvious fit or compatibility problems. • reject unsuitable or visibly defective PPE. • explain why PPE does not replace a missing higher-level control. • complete the required PPE-selection record.
Required evidence	You will produce: • completed **PPE Selection, Fitting and Inspection Record** • direct assessor observation • PPE marking/performance interpretation • accept/reject decision • fit and compatibility checks • short workplace/equipment-safety checklist • defective-PPE/equipment reporting statement where applicable • one verbal PPE-selection justification

PREPARATION

PPE TRAINING ITEMS	RESOURCES	PRE-START CHECKS
The trainer may provide controlled examples such as: ☐ Safety glasses/goggles ☐ Face shield ☐ Protective gloves ☐ Hearing protection ☐ Safety helmet ☐ Safety footwear ☐ Protective clothing ☐ Other authorized training PPE: _____	☐ Session 6 risk-assessment extracts ☐ PPE-selection record ☐ PPE markings/reference card ☐ Manufacturer fitting/use instructions ☐ Equipment-safety checklist ☐ Training equipment or photographs ☐ Pen or digital recording device where authorized	☐ I understand the residual risks in the assigned tasks. ☐ I know that PPE is selected after considering higher-level controls. ☐ I know which PPE items I am authorized to handle. ☐ I have the manufacturer's/trainer fitting instructions. ☐ I know how to report visibly damaged PPE.

WORKFLOW

Step	Action	Evidence/checkpoint
1	Read the task and identify the residual hazard, exposure, and body part requiring protection.	PPE need is linked to the risk assessment.
2	Choose an appropriate PPE type from the authorized items.	PPE type matches the identified protection need.
3	Read available category, CE, pictogram, standard, or performance information relevant to the item.	Marking/performance information recorded accurately.
4	Examine the PPE's visible condition before use according to the authorized procedure.	Accept/reject decision supported by observation.
5	Fit or position the PPE according to the provided instructions.	Correct fitting demonstrated.
6	Where several items are required, check that they can be worn together without obvious interference.	Compatibility confirmed or problem identified.
7	Accept the PPE for the simulated task or reject/refer it where the requirements are not met.	Decision justified.
8	Complete the PPE and equipment-safety record and report any defect or discrepancy.	Traceable practical evidence.
9	Give one verbal justification linking risk ➡ PPE ➡ performance ➡ fit ➡ workplace/equipment controls.	Oral evidence.

DEMONSTRATE, PRACTISE AND ASSESS

DEMONSTRATION SEQUENCE

I DO	WE DO	YOU DO
The trainer demonstrates one PPE-selection case. Scenario: A worker must handle sheet steel where residual sharp-edge risk remains after higher-level controls have been considered. Several glove types are available. The trainer verbalizes: **1.** "I start with the residual risk, not with the PPE." **2.** "The hands require protection." **3.** "I select gloves intended for the relevant mechanical risk." **4.** "I check the marking and required performance information." **5.** "I inspect the condition before use." **6.** "I fit the gloves according to the supplied instructions." **7.** "I check compatibility with other required PPE." **8.** "If the gloves are damaged, unsuitable or incorrectly fitted, I reject or refer them." **9.** "PPE does not replace required guarding or another higher-level control." **10.** "I record and explain my decision."	The group analyses controlled examples involving: • eye and face protection • hearing protection • hand protection • defective PPE	You will rotate through PPE and equipment-safety stations. You must demonstrate: • at least four PPE selection and inspection decisions • one fit check • one compatibility check • one reject or refer decision • one workplace or equipment-safety finding • one verbal justification linking the residual risk to the selected PPE and any required higher-level control

OBSERVATION CHECKLIST

Performance criterion	Observed	Not yet	Assessor evidence/ comment
Linked PPE selection to an identified residual risk.	☐	☐	
Identified the body part requiring protection.	☐	☐	
Selected an appropriate PPE type.	☐	☐	
Checked relevant marking/performance information.	☐	☐	
Inspected visible condition before use.	☐	☐	
Fitted PPE according to authorised instructions.	☐	☐	
Checked compatibility where several PPE items were required.	☐	☐	
Accepted, rejected or referred PPE appropriately.	☐	☐	
Recognised that PPE does not replace higher-level controls.	☐	☐	
Identified guarding or protective-device concerns where present.	☐	☐	
Remained within my authority and did not test PPE through hazardous exposure.	☐	☐	
Completed the PPE-selection record accurately.	☐	☐	
Explained one PPE decision clearly using technical language.	☐	☐	

PERFORMANCE RUBRIC

Dimension	4 • Exceeds	3 • Competent	2 • Developing	1 • Not yet
PPE selection	Precisely links residual risk, body part, and required performance and considers alternative options.	Selects suitable PPE for the identified risk.	Generally suitable selection, but some reasoning is incomplete.	Selects PPE unrelated to the risk or relies on appearance alone.
Marking/ inspection	Accurately interprets relevant information and identifies less-obvious condition issues.	Checks required markings and visible condition correctly.	Requires support interpreting information or identifying defects.	Ignores markings or accepts clearly unsuitable PPE.
Fit/ compatibility	Fits PPE confidently and identifies subtle compatibility issues.	Demonstrates correct fit and compatibility checks.	Requires occasional fitting guidance.	Uses PPE incorrectly or ignores incompatible combinations.
Workplace/ equipment safety	Clearly explains why PPE cannot replace required higher-level controls.	Identifies relevant access, guarding or equipment concerns.	Recognizes obvious concerns, but needs support identifying the required response.	Uses PPE to justify an unsafe workplace/ equipment condition.
Records/ communication	Produces concise, technically precise and action-oriented evidence.	Completes records and explains decisions clearly.	Record contains omissions or vague reasoning.	Evidence is incomplete, unsupported or unsafe.

LESSON 8

UNIT REVISION

BRING THE WHOLE SAFETY PROCESS TOGETHER



You have now completed Unit 1. During the previous lessons, you learned how to move from recognizing a workplace condition to making a safe, evidence-based decision. You have learned how to define hazards and risks, interpret workplace safety requirements, understand responsibilities, interpret European directives, technical standards and product-safety information, recognize occupational-health and work-related stress concerns, identify industrial hazards, assess likelihood and severity, select preventive and protective measures, and select, fit and inspect PPE. In this revision, you will bring those skills together.

The professional workflow behind the revision is:

INTERPRET → OBSERVE → IDENTIFY → ASSESS → CONTROL → CHECK RESIDUAL RISK → SELECT PPE → CHECK WORKPLACE / EQUIPMENT → RECORD → REPORT

As a future technician, you should be able to move through this process without treating each stage as a completely separate task.

REVISION INFORMATION:

Item	Information
Duration	**2 hours / 120 minutes**
Sections	**5 sections:** Section A: Safety foundations, responsibilities and technical requirements Section B: Occupational health and work-related stress Section C: Risk assessment and preventive measures Section D: Practical hazard identification, risk and control Section E: Practical PPE and workplace/equipment safety
Questions	16 questions

Section A: SAFETY FOUNDATIONS, RESPONSIBILITIES AND TECHNICAL REQUIREMENTS

QUESTION 1:

Define the following terms:

a. Hazard

.....

b. Risk

.....

c. Prevention

.....

QUESTION 2:

WORKPLACE RESPONSIBILITIES

A new maintenance trainee is instructed to use a machine, but has not yet received the required task-specific training.

a. What safety requirement should be recognised?

.....

b. What should the trainee do?

.....

c. Why should the trainee not simply begin because a more experienced worker says, "It will be fine"?

.....

.....

QUESTION 3:

DIRECTIVES, STANDARDS AND CE MARKING

Complete the following statements:

a. A European product directive establishes:

.....

b. A technical standard provides:

.....

c. What does CE marking communicate?

.....

d. State one thing CE marking does NOT mean.

.....

QUESTION 4:

EQUIPMENT INFORMATION

A machine nameplate identifies the equipment as:

Model MX-200

The maintenance instructions supplied beside the machine are marked:

Model MX-200A

a. What discrepancy should you record?

.....

b. What should happen next?

.....\

Section B: OCCUPATIONAL HEALTH AND WORK-RELATED STRESS

QUESTION 5:

CLASSIFY THE OCCUPATIONAL RISK

Classify each condition as mainly:

- Safety risk
- Physical occupational-health risk
- Psychosocial/organisational risk

Condition	Classification
• A sharp metal edge projects into a work route	
• Workers experience repeated high workplace noise	
• A team receives conflicting priorities from two supervisors	

QUESTION 6:

WORK CONTEXT OR JOB CONTENT?

Classify each factor as Work Context or Job Content.

Condition	Work Context / Job Content
• Conflicting instructions and unclear responsibilities	
• Repeated extended and unpredictable working hours	
• Excessive workload and time pressure	

QUESTION 7:

FACTUAL OCCUPATIONAL-HEALTH REPORTING

A worker has completed several extended shifts while managing several urgent deadlines.

A colleague writes:

"The worker has a stress disorder."

Explain why this statement is inappropriate.

.....\

Rewrite it as a factual workplace observation.

.....

Section C: RISK ASSESSMENT AND PREVENTIVE MEASURES

QUESTION 8:

LIKELIHOOD AND SEVERITY

For each item, state whether it mainly affects likelihood or severity.

Evidence	Likelihood / Severity
• Workers pass the hazard many times during each shift	
• Contact could cause a serious crushing injury	
• A physical barrier prevents normal access to the hazard	
• The credible consequence could be a minor cut	

QUESTION 9:

SELECT THE STRONGER CONTROL

A machine danger zone is accessible because the expected physical guard is absent. Three actions are suggested:

- place another warning sign
- provide thicker gloves
- restore a suitable engineering control that prevents access to the danger zone

a. Which action should normally be considered the strongest feasible response?

.....\

b. Which level of the hierarchy does it represent?

.....\

c. Explain why PPE alone is not an adequate response.

.....\

QUESTION 10:

CORRECTIVE ACTION AND RESIDUAL RISK

A required guard is missing. A replacement guard has been ordered, but has not yet been installed.

a. Is this mainly a corrective action or an improvement action?

.....

b. Can the residual-risk rating be reduced simply because the replacement guard has been ordered?

☐ **Yes**

☐ **No**

c. Explain your answer.

.....

.....

Section D: PRACTICAL PERFORMANCE — HAZARD IDENTIFICATION, RISK AND CONTROL

Complete this section in the controlled steel-workshop or simulated industrial area provided by the trainer.

QUESTION 11:

IDENTIFY AND RECORD FOUR HAZARDS

Inspect the authorised area. Do not touch, operate or alter equipment merely to identify a hazard.

For four credible findings, complete the record.

No.	Location	Observed condition/hazard	Who may be exposed?	Potential consequence	Immediate safe action/report
1					
2					
3					
4					

QUESTION 12:

ASSESS ONE SIGNIFICANT HAZARD

Select one of your findings.

Hazard/record number

.....

Existing controls

.....

Likelihood rating/description

.....

Evidence supporting the likelihood

.....

Severity rating/description

.....

Credible consequence

.....

Initial risk priority

.....

Selected additional control

.....

Hierarchy level

☐ Elimination

☐ Substitution

☐ Engineering

☐ Administrative

☐ PPE

Corrective action/action owner

.....

.....

Residual-risk status

☐ Reassessed after confirmed control

☐ Action pending — not yet reassessed

☐ Requires authorized specialist review

QUESTION 13:

VERBAL HAZARD REPORT

Give the trainer a concise verbal report containing:

location → hazard → exposure/consequence → risk priority → required control/action

The report should be:

☐ Clear and accurate

☐ Partially complete

☐ Further evidence required

Section E: PRACTICAL PERFORMANCE — PPE AND WORKPLACE / EQUIPMENT SAFETY

Complete the PPE task provided by the trainer.

QUESTION 14:

SELECT, INSPECT AND FIT PPE

Residual risk

.....

Body part requiring protection

.....

PPE selected

.....

Relevant marking/performance information checked

.....

Why is this PPE appropriate for the residual risk?

.....

Visible condition

☐ Acceptable

☐ Not acceptable

☐ Requires authorized review

Correct fit demonstrated according to supplied instructions

☐ **Yes** ☐ **No** ☐ **Further support required**

Compatibility with other required PPE

☐ Acceptable

☐ Issue identified

☐ Not applicable

Final decision

☐ Accept

☐ Reject

☐ Refer

Reason

.....

QUESTION 15:

WORKPLACE / EQUIPMENT-SAFETY CHECK

Inspect the associated controlled workplace/equipment scenario. Record one finding for each relevant area.

Check	Observation/decision
• Guard / protective device / danger-zone control	
• Defective equipment or other safety condition	

Does the selected PPE make a missing higher-level control acceptable?

☐ **Yes** ☐ **No**

QUESTION 16:

VERBAL PPE / SAFETY JUSTIFICATION

Explain your decision using the sequence:

residual risk → body part → PPE → performance → condition / fit → higher-level control